

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST- BUCKLING PHENOMENA A COLUMN WITH SEMI-RIGID CONNECTIONS: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

INCREMENTAL DISPLACEMENT

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$



m

N+1

Initial displacement
Initial velocity
Initial acceleration

NONLINEAR
SPRING

$EI, \rho A$

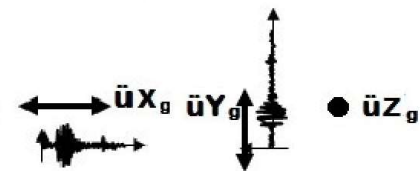
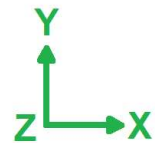
Initial
Imperfection



m

1

NONLINEAR
SPRING



$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

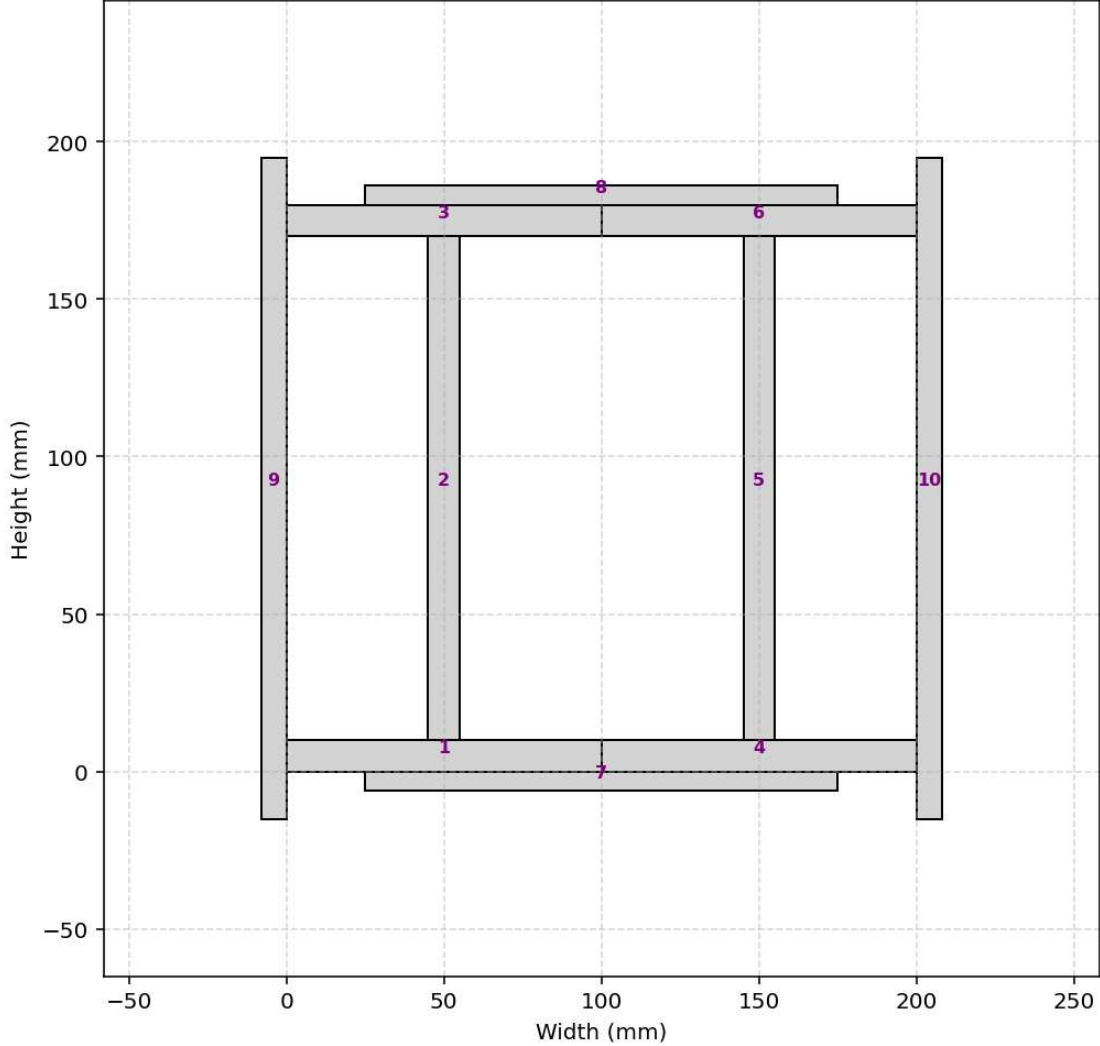
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

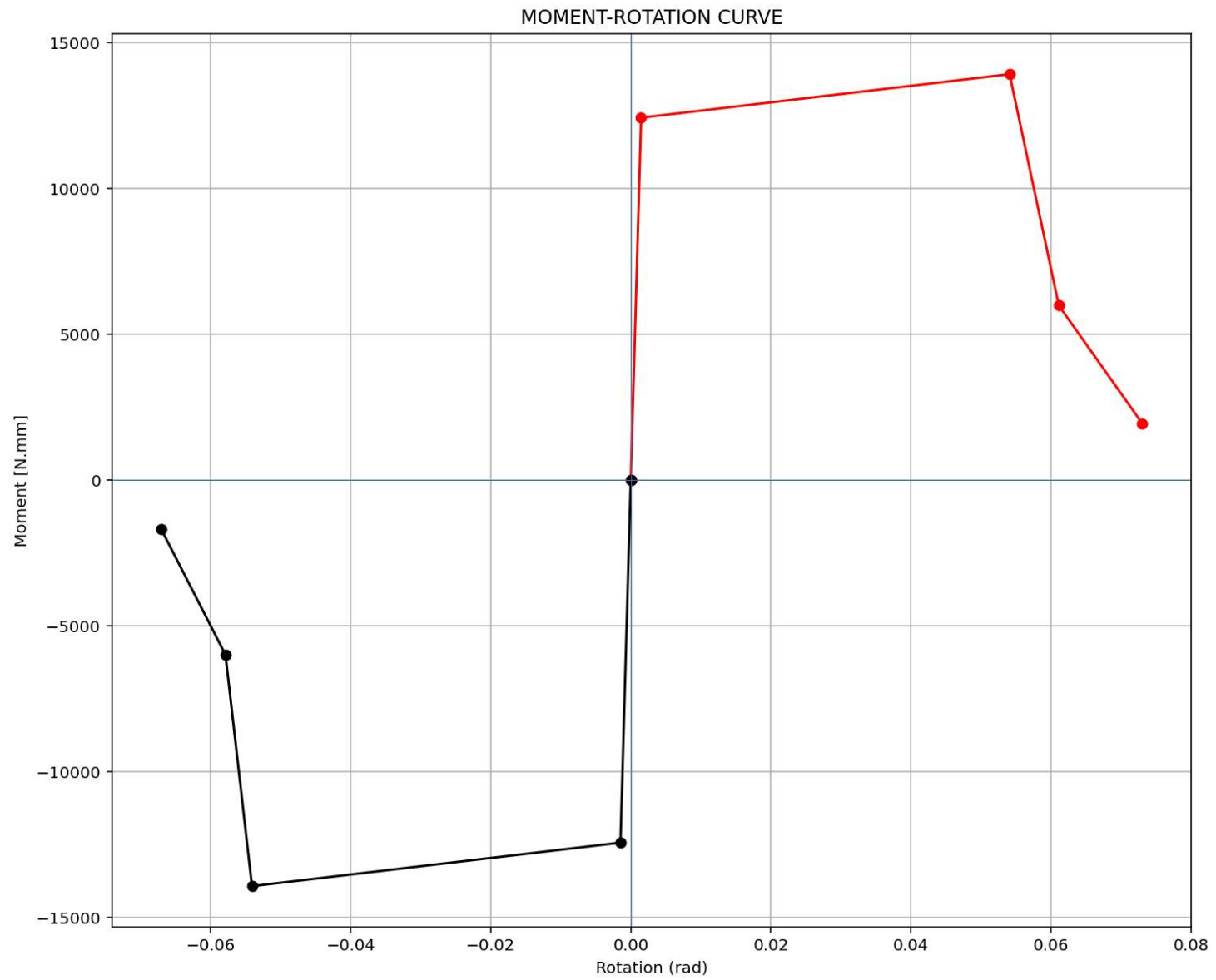
$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$

boundary conditions

$$v(0,t) = v(L,t) = 0, \quad \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=0} = \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=L} = 0.$$

Steel section (10 layers)





Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\HEIGHT_ANALY...P(t)_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_STEEL_CONCRET...-BUCKLING_8_ANA.py x PLATE_DOUBLE_I_STE...HEIGHT_FUN_EXTRA.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF MULTI-MODE POST-BUCKLING PH
4 # AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STA
5 # DYNAMIC TIME-HISTORY ANALYSIS
6 #-----
7 # P(t) = P0 sin(wt)
8 # P(t) = P0 exp(-0.05wt) sin(wt)
9 #-----
10 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
11 # OF ENERGY DISSIPATION CAPACITY IND
12 #-----
13 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
14 # EMAIL: salar.d.ghashghaei@gmail.com
15 #####
16 """
17 Nonlinear Seismic Performance Assessment of a column:
18 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
19
20 This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
21 for performance-based earthquake engineering. The 2D model incorporates bo
22 (elastic-perfectly plastic or hysteretic steel with strain hardening)
23 and geometric nonlinearity to capture P-delta effects
24 and large displacements-essential for collapse assessment.
25
26 Eight analysis protocols are implemented:
27 (1) [PERIOD] : Structural Period
28 (2) [STATIC] : Gravity load analysis establishing dead load state
29 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
30 and plastic mechanism identification
31 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
32 pinching behavior, and energy dissipation degradation
33 (5) [STATIC_EXTERNAL_TIME-DEPENDENT LOADING] : Static Analysis of External
34 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT LOADING] : Dynamic Analysis of Externa
```

Steel section (10 layers)

The diagram shows a rectangular cross-section of a steel section with 10 layers. The vertical axis is labeled 'Height (mm)' and ranges from -50 to 200. The horizontal axis is labeled 'Width (mm)' and ranges from -50 to 250. The section is divided into 10 layers, numbered 1 through 10. Layer 1 is at the bottom, and layer 10 is at the top. The section is composed of two vertical plates and two horizontal plates. The vertical plates are labeled 2 and 10, and the horizontal plates are labeled 3 and 6. The section is shown in a perspective view, with the top and bottom plates being slightly offset from each other.

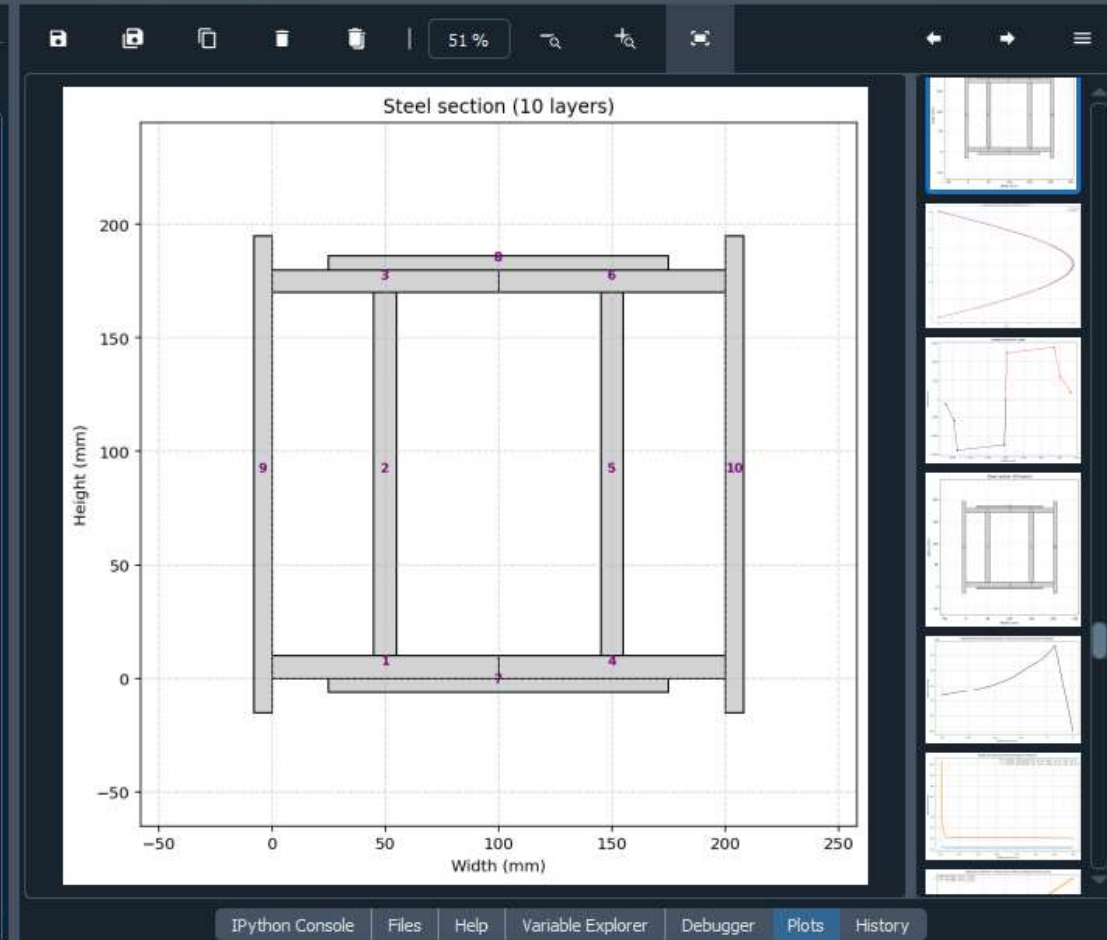
IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 811, Col 44 UTF-8 CRLF RW Mem 48%

```

21 # Define parameters (units: mm, N)
22 # -----
23 # STEEL
24 fy = 240      # [N/mm²] Steel Yield Strength
25 Es = 2e5      # [N/mm²] Modulus of Elasticity
26 ey = fy/Es    # [mm/mm] Steel Yield Strain
27 fu = 1.1818*fy # [N/mm²] Steel Ultimate Strength
28 esu = 0.15    # [mm/mm] Steel Ultimate Strain
29 Esh = (fu - fy)/(esu - ey)
30 Bs = Esh / Es
31
32 steelTag = secTag + 100
33 if STEEL_TYPE == 'ELASTIC':
34     ops.uniaxialMaterial('Elastic', steelTag, Es)
35 if STEEL_TYPE == 'INELASTIC':
36     pinchX = 0.8 # Pinching factor in X direction
37     pinchY = 0.5 # Pinching factor in Y direction
38     damage1 = 0.0 # Damage due to ductility
39     damage2 = 0.0 # Damage due to energy
40     beta = 0.1   # Stiffness degradation parameter
41     ops.uniaxialMaterial('Hysteretic', steelTag,
42                          fy, ey,
43                          fu, esu,
44                          0.2*fu, 1.1*esu,
45                          -fy, -ey,
46                          -fu, -esu,
47                          -0.2*fu, -1.1*esu,
48                          pinchX, pinchY,
49                          damage1, damage2, beta)
50     # INFO LINK: https://opensees.berkeley.edu/wiki/index.php/Hysteretic
51
52 # ----- Fiber section -----
53 ops.section('Fiber', secTag)
54

```



STATIC ANALYSIS

SPYDER (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\EIGHT_ANALY...P(t)_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_STEEL_CONCRET...-BUCKLING_8_ANA.py x PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py x

```
1183 #####
1184 # STATIC ANALYSIS
1185 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1186 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1187 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1188 ANAL_TYPE = 'STATIC'
1189
1190 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS,
1191 (reactionX, reactionY, reactionZ, disp_mid,
1192 ele_axialforce, ele_shearforce, ele_momentforce,
1193 node_displacementsX, node_displacementsY,
1194 dispX, dispY) = DATA
1195
1196 S01.PLOT_2D_FRAME(deformed_scale=1.0)
1197 #####
1198 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1199 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1200 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1201 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1202 ANAL_TYPE = 'PUSHOVER'
1203
1204 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS,
1205 (reactionX_PUSH, reactionY_PUSH, reactionZ_PUSH, disp_mid_PUSH,
1206 ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1207 node_displacementsX_PUSH, node_displacementsY_PUSH,
1208 dispX_PUSH, dispY_PUSH,
1209 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1210 STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINT_PUSH, CURVATURE_PUSH, NA
1211
1212
1213 XDATA = disp_mid_PUSH
1214 YDATA = reactionY_PUSH
1215 XLABEL = 'Displacement [mm]'
1216 YLABEL = 'Base Reaction [N]'
```

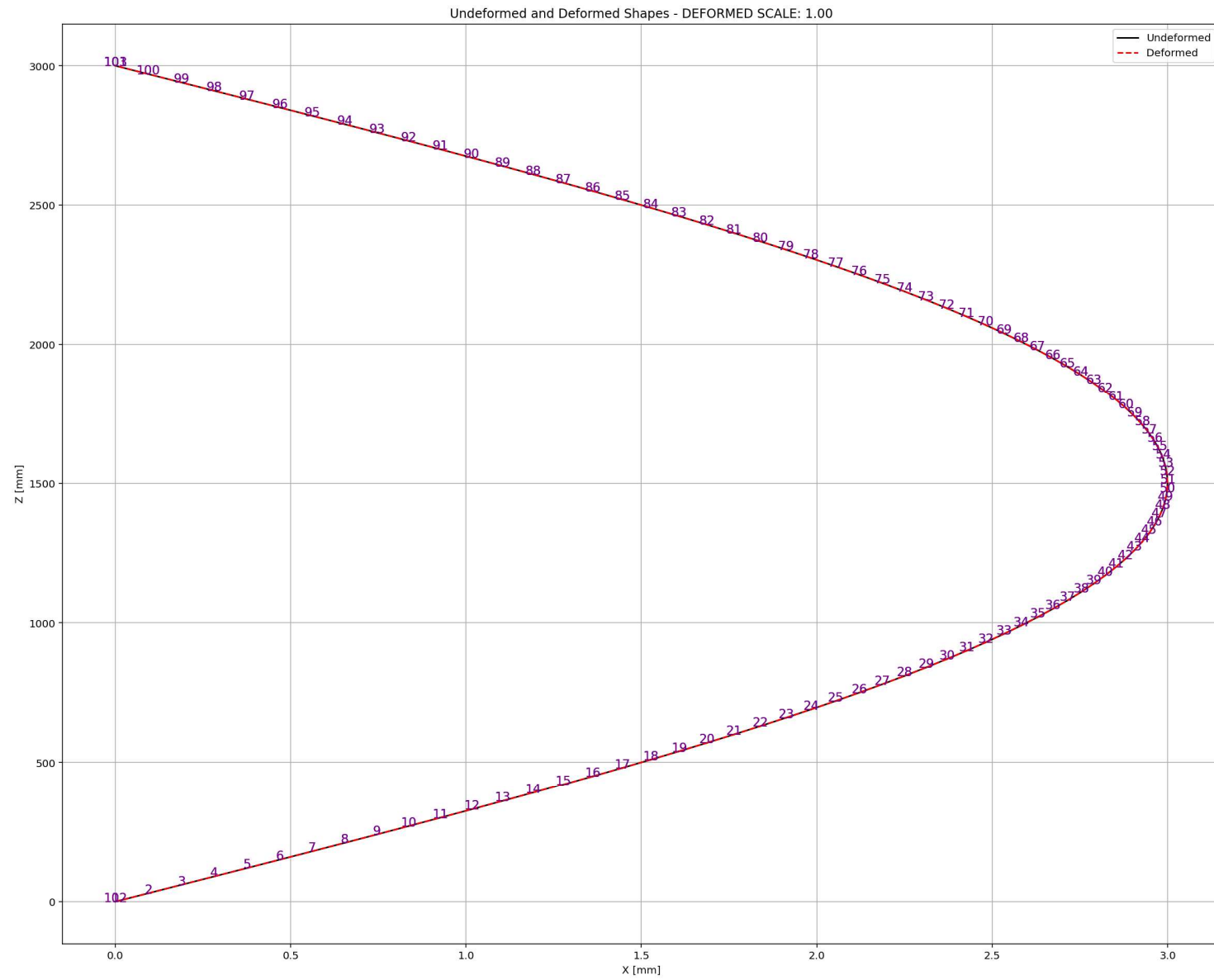
ON_CAPACITY_INDEX\55_STEEL_CONCRETE_COLUMN_POST-BUCKLING +

23 %

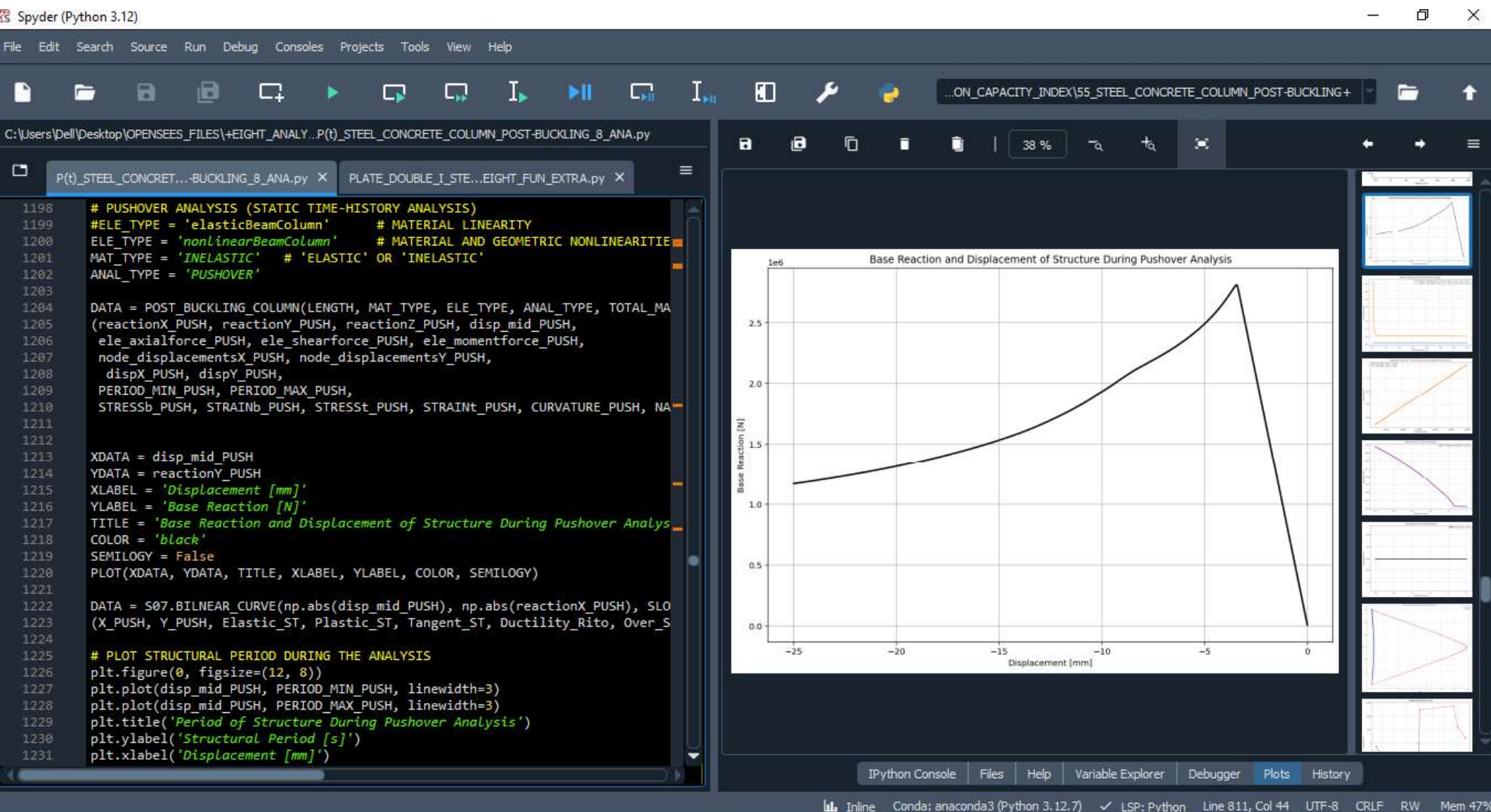
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00

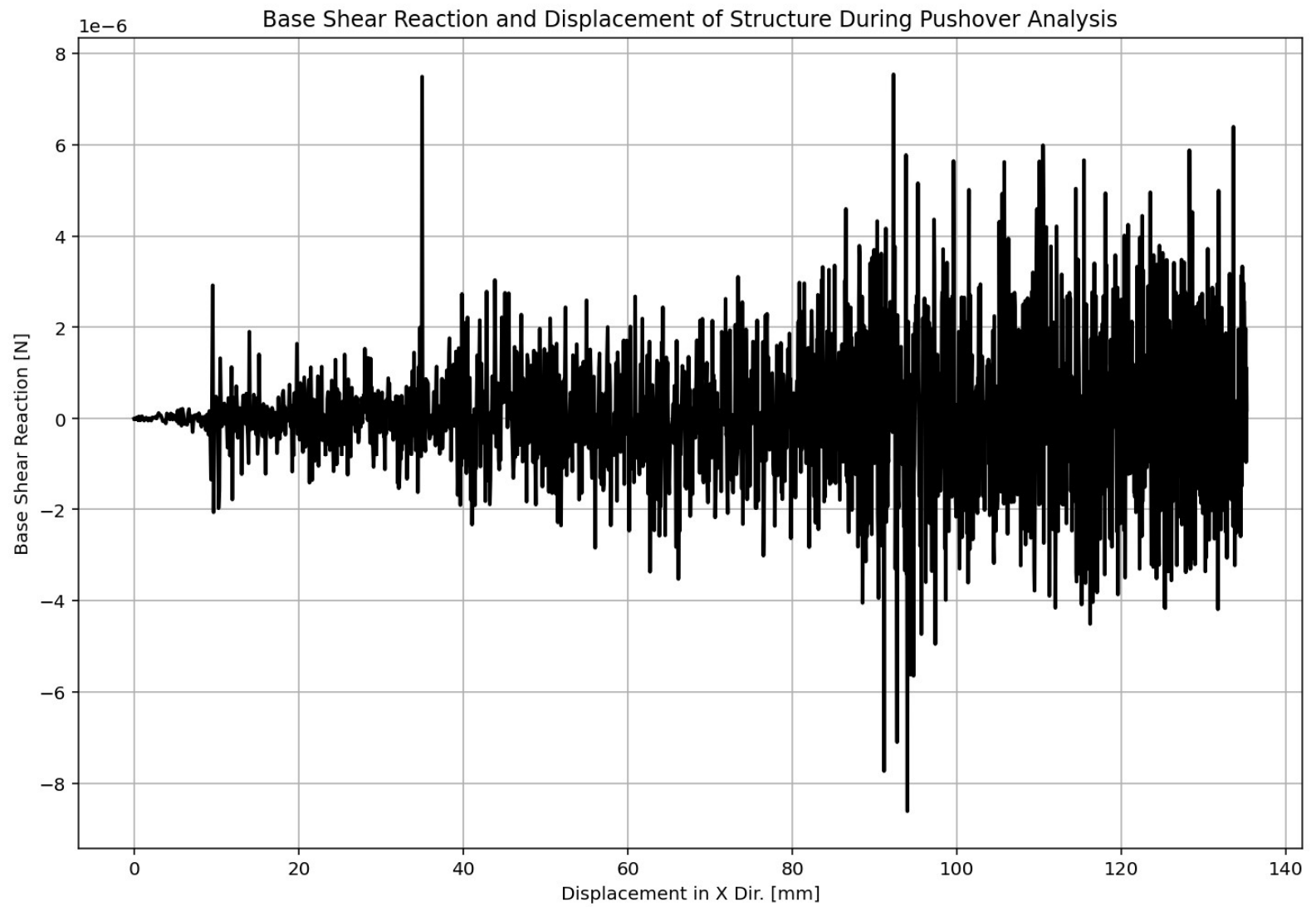
IPython Console Files Help Variable Explorer Debugger Plots History

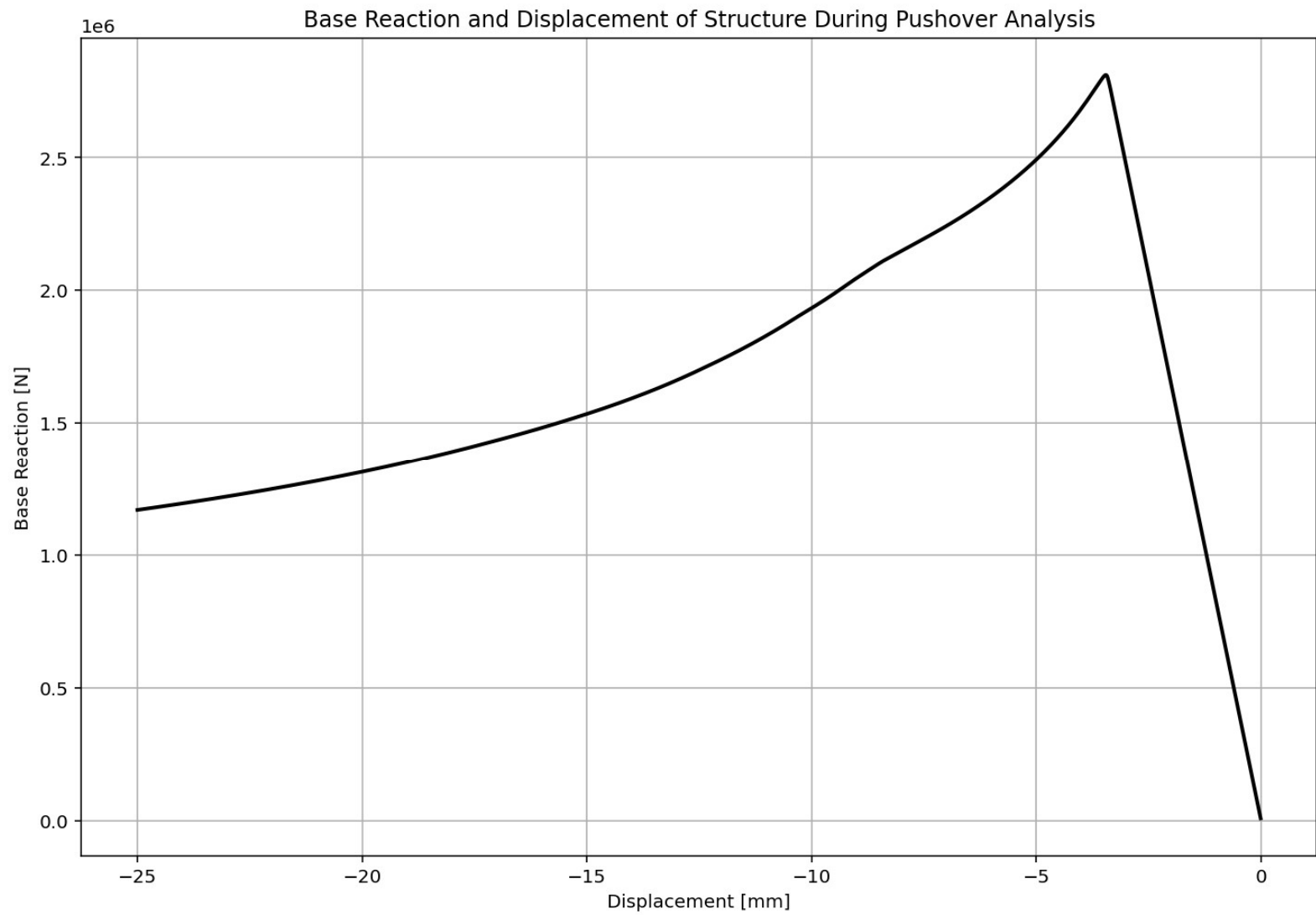
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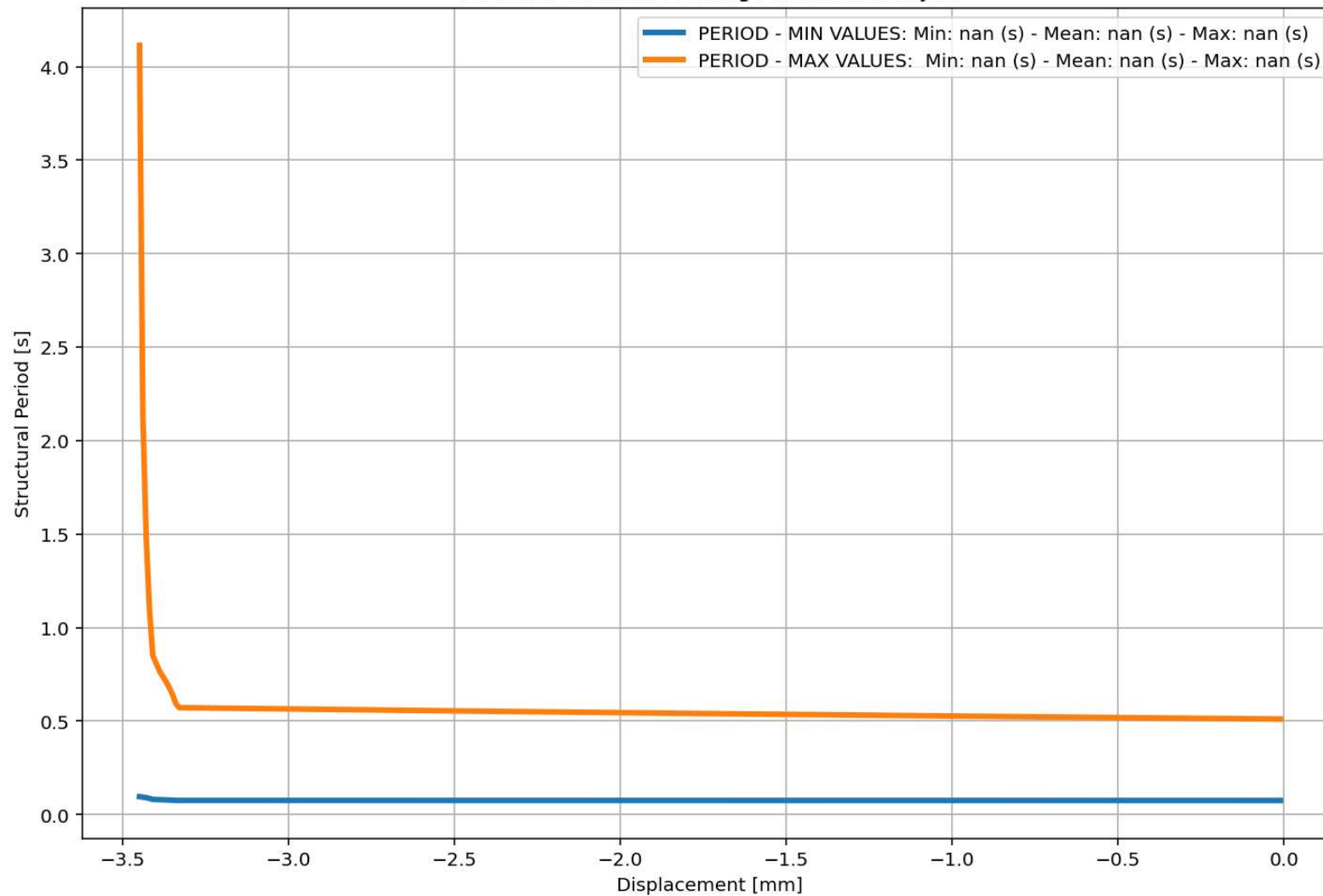
PUSHOVER ANALYSIS

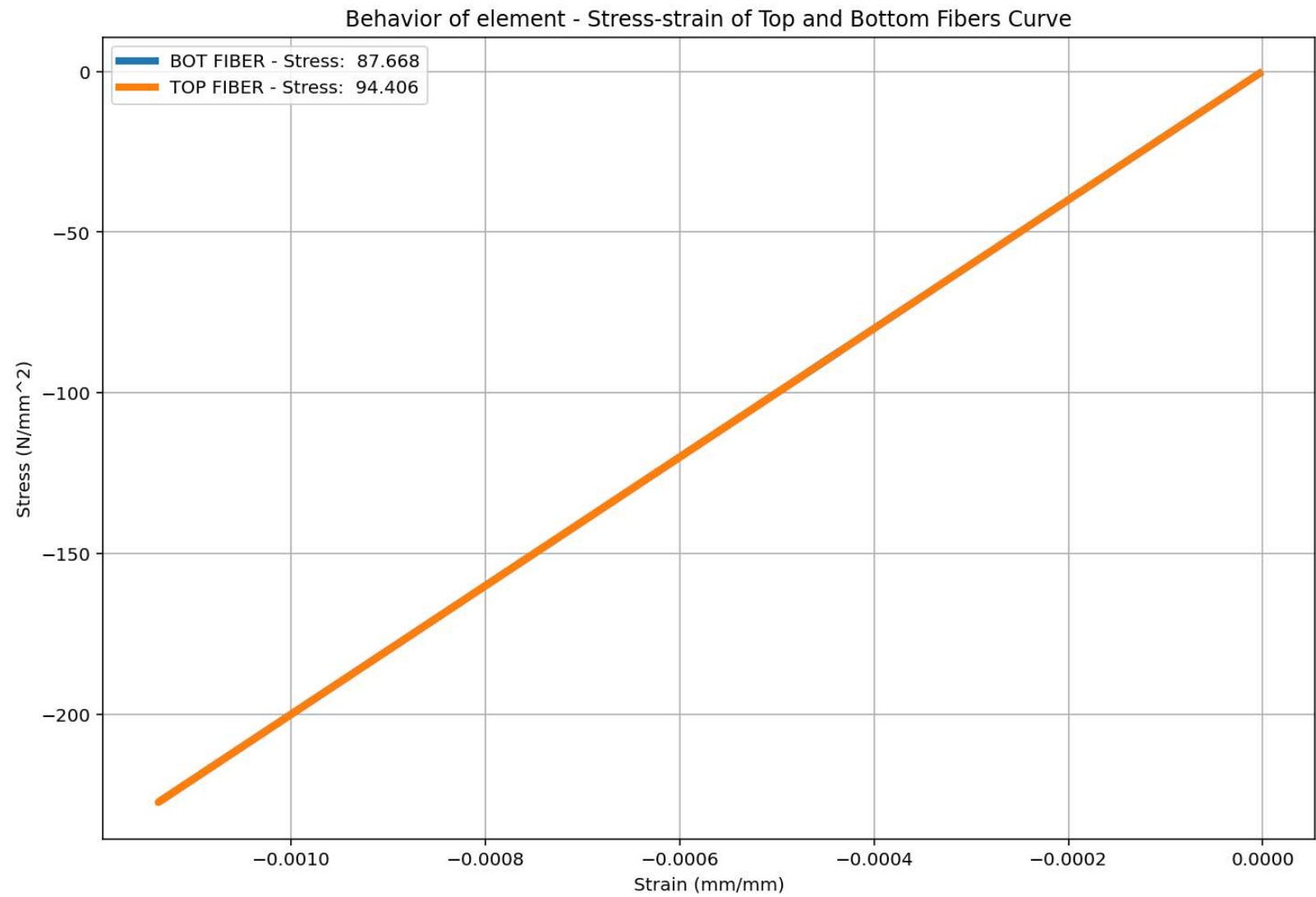


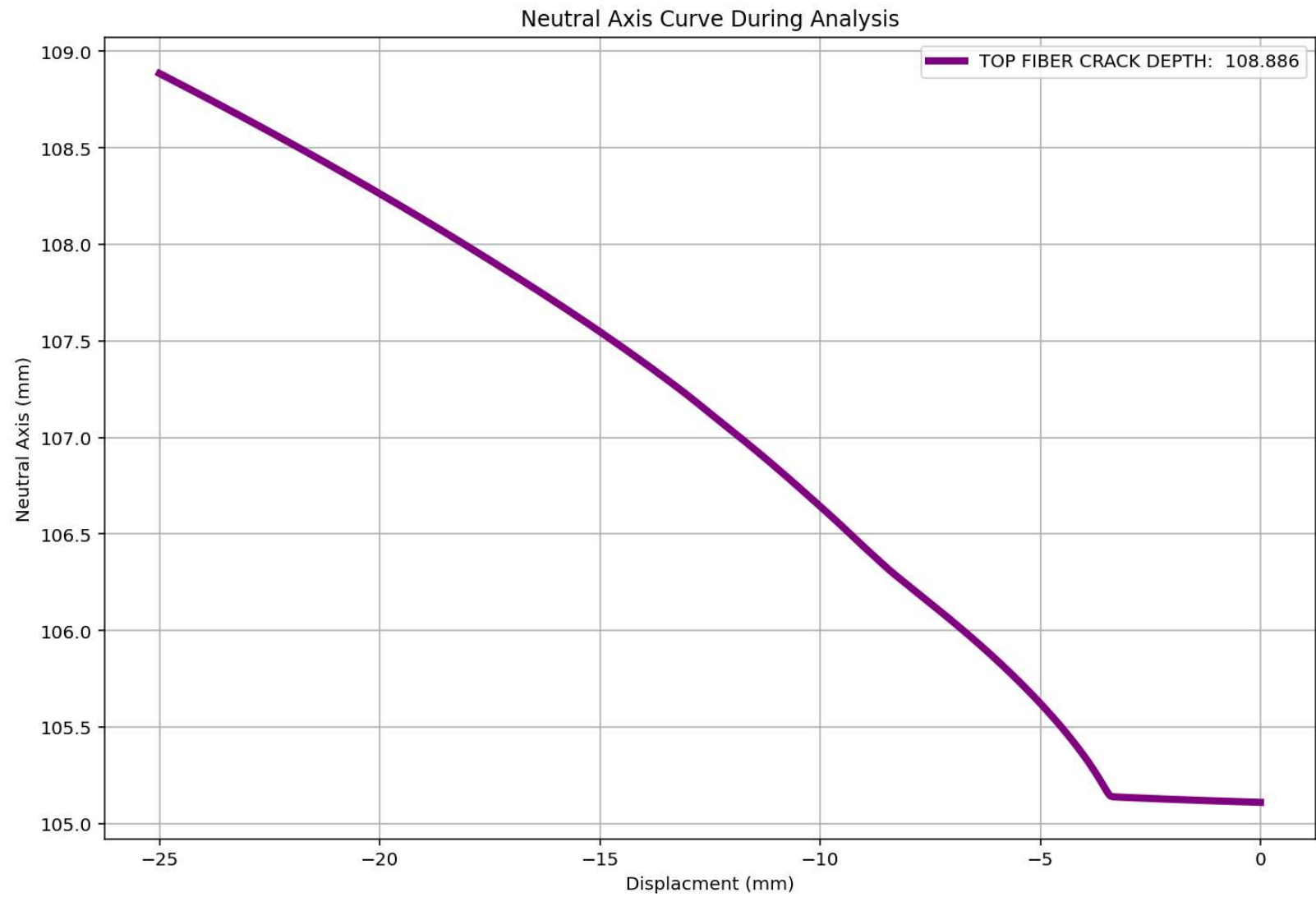


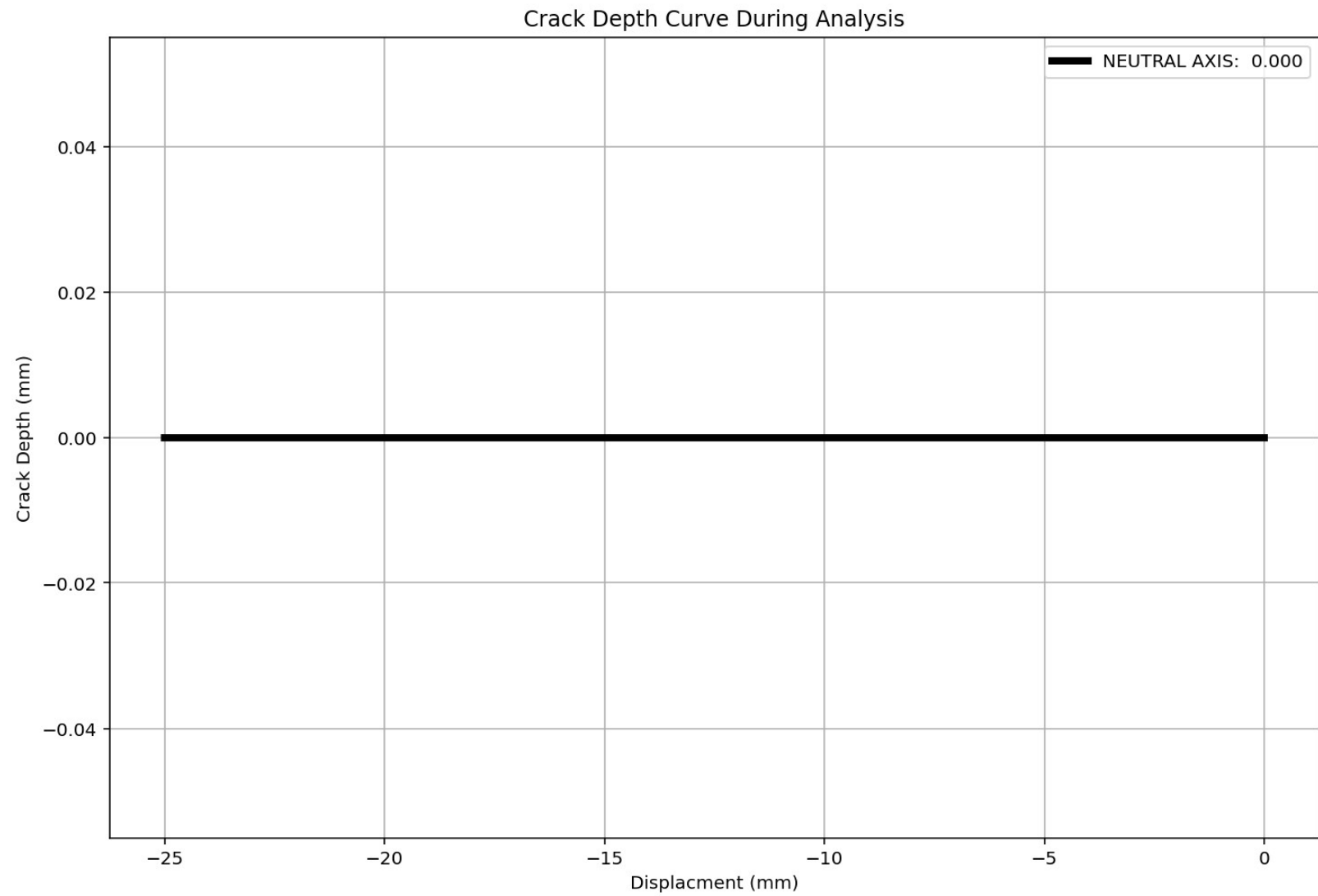


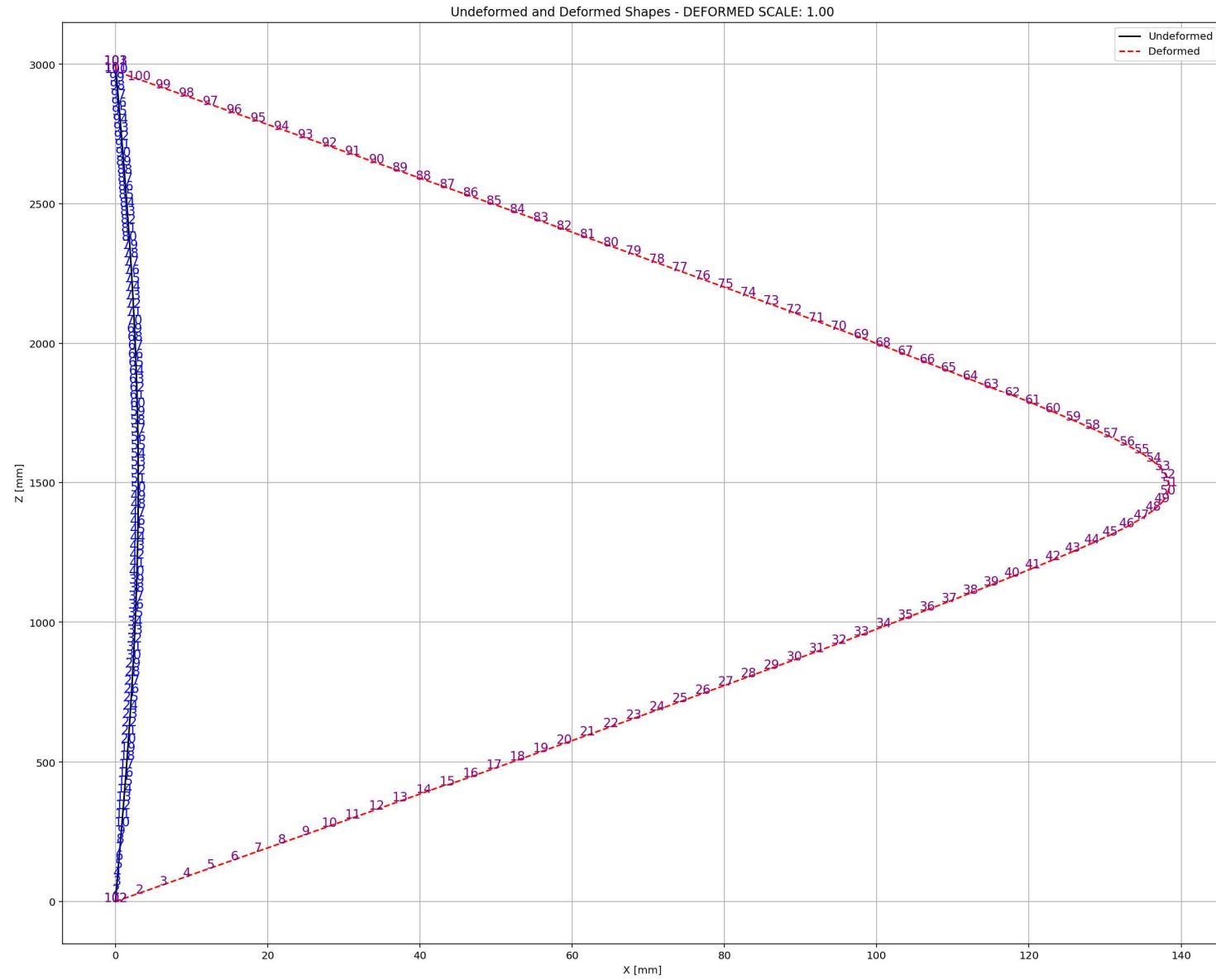
Period of Structure During Pushover Analysis











CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\EIGHT_ANALY..P(t)_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

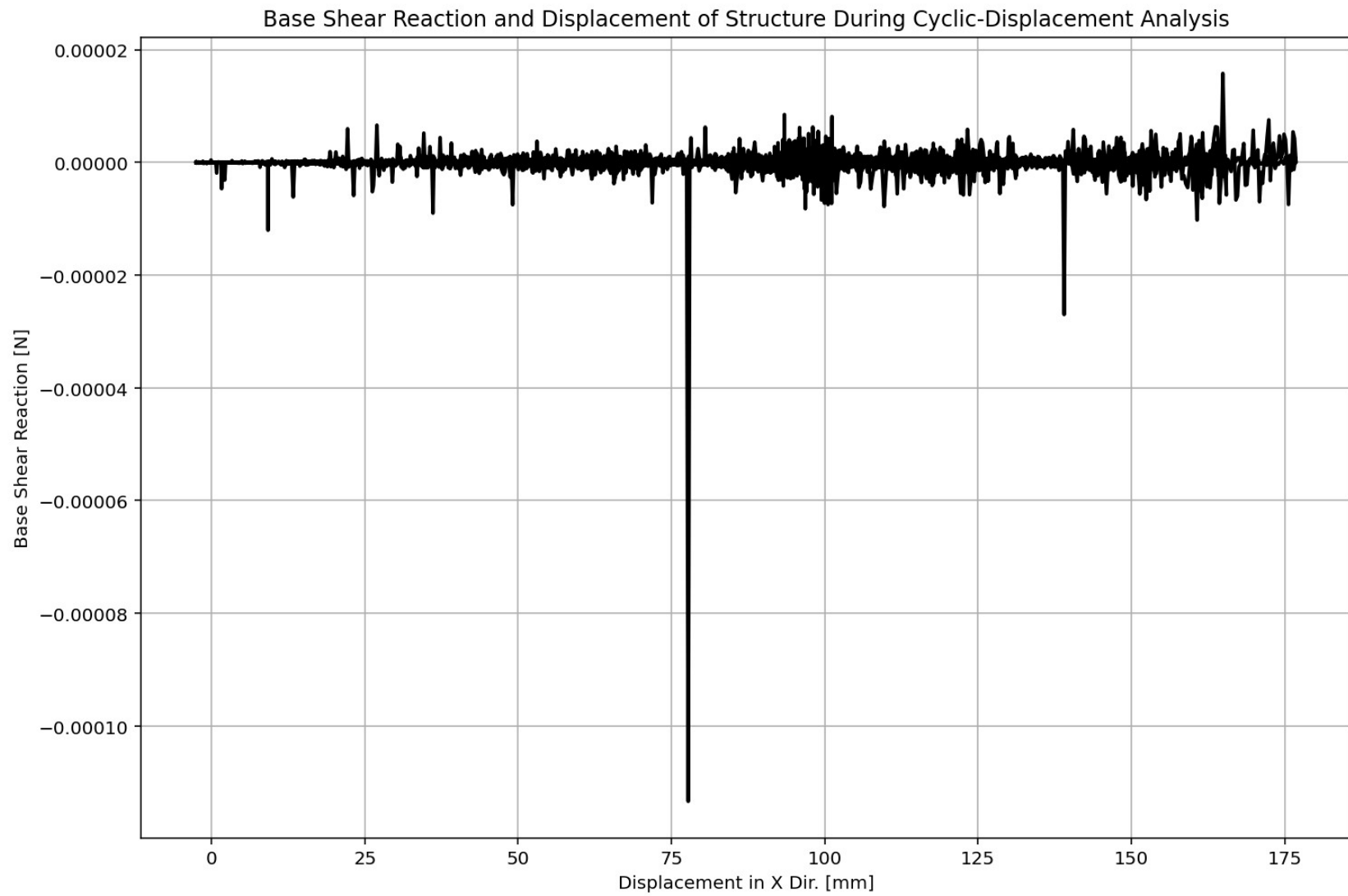
P(t)_STEEL_CONCRET...-BUCKLING_8_ANA.py x PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py x

```
1272 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1273 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1274 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1275 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1276 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1277
1278 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1279 (reactionX_CP, reactionY_CP, reactionZ_CP, disp_mid_CP,
1280 ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1281 node_displacementsX_CP, node_displacementsY_CP,
1282 dispX_CP, dispY_CP,
1283 PERIOD_MIN_CP, PERIOD_MAX_CP,
1284 STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINT_CP, CURVATURE_CP, NA_CP, CD_CP
1285
1286
1287 XDATA = disp_mid_CP
1288 YDATA = reactionY_CP
1289 XLABEL = 'Displacement [mm]'
1290 YLABEL = 'Base Reaction [N]'
1291 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displace
1292 COLOR = 'black'
1293 SEMILOGY = False
1294 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1295
1296 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1297 plt.figure(0, figsize=(12, 8))
1298 plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1299 plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1300 plt.title('Period of Structure During Cyclic Displacement Analysis')
1301 plt.ylabel('Structural Period [s]')
1302 plt.xlabel('Displacement [mm]')
1303 #plt.semilogy()
1304 plt.grid()
1305 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - M
```

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

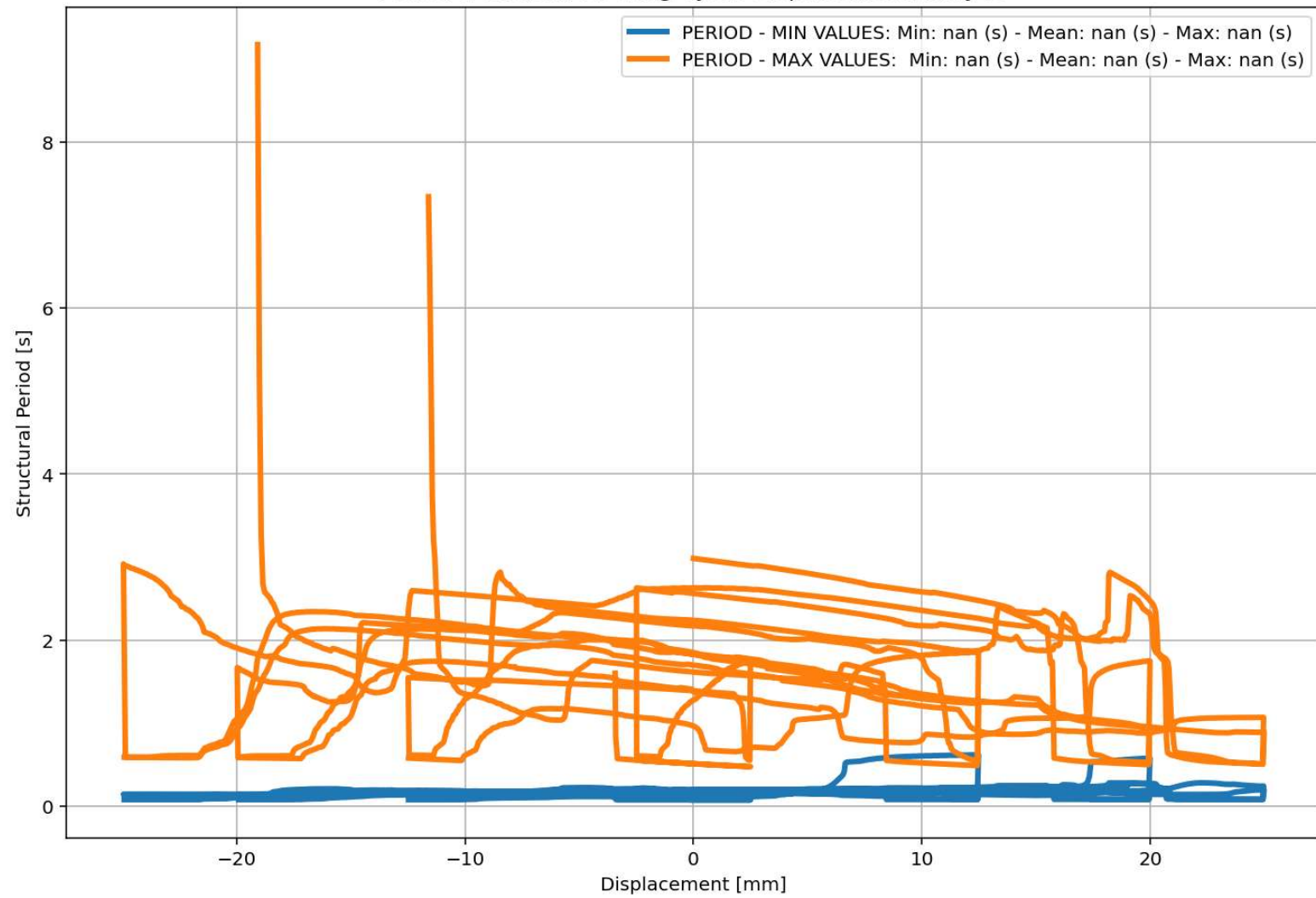
IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 811, Col 44 UTF-8 CRLF RW Mem 46%

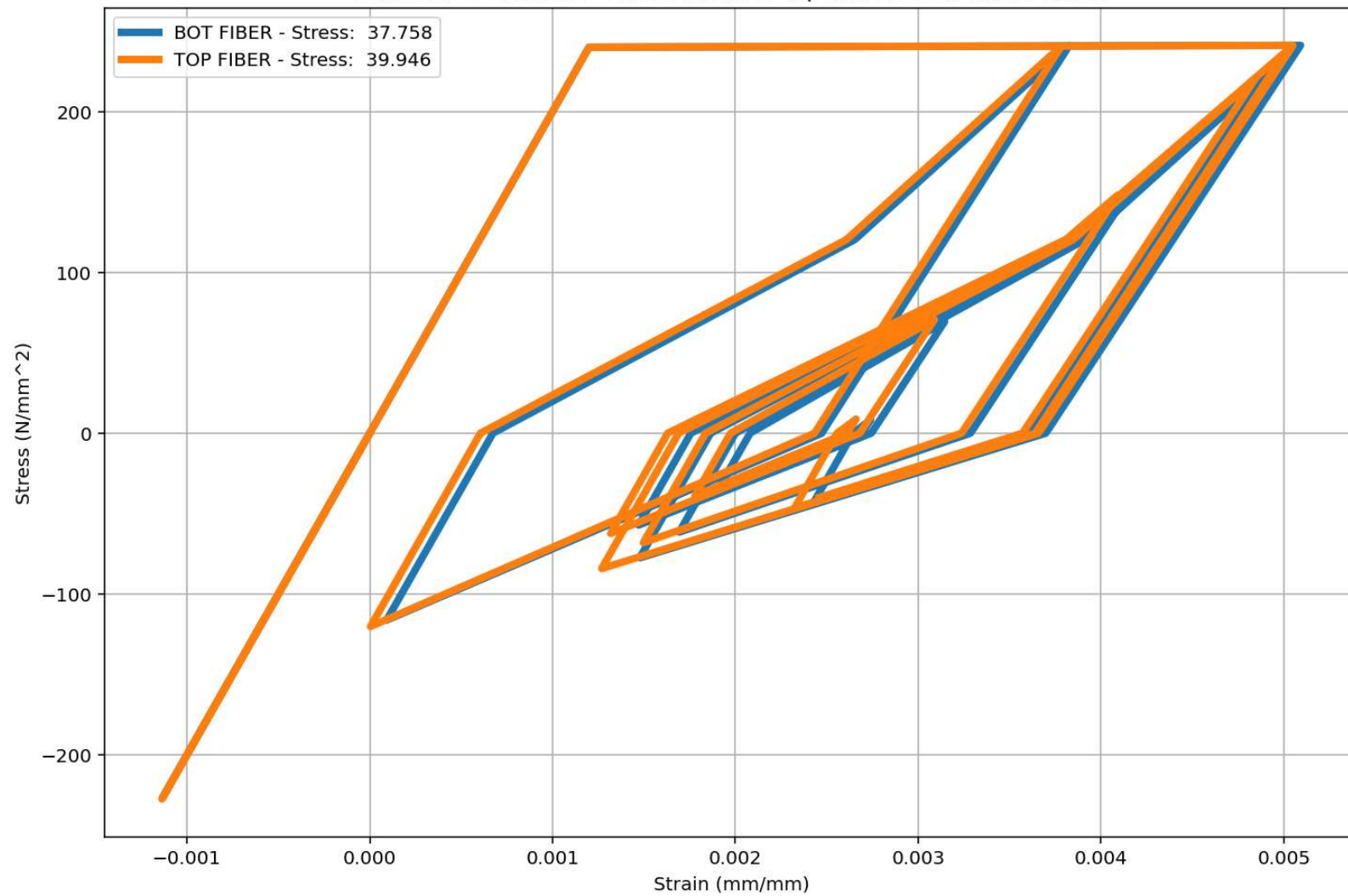


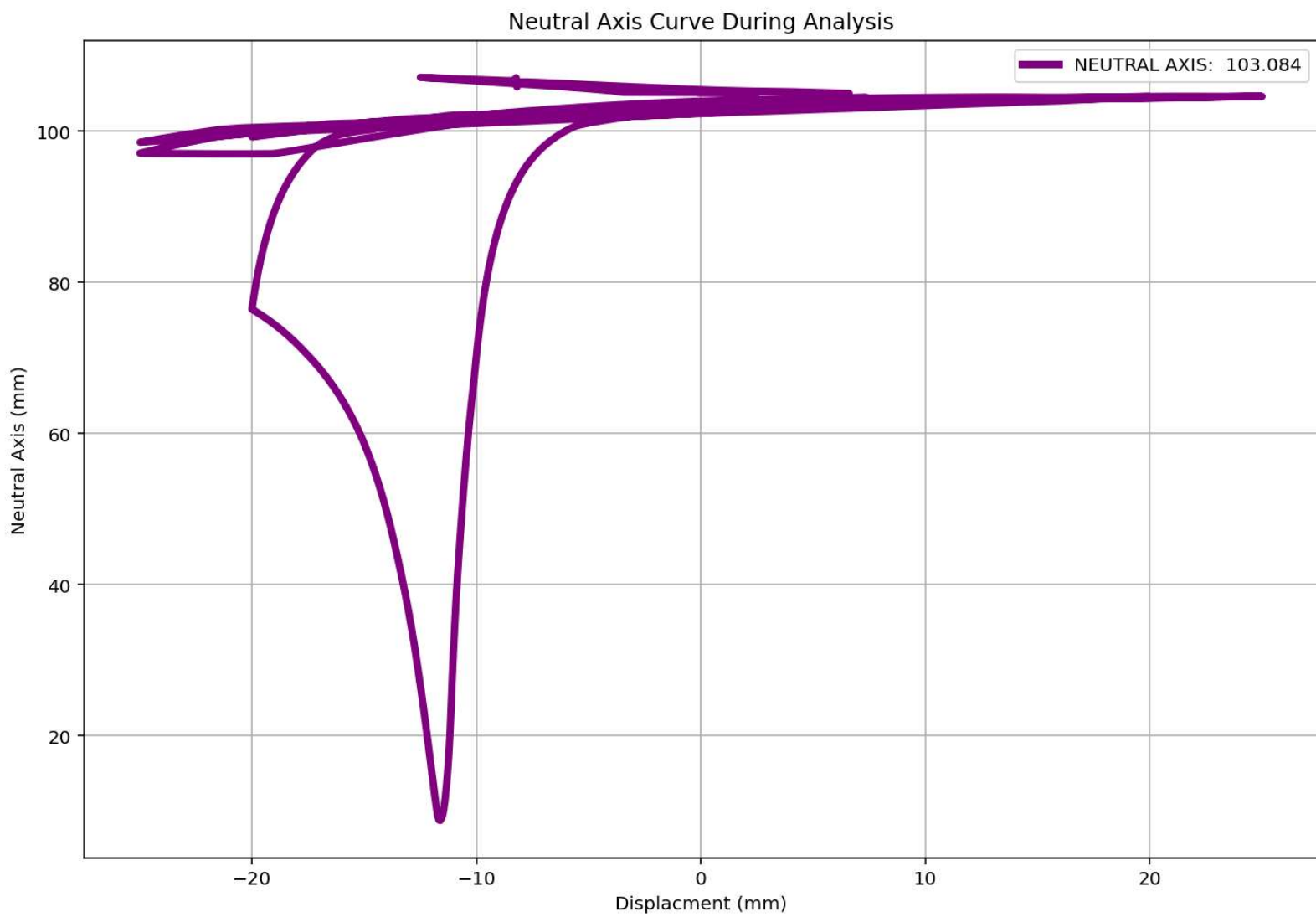


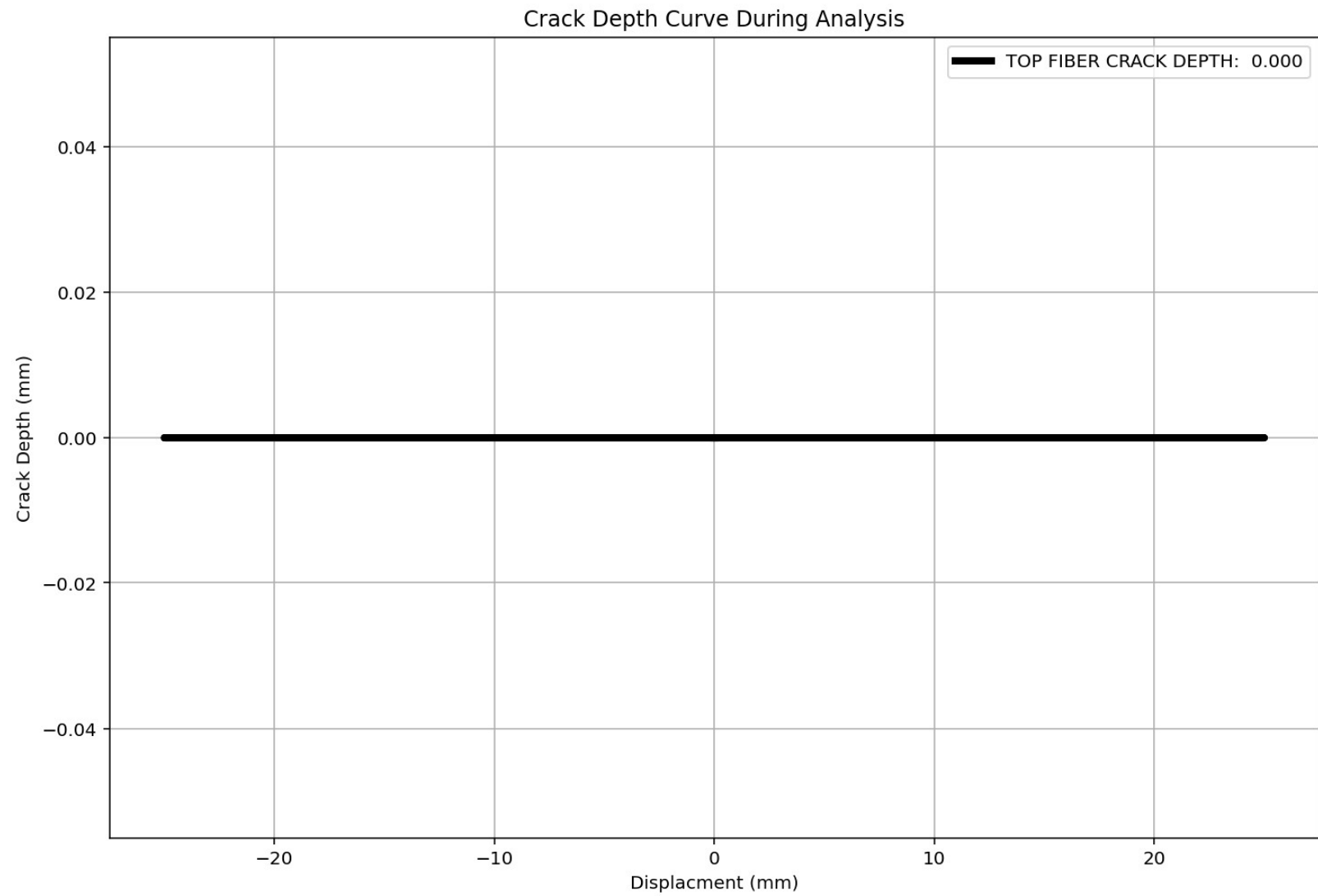
Period of Structure During Cyclic Displacement Analysis

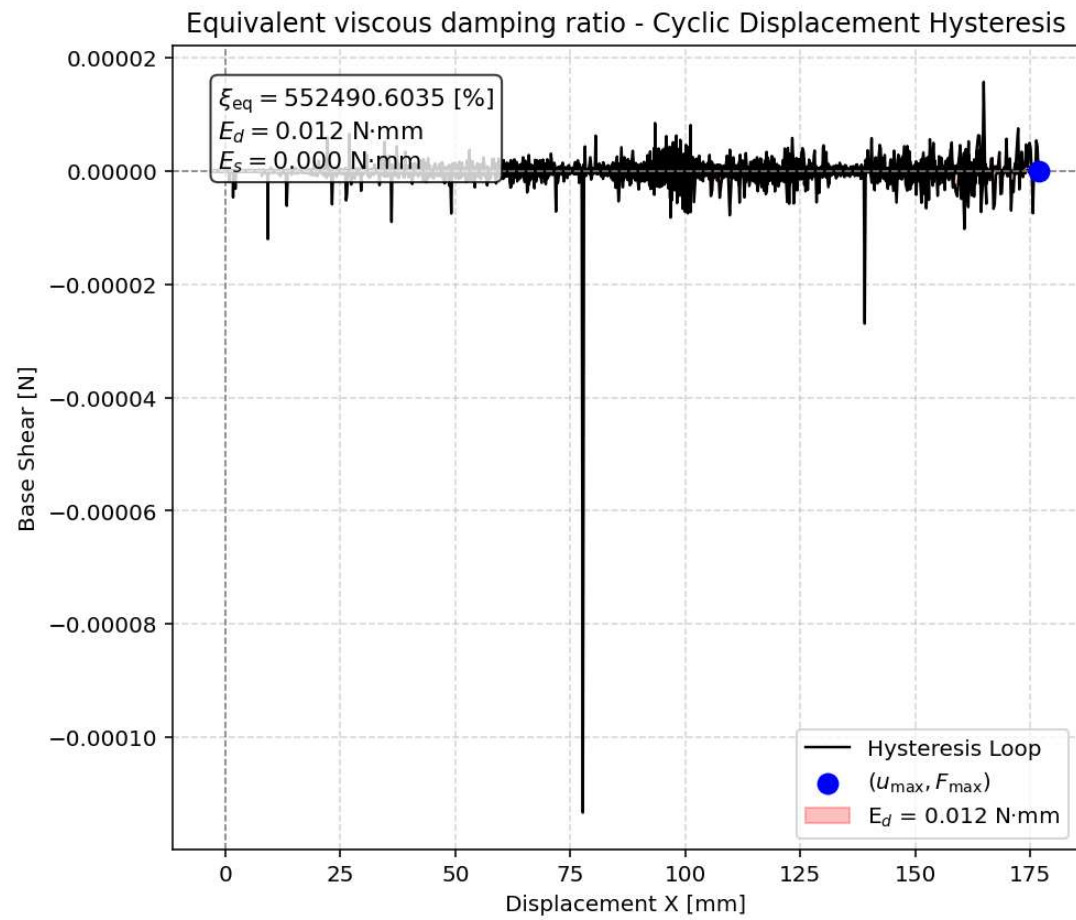


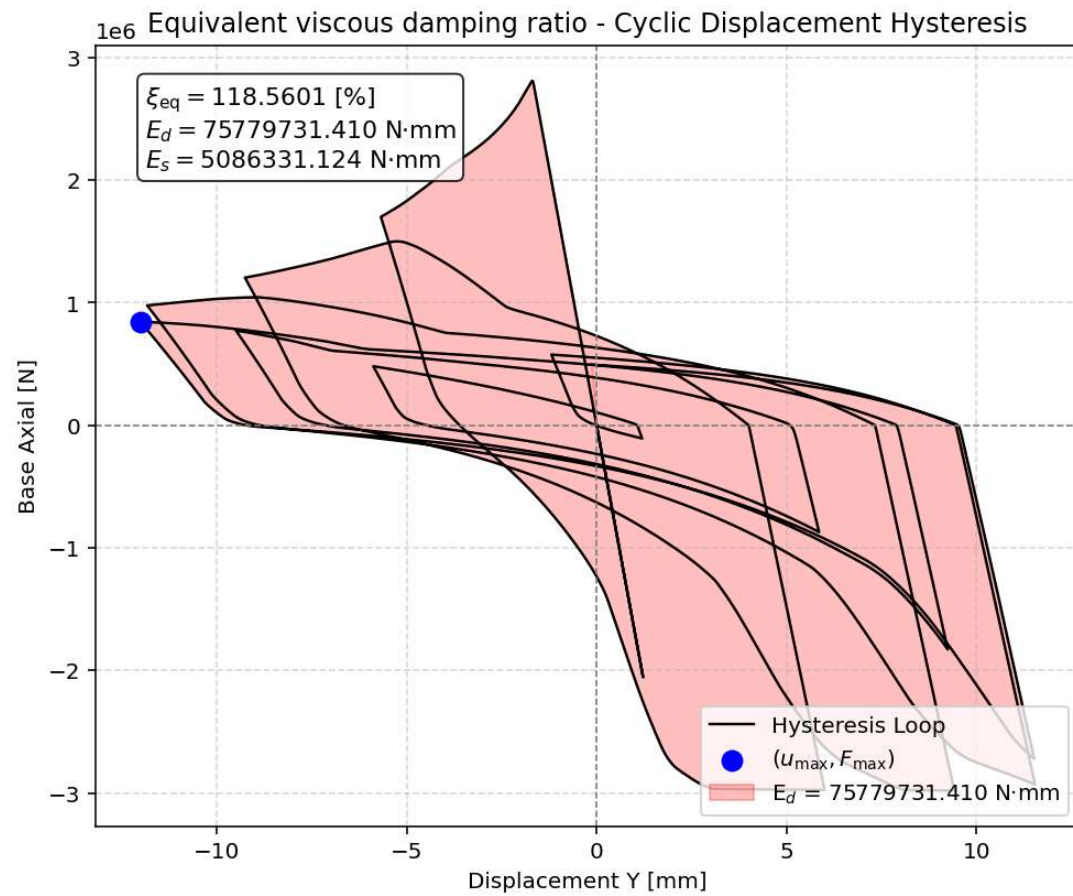
Behavior of element - Stress-strain of Top and Bottom Fibers Curve

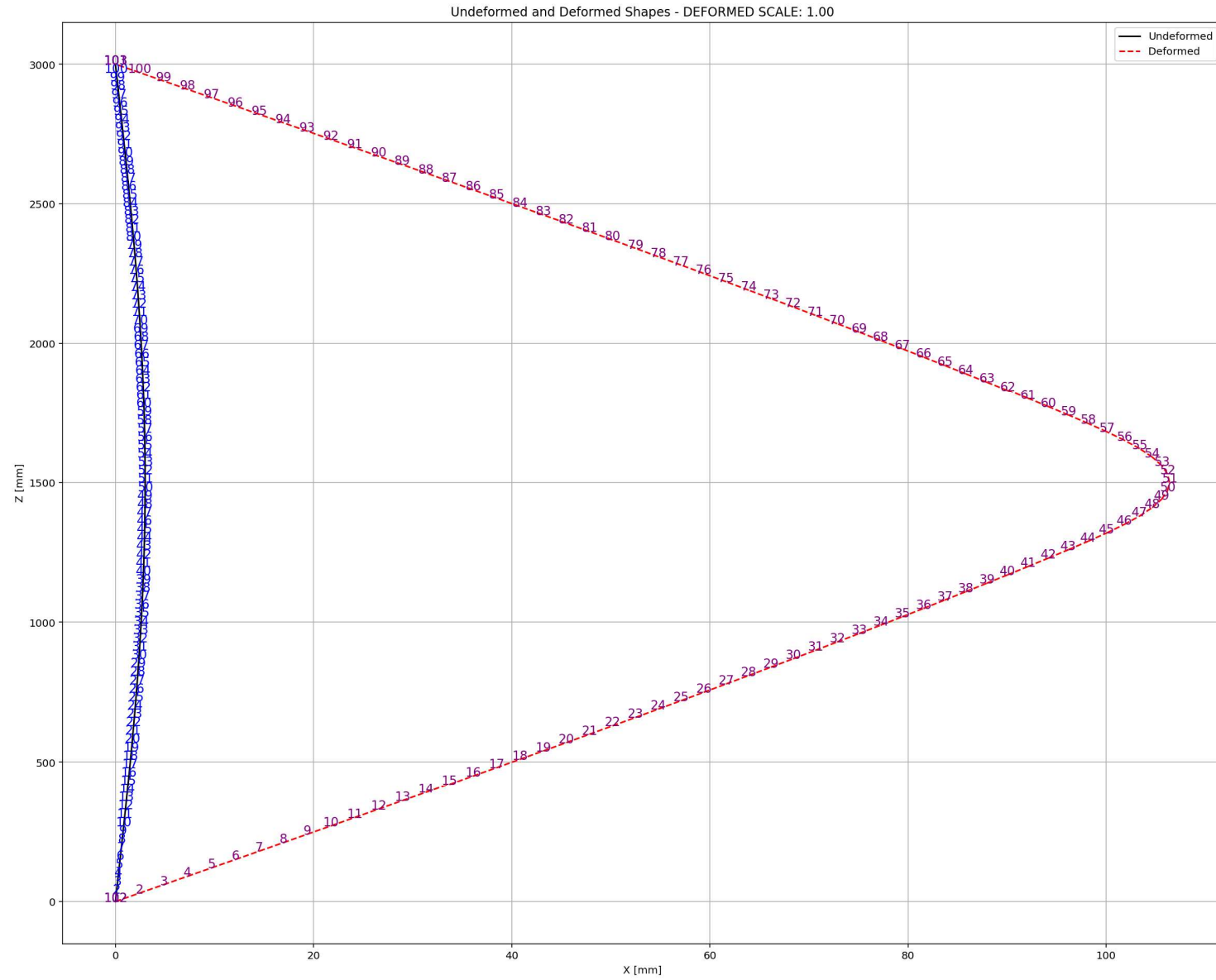












STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...P(t)_STEEL_CONCRETE_COLUMN_POST-BUCKLING_8_ANA.py

P(t)_STEEL_CONCRET...BUCKLING_8_ANA.py x PLATE_DOUBLE_I_STE...EIGHT_FUN_EXTRA.py x

```
1343 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1344 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1345 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIE
1346 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1347 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1348
1349 DATA = POST_BUCKLING_COLUMN(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MA
1350
1351 (reactionX_ETDLS, reactionY_ETDLS, reactionZ_ETDLS, disp_mid_ETDLS,
1352 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
1353 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
1354 dispX_ETDLS, dispY_ETDLS,
1355 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
1356 STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINT_ETDLS, CURVATURE_ETDL
1357
1358
1359 XDATA = disp_mid_ETDLS
1360 YDATA = reactionY_ETDLS
1361 XLABEL = 'Displacement [mm]'
1362 YLABEL = 'Base Reaction [N]'
1363 TITLE = 'Base Reaction and Displacement of Structure During Static External
1364 COLOR = 'black'
1365 SEMILOGY = False
1366 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1367
1368 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1369 plt.figure(0, figsize=(12, 8))
1370 plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1371 plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1372 plt.title('Period of Structure During Static External Time-dependent Loadin
1373 plt.ylabel('Structural Period [s]')
1374 plt.xlabel('Displacement [mm]')
1375 #plt.semilogy()
1376 plt.grid()
```

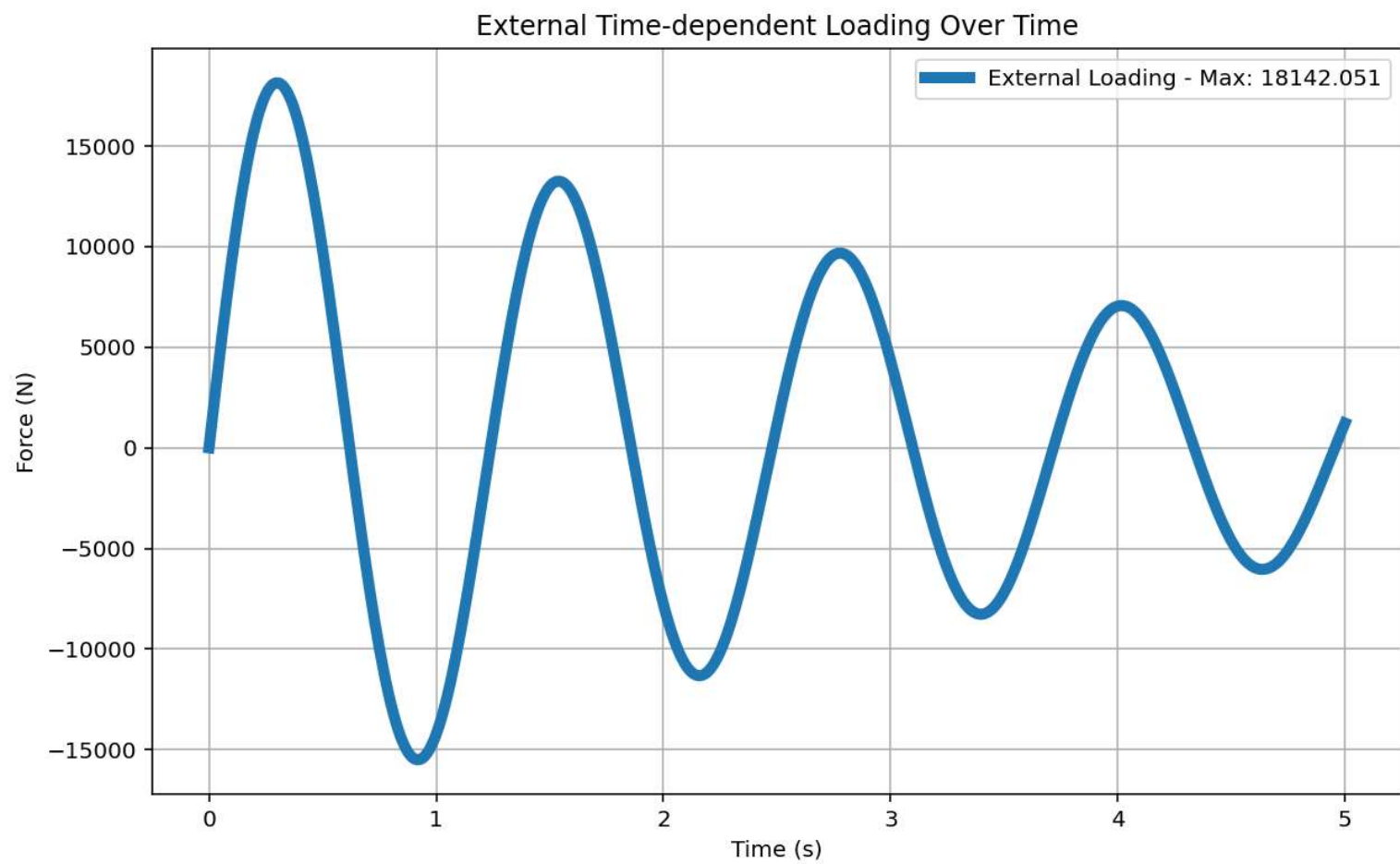
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

Base Reaction [N]

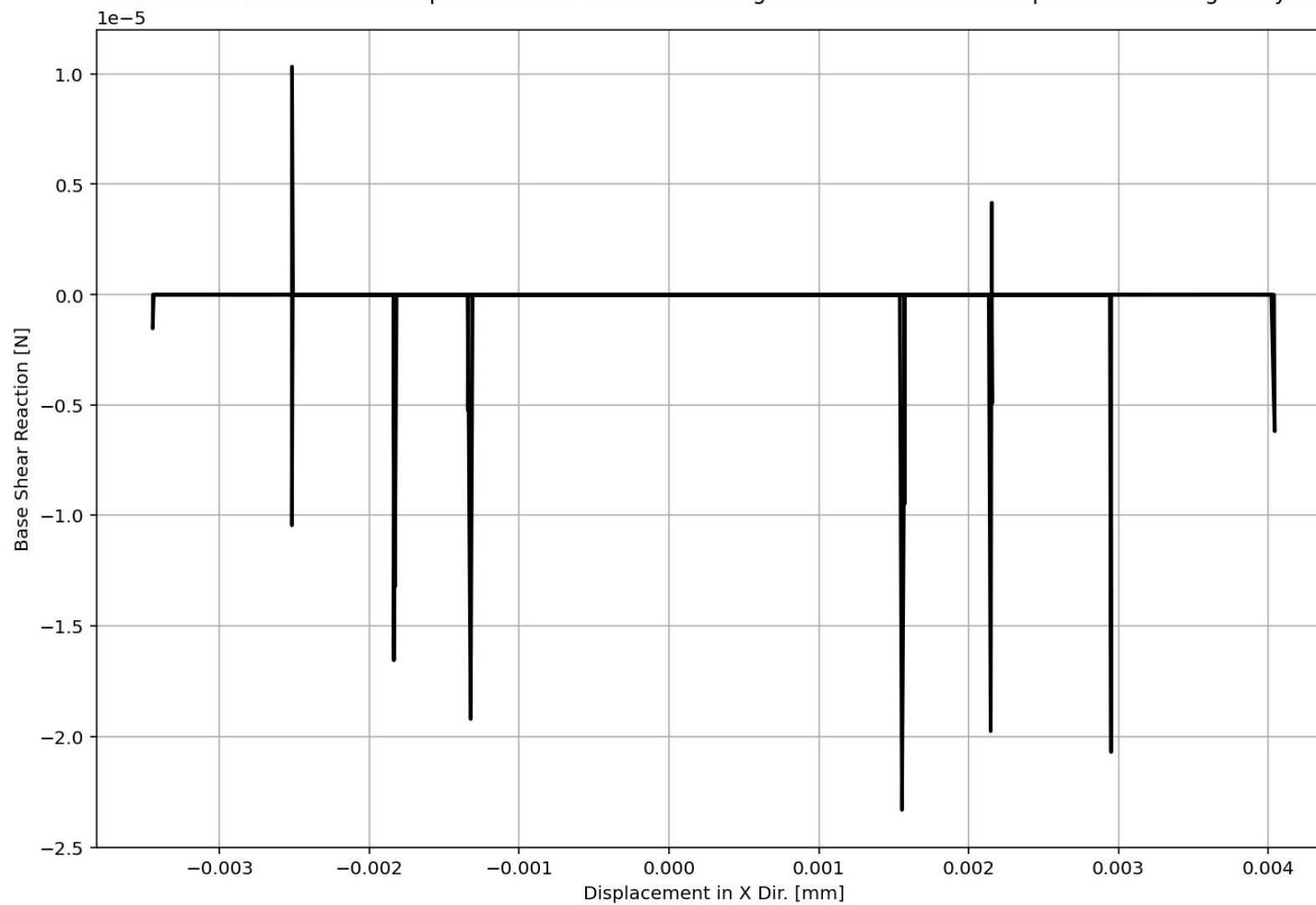
Displacement [mm]

IPython Console Files Help Variable Explorer Debugger Plots History

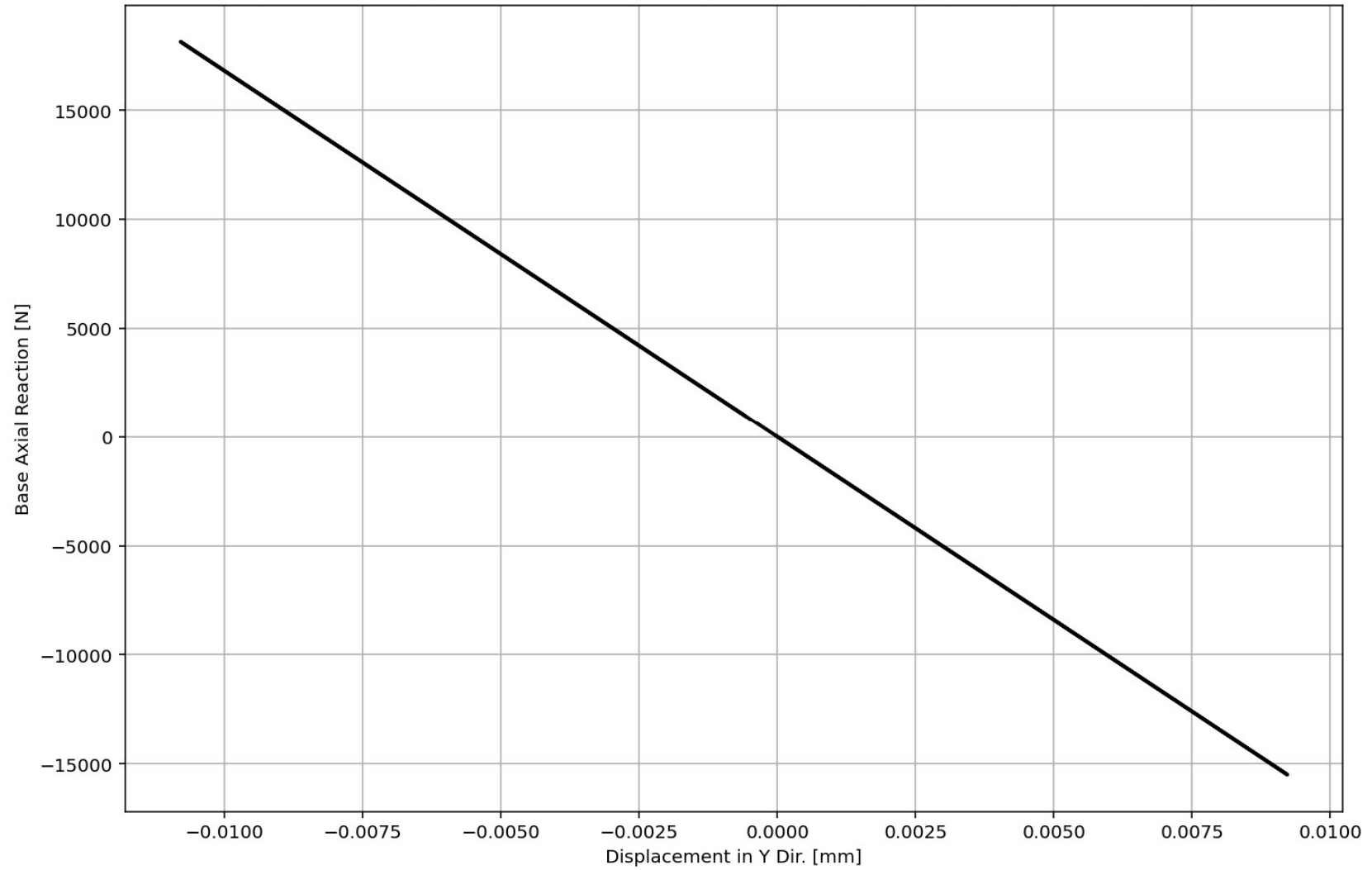
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 811, Col 44 UTF-8 CRLF RW Mem 47%



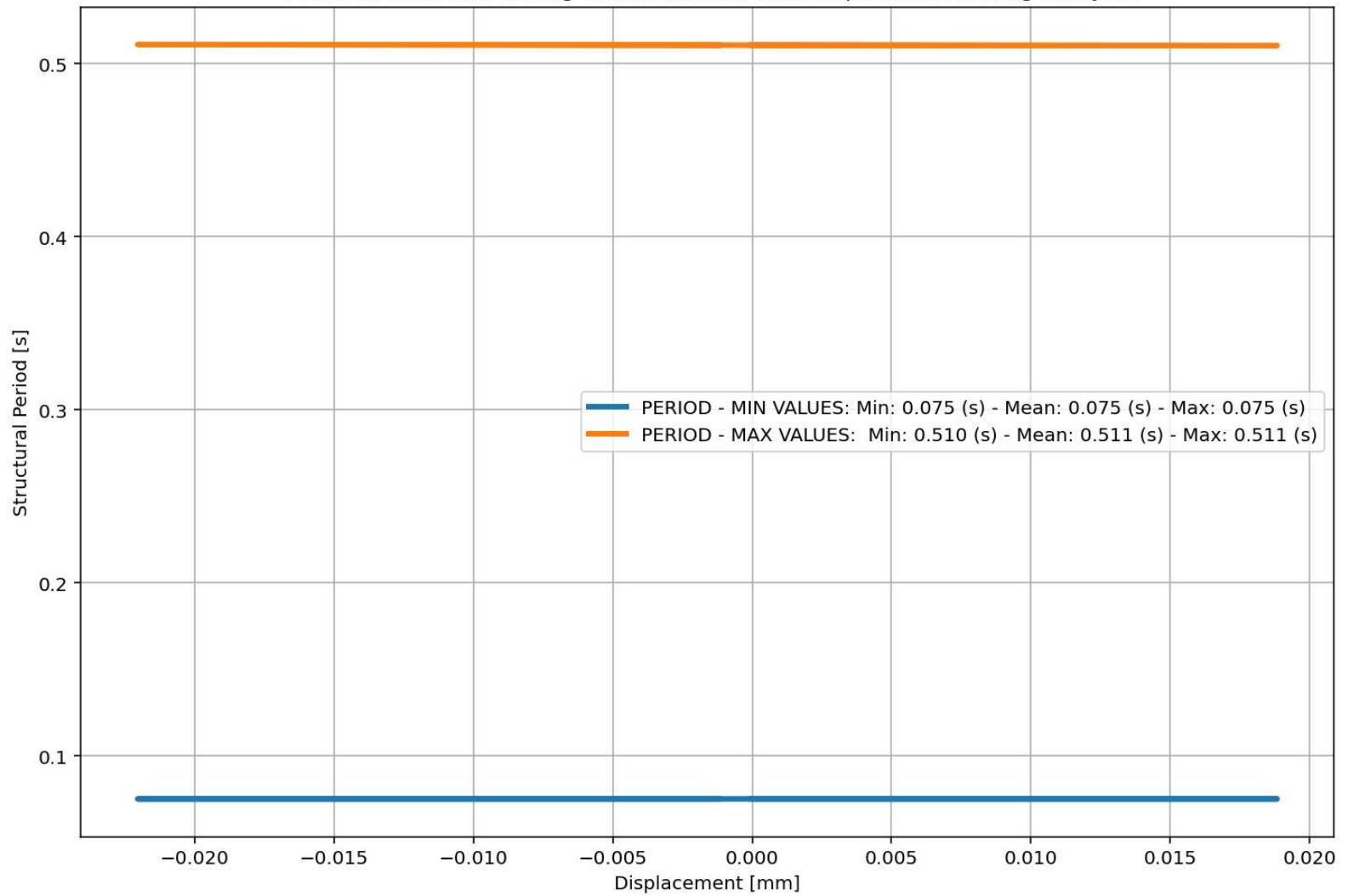
Base Shear Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis



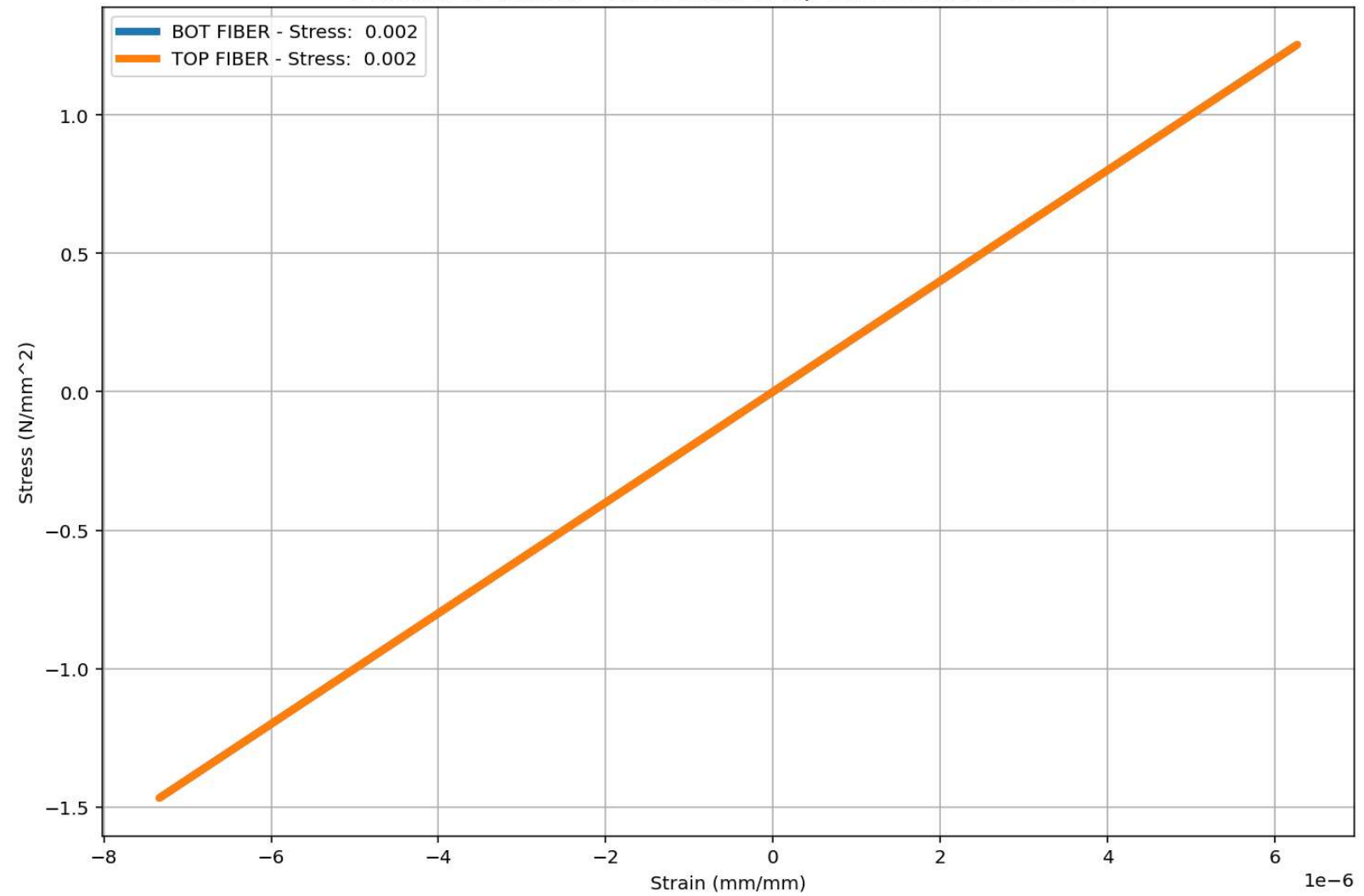
Base Axial Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

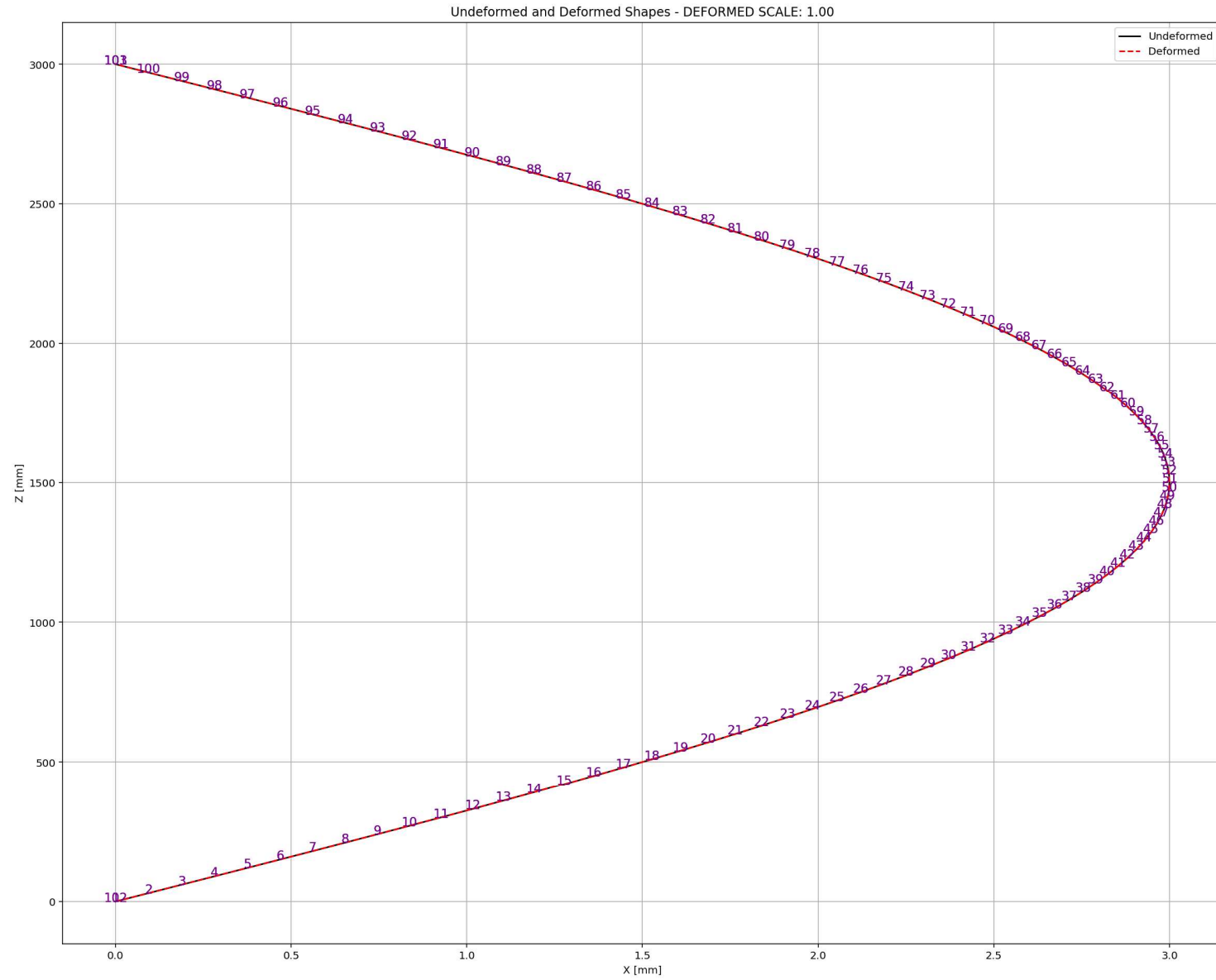


Period of Structure During Static External Time-dependent Loading Analysis

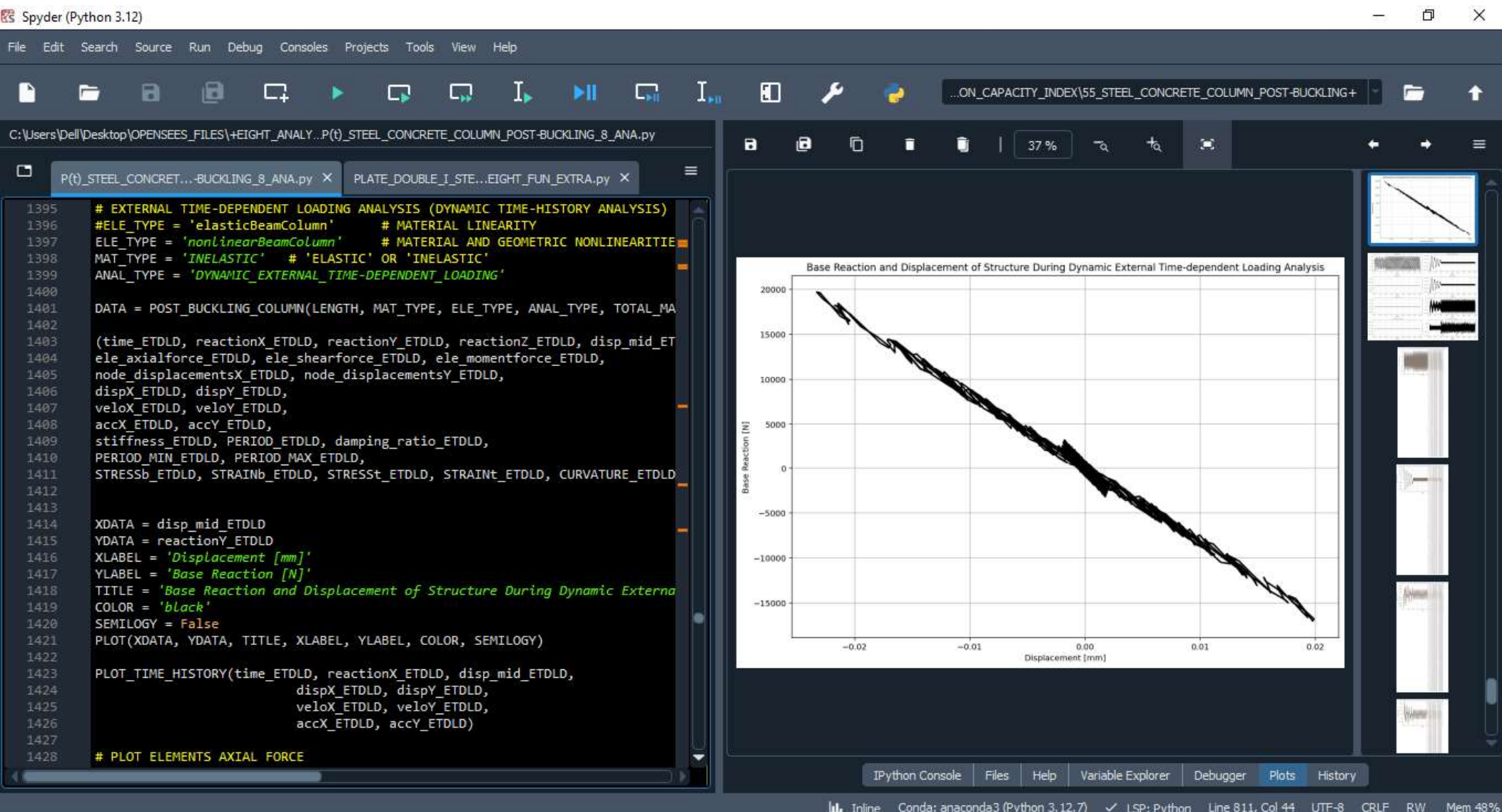


Behavior of element - Stress-strain of Top and Bottom Fibers Curve

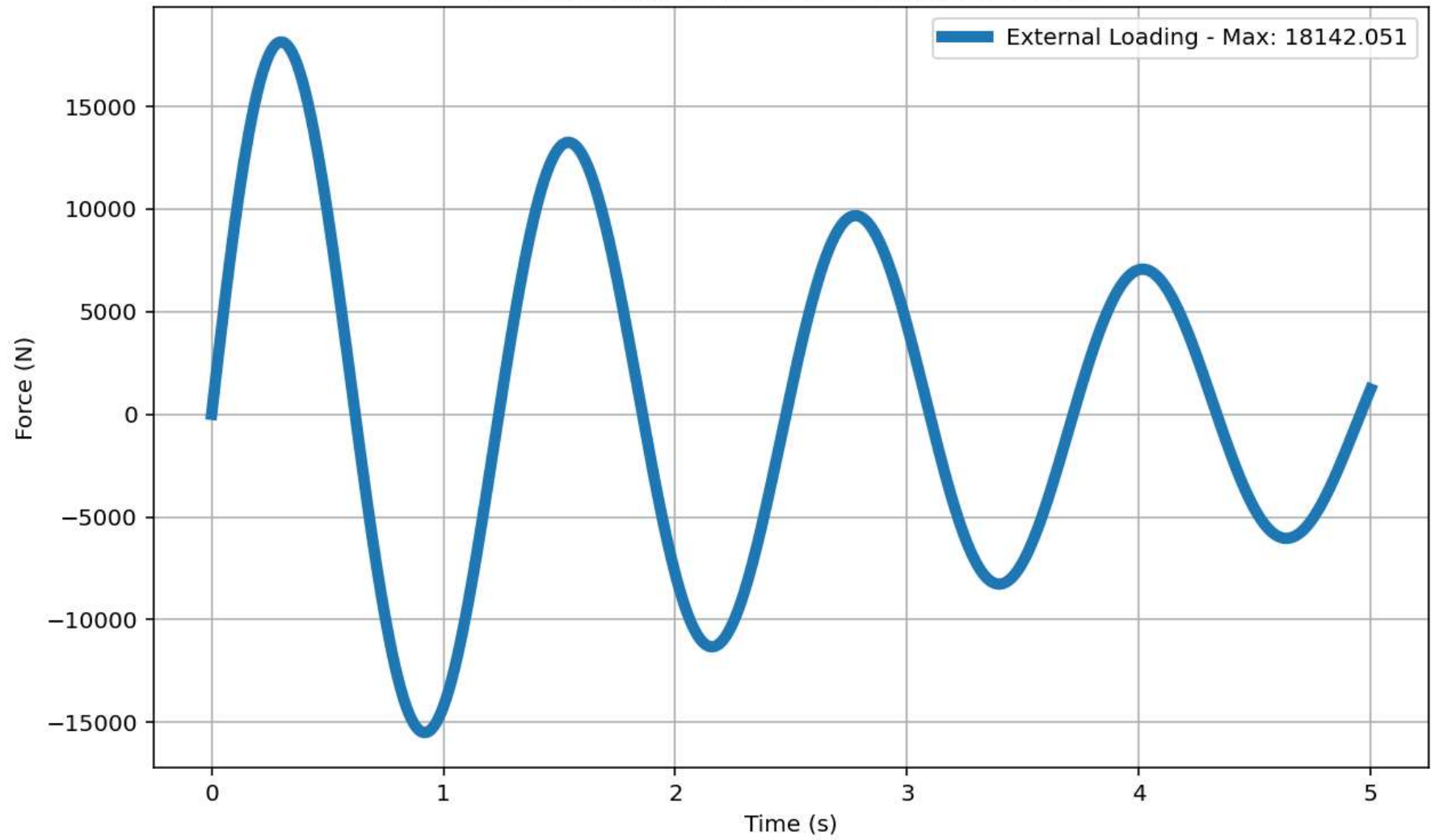




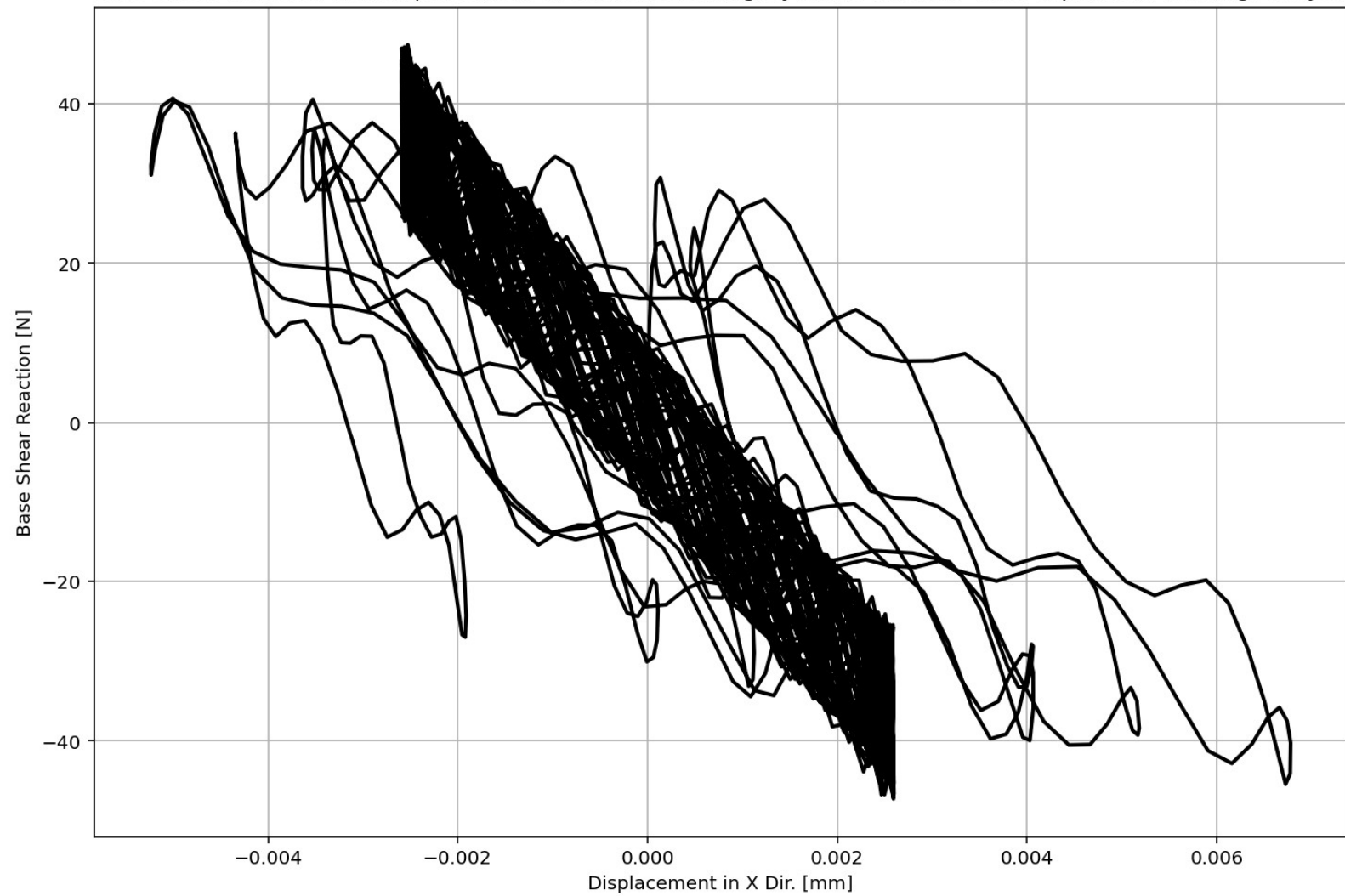
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

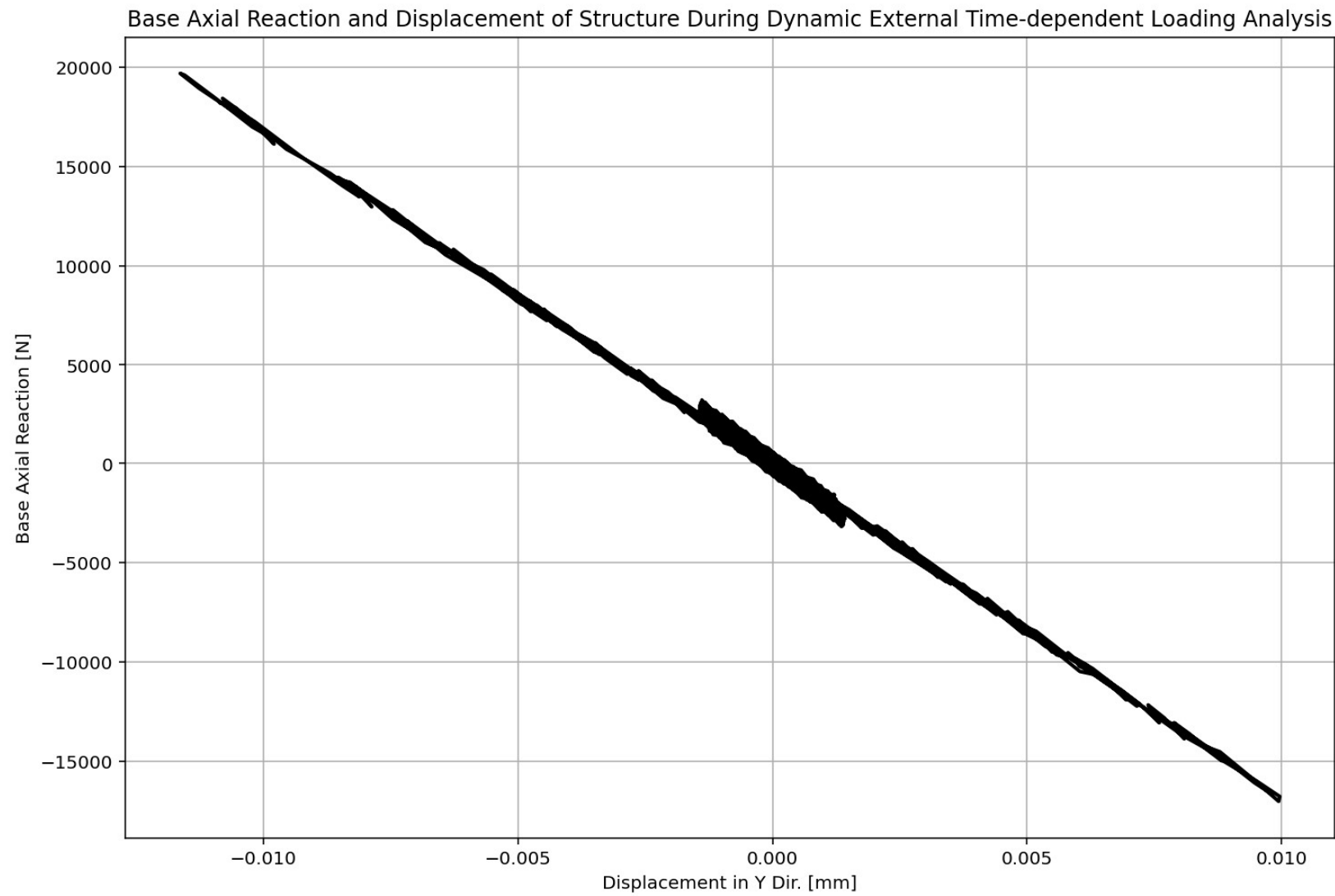


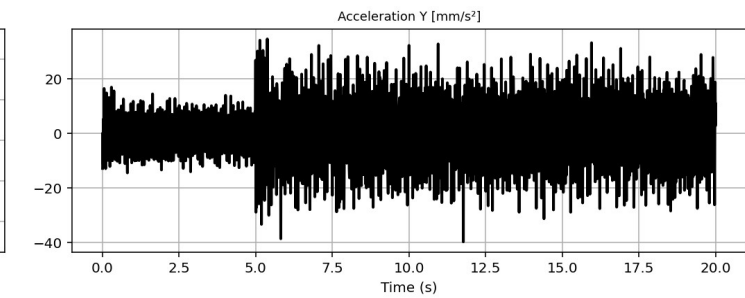
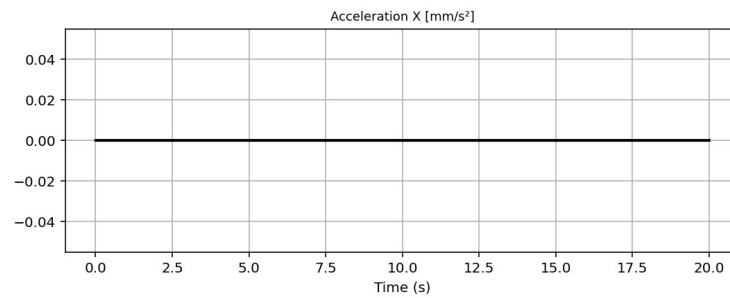
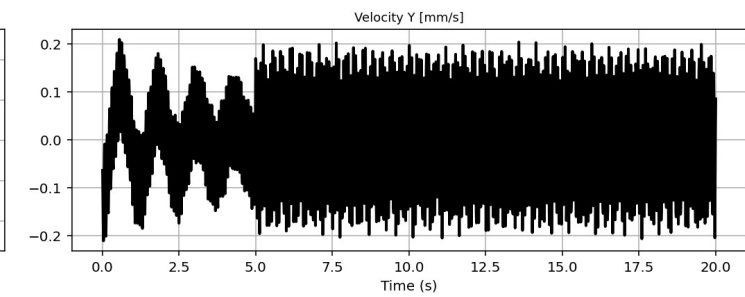
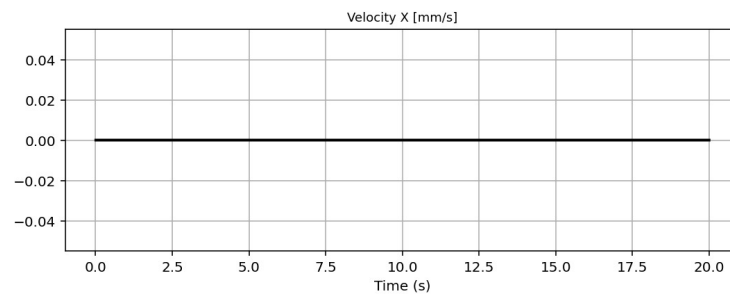
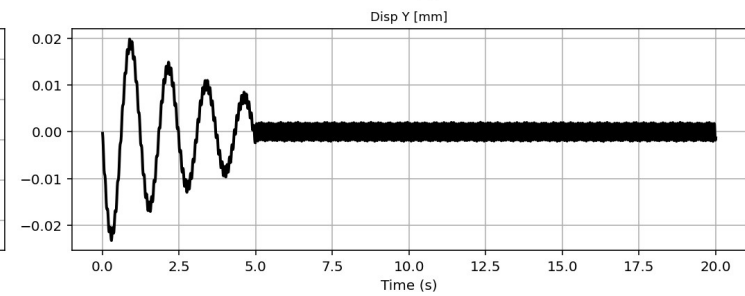
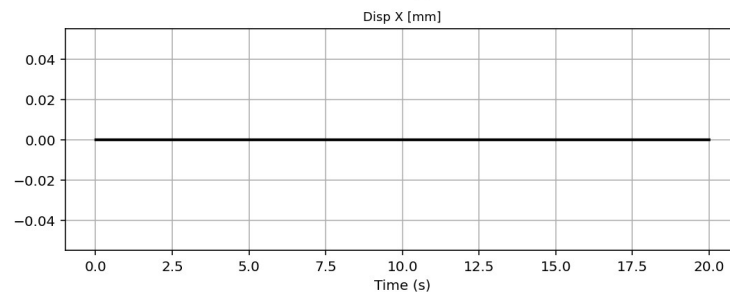
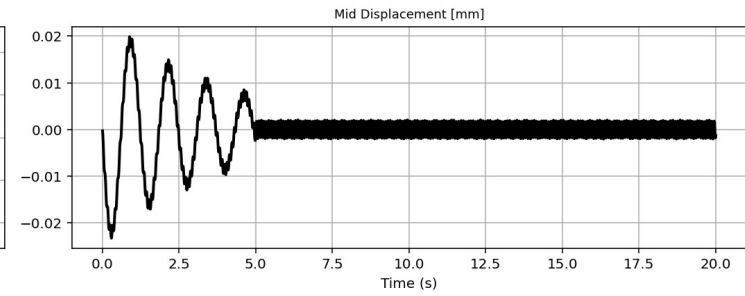
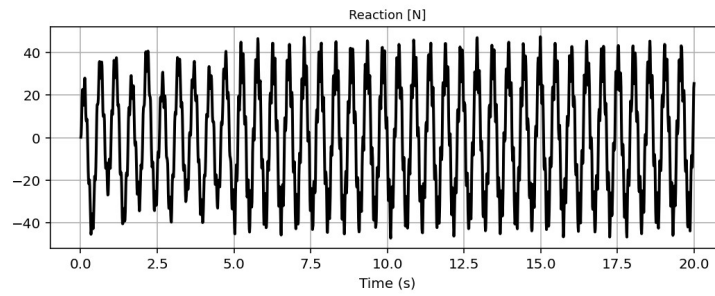
External Time-dependent Loading Over Time



Base Shear Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



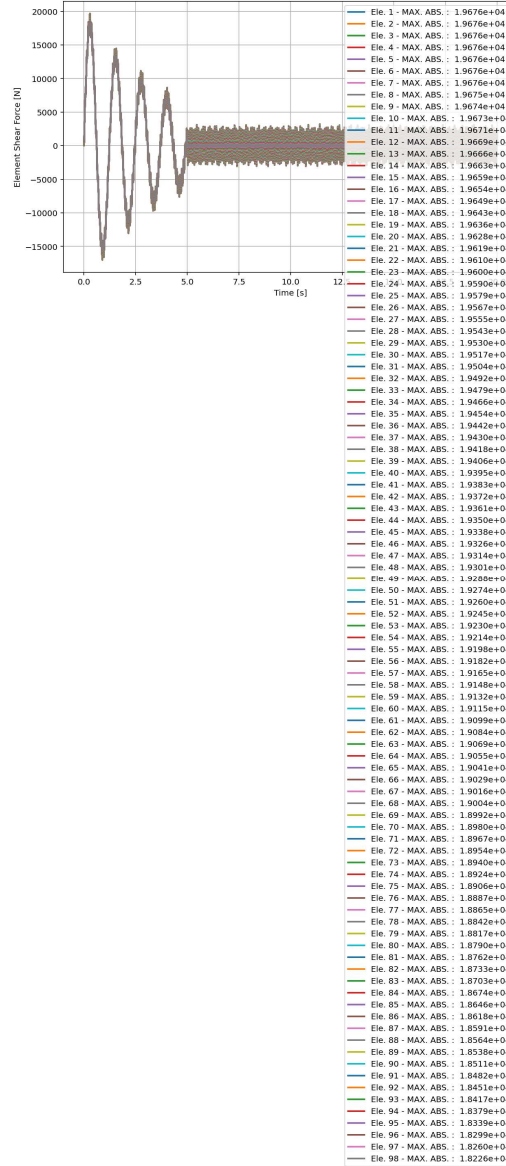




elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



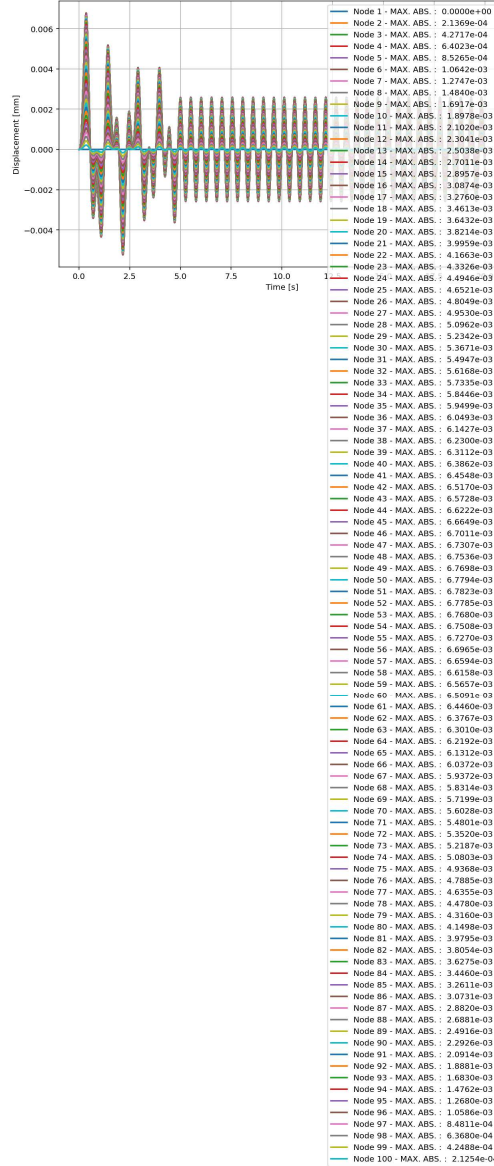
elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



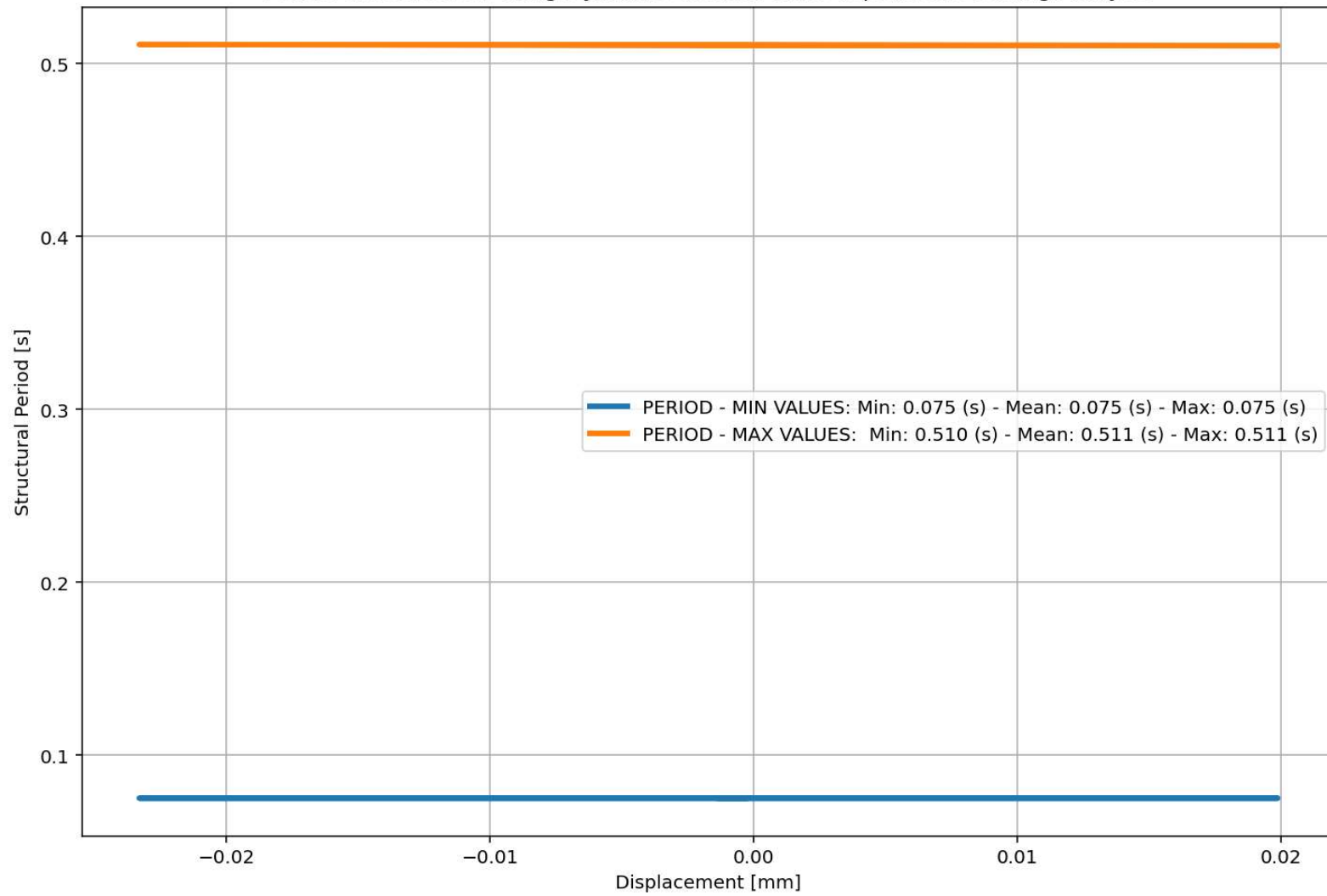
Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



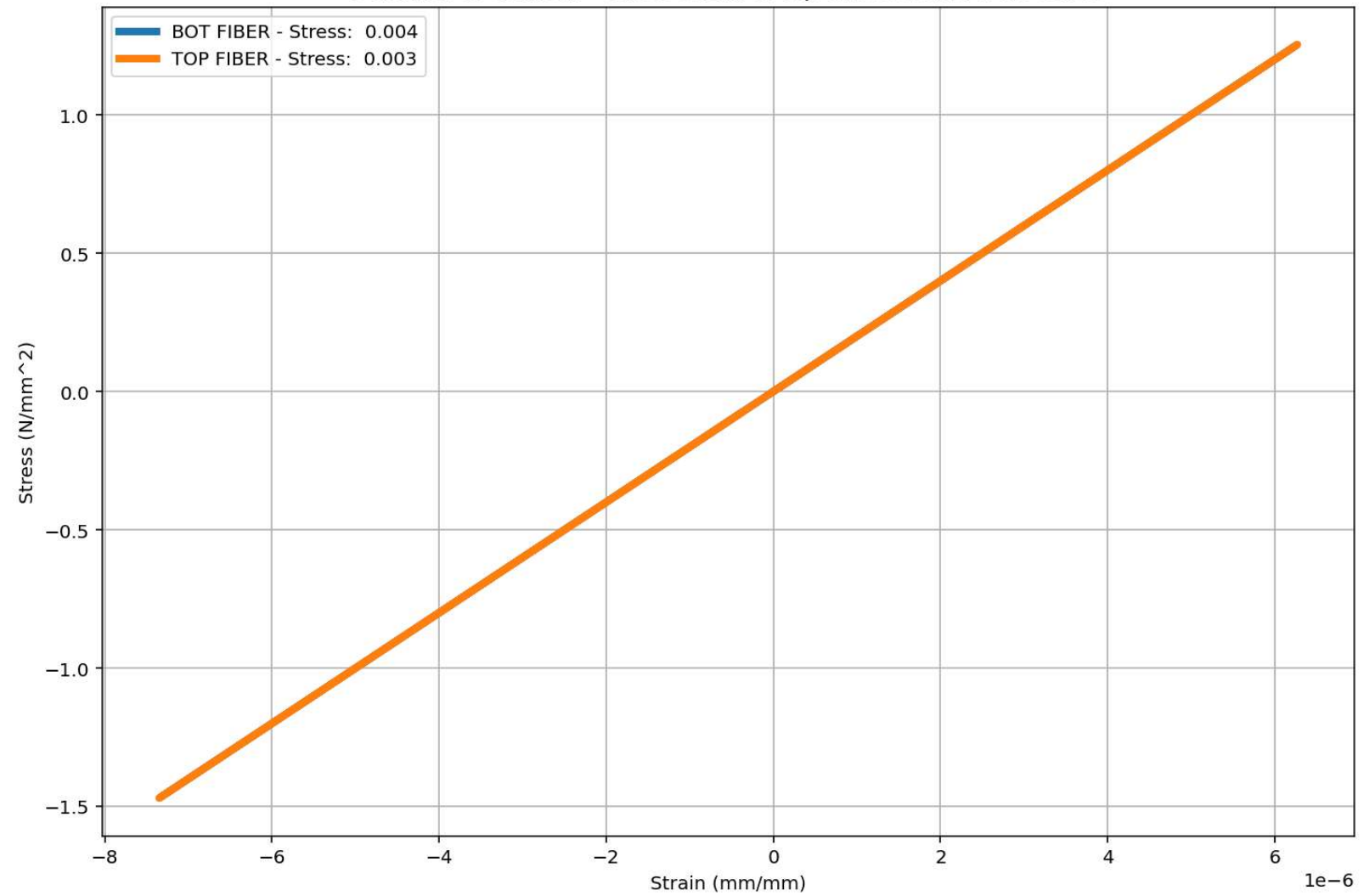
Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

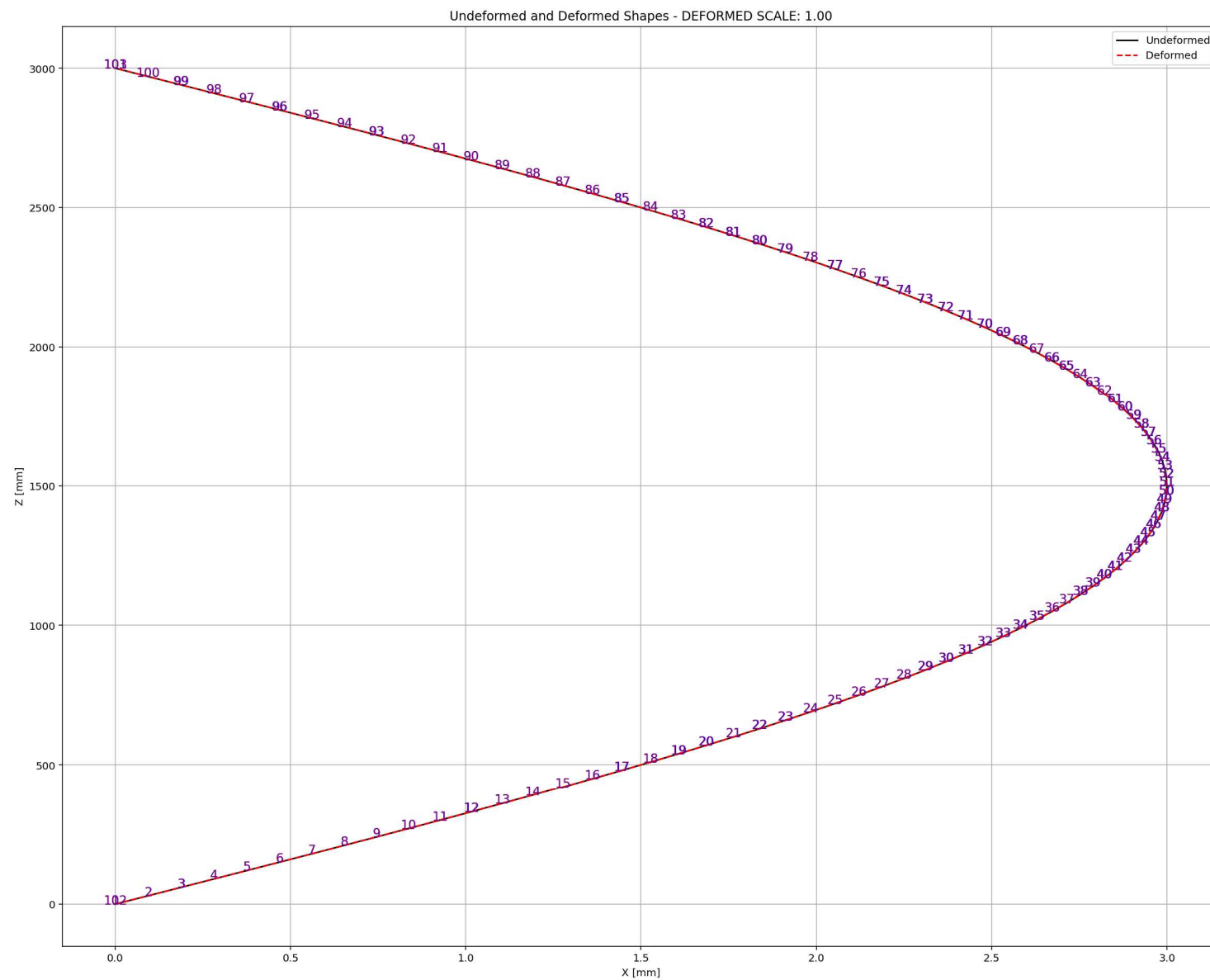


Period of Structure During Dynamic External Time-dependent Loading Analysis

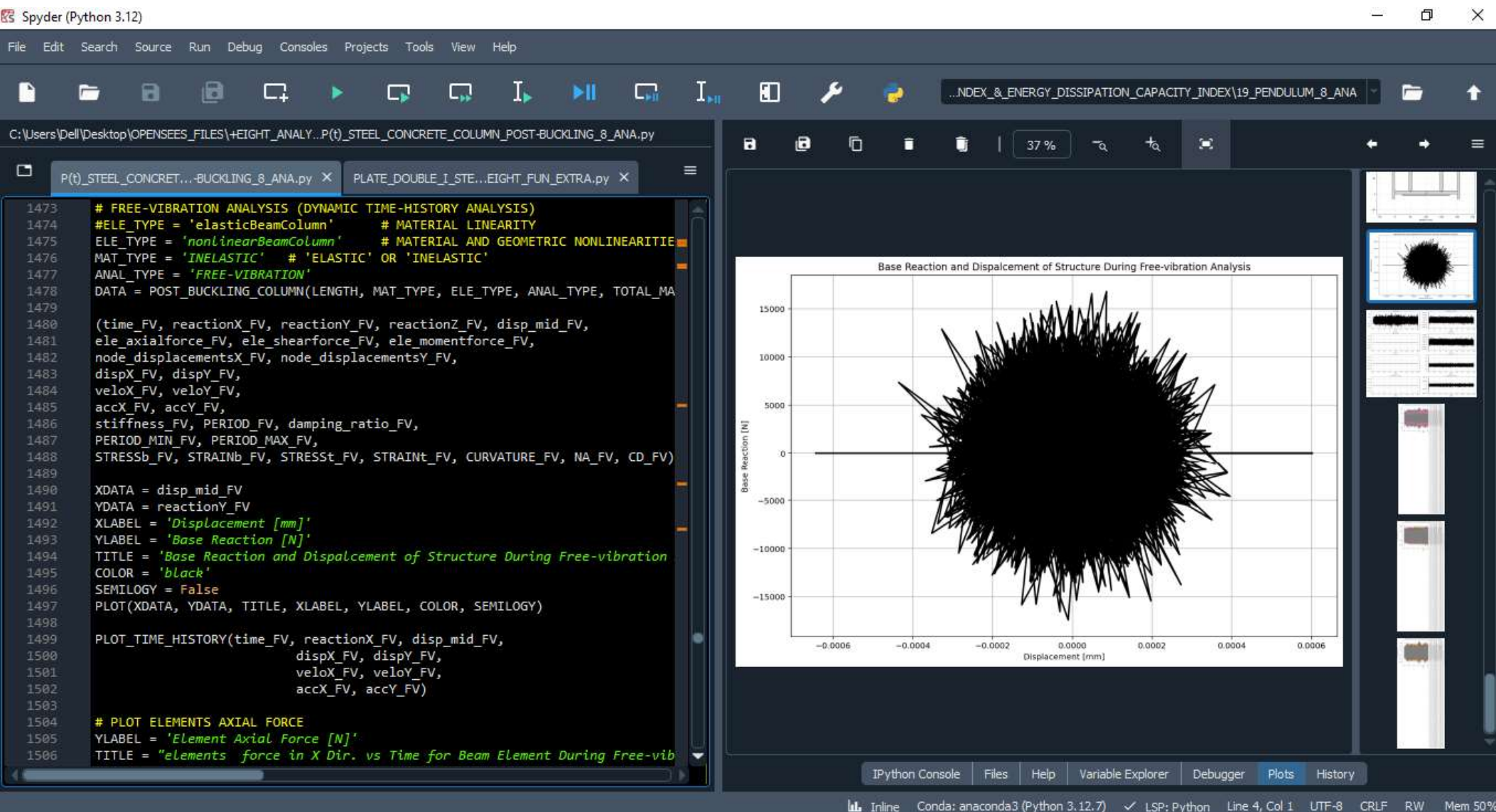


Behavior of element - Stress-strain of Top and Bottom Fibers Curve

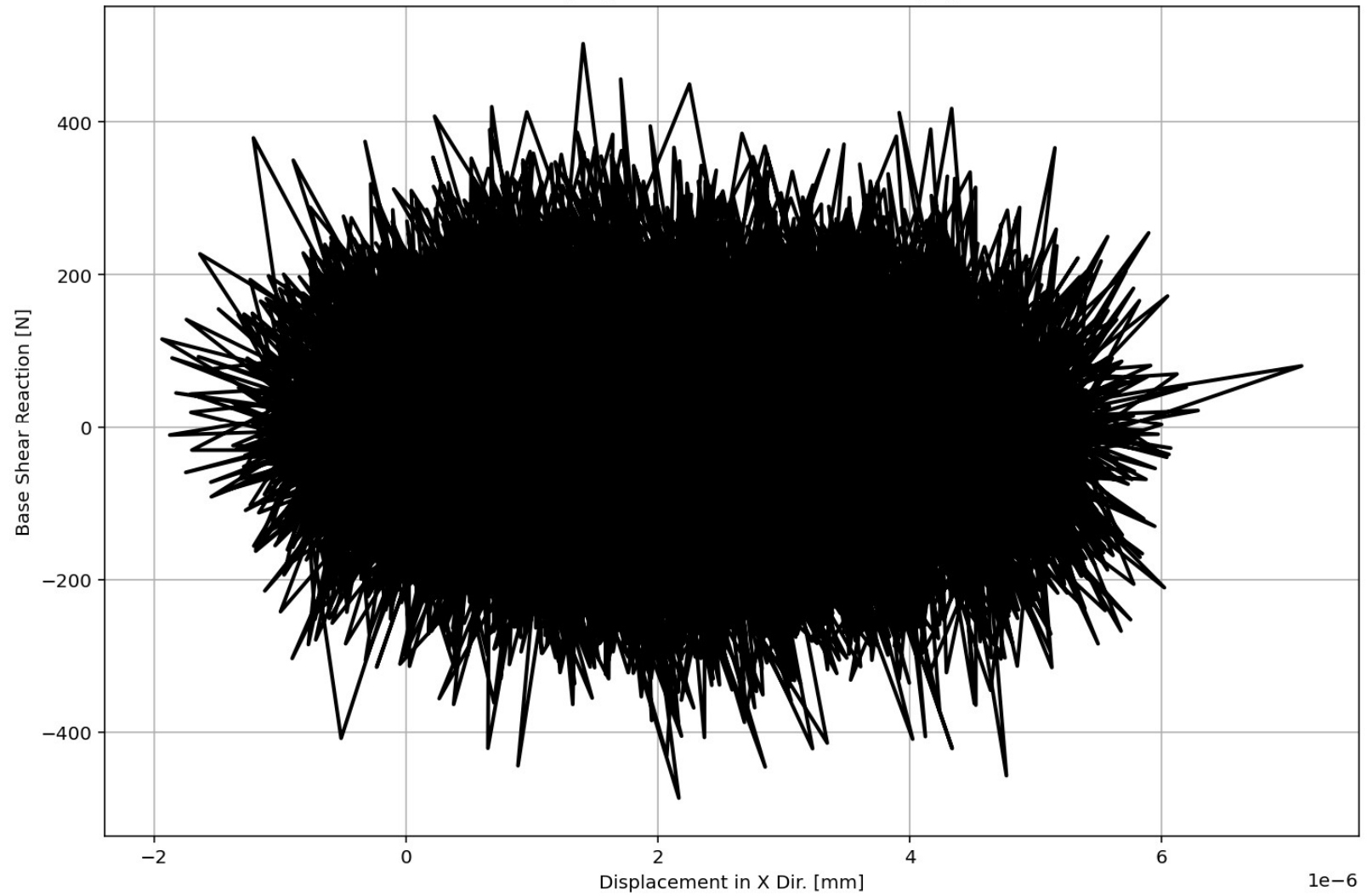




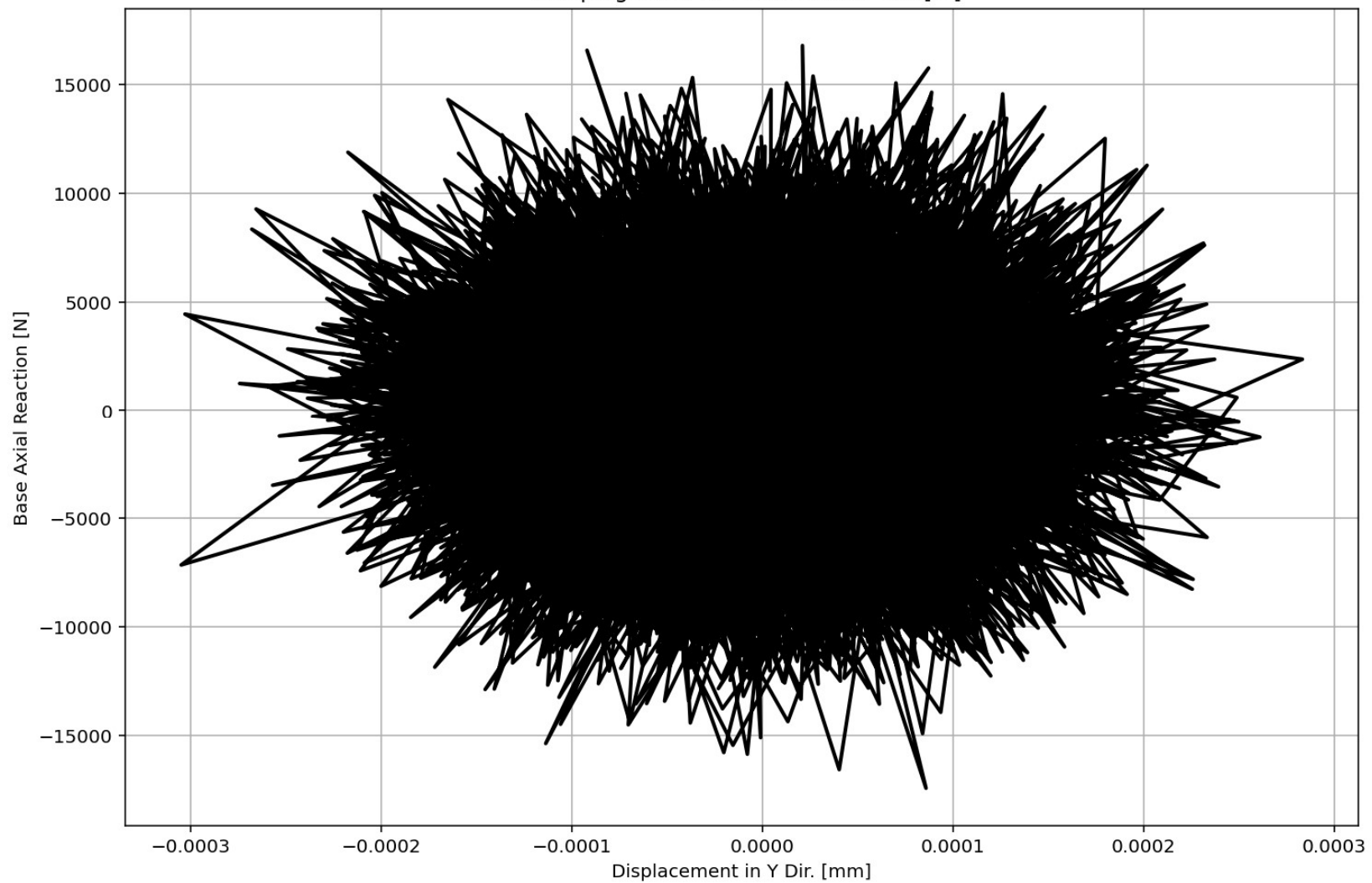
FREE-VIBRATION ANALYSIS

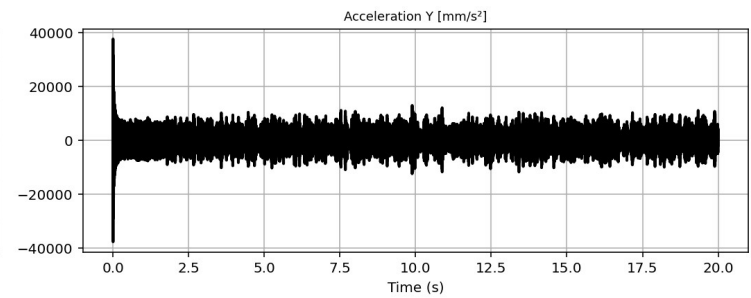
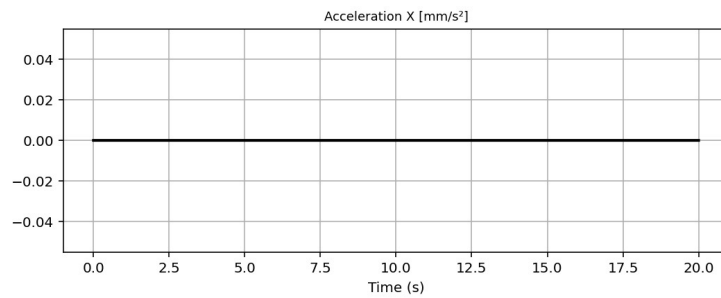
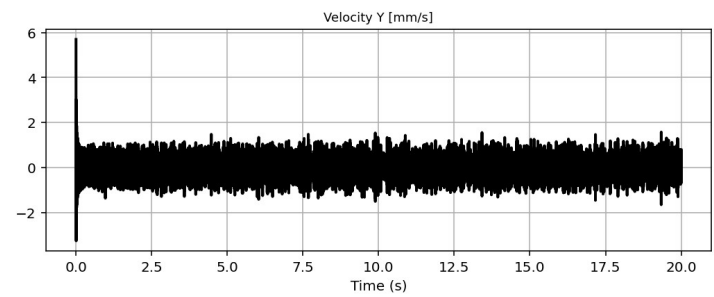
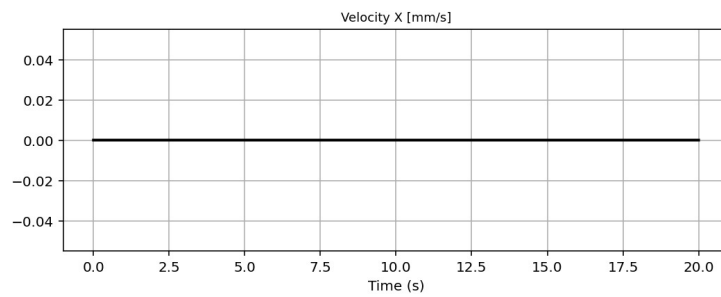
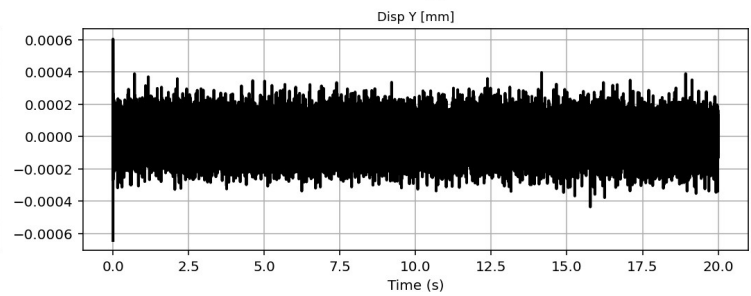
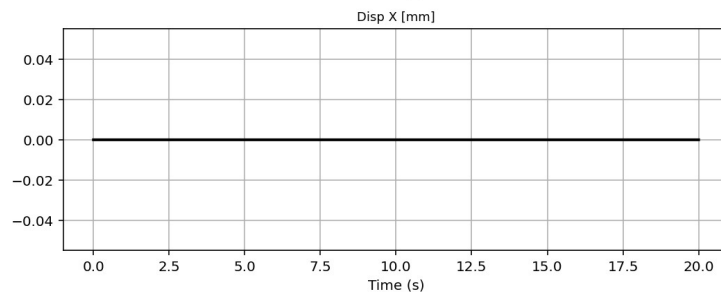
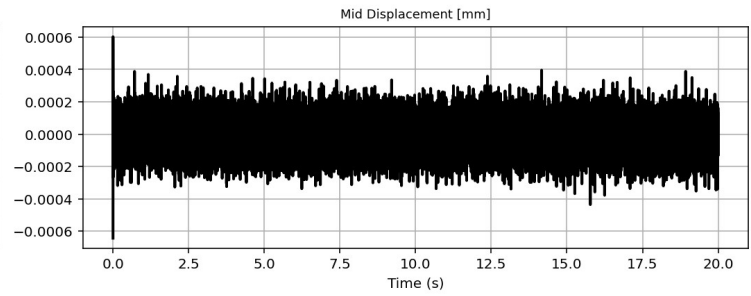
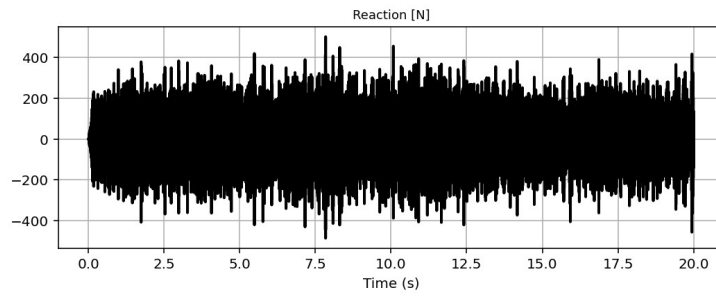


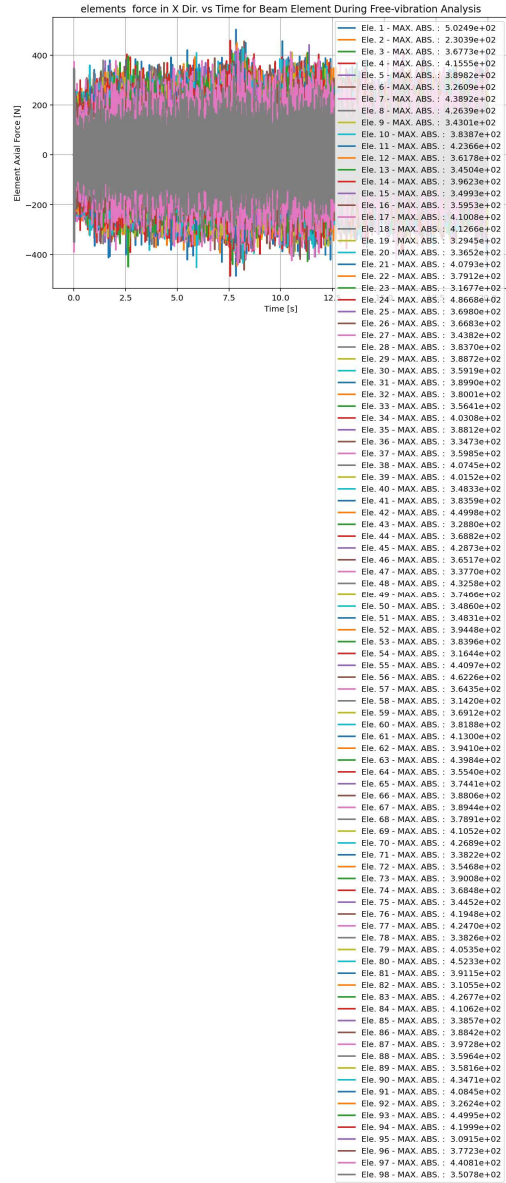
Base Shear Reaction and Displacement of Structure During Free-vibration Analysis
Damping Ratio in X Dir: 0.000e+00 [%]



Base Axial Reaction and Dispalcement of Structure During Free-vibration Analysis
Damping Ratio in Y Dir: 0.000e+00 [%]

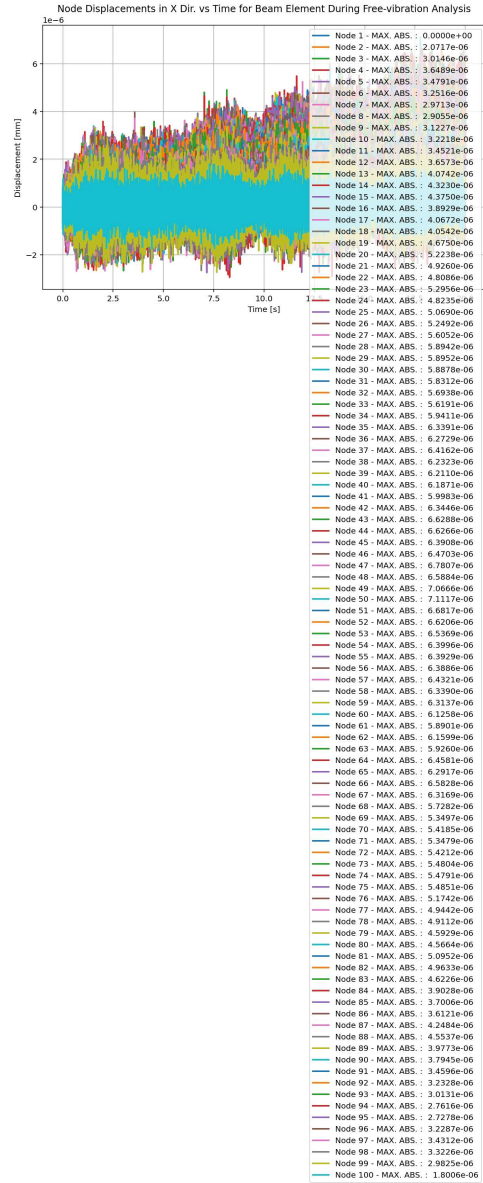


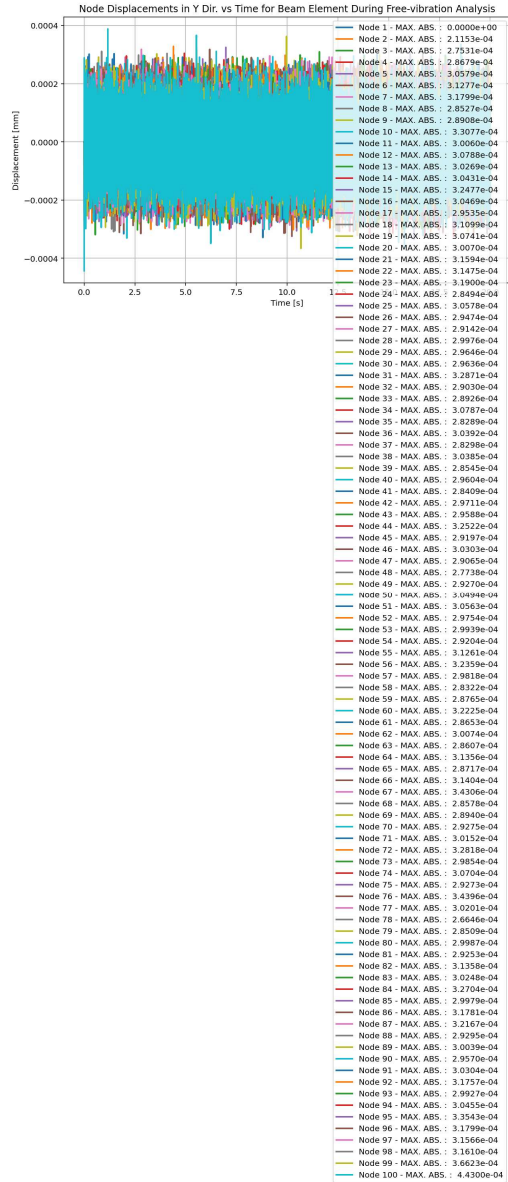


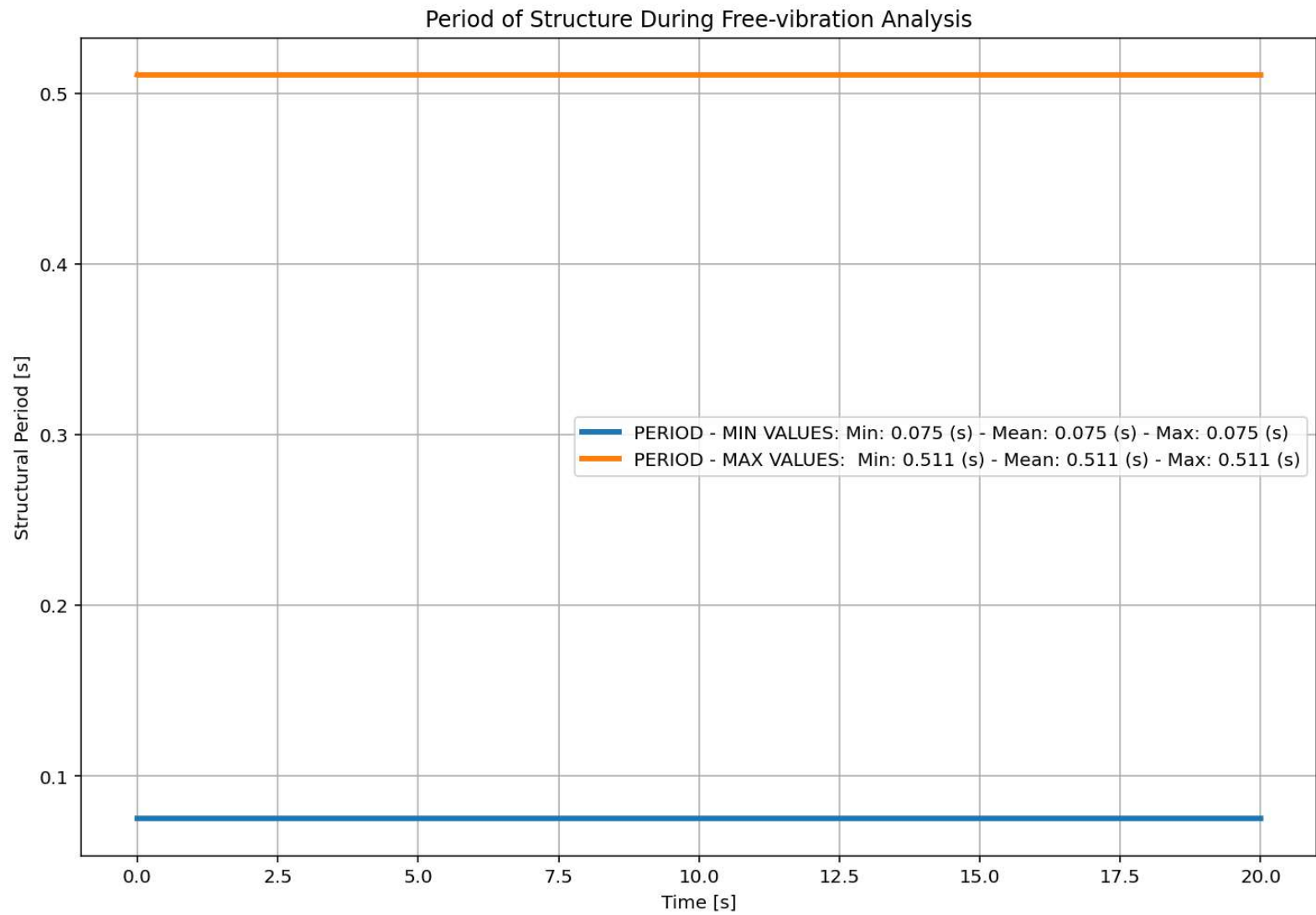


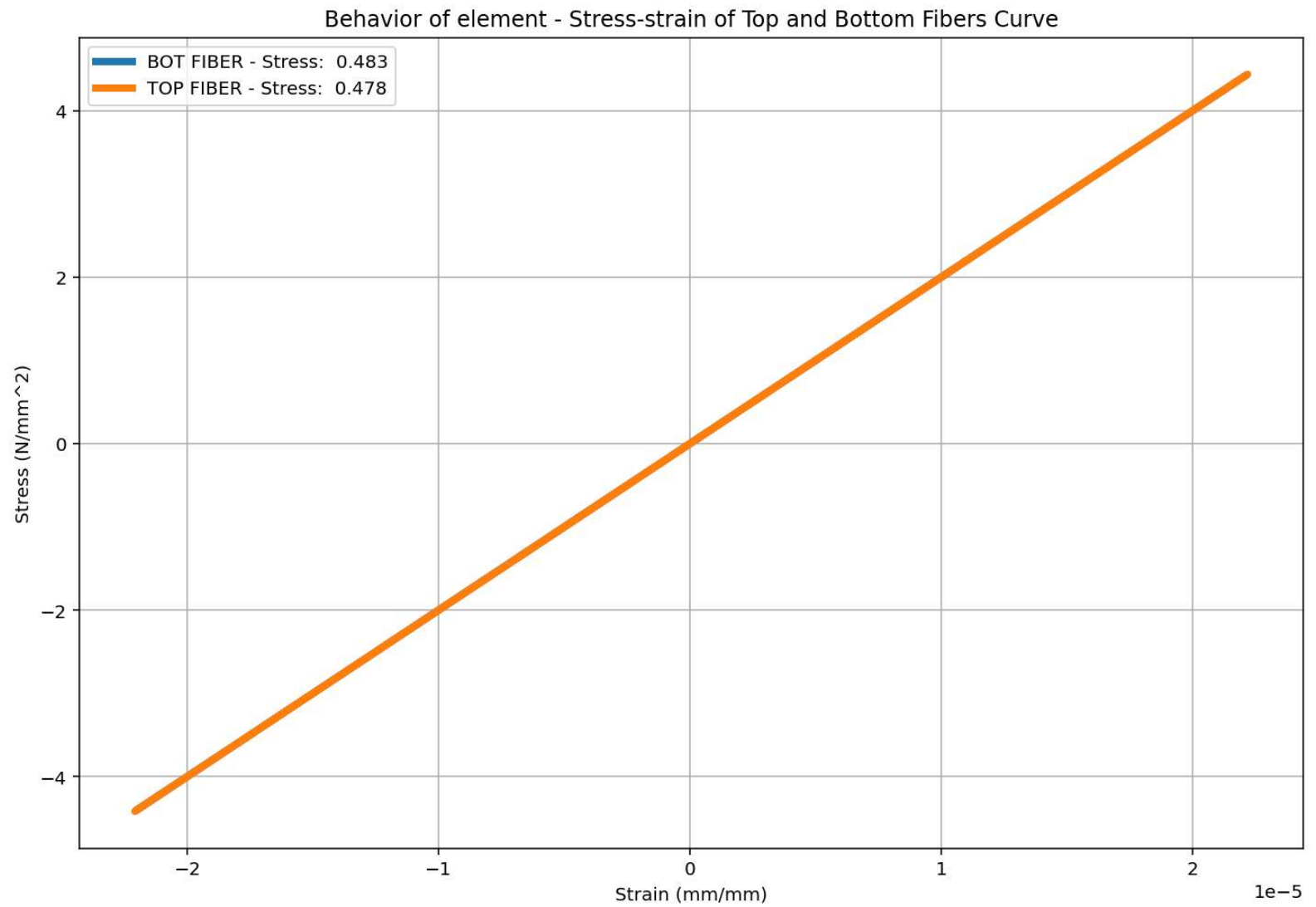


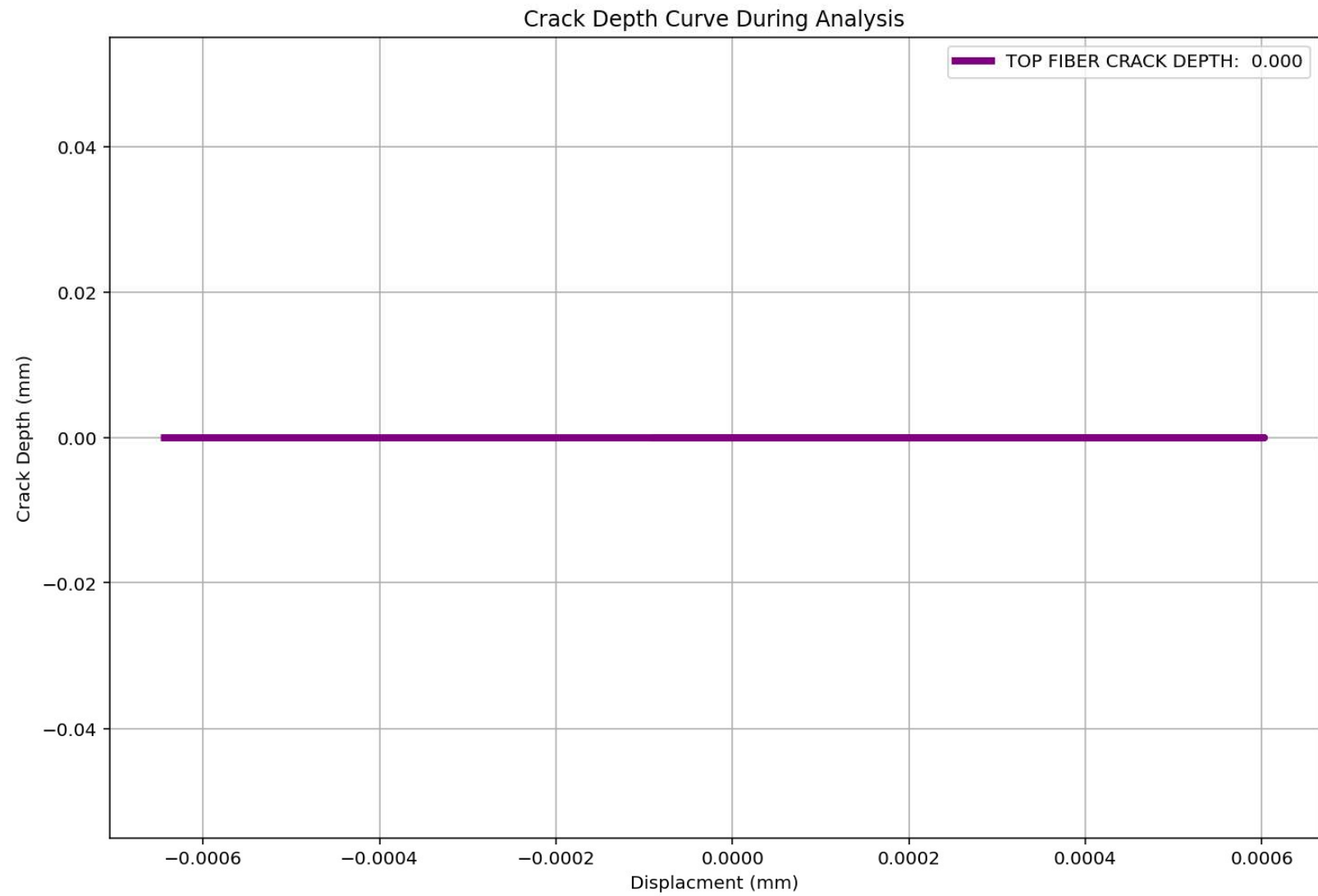


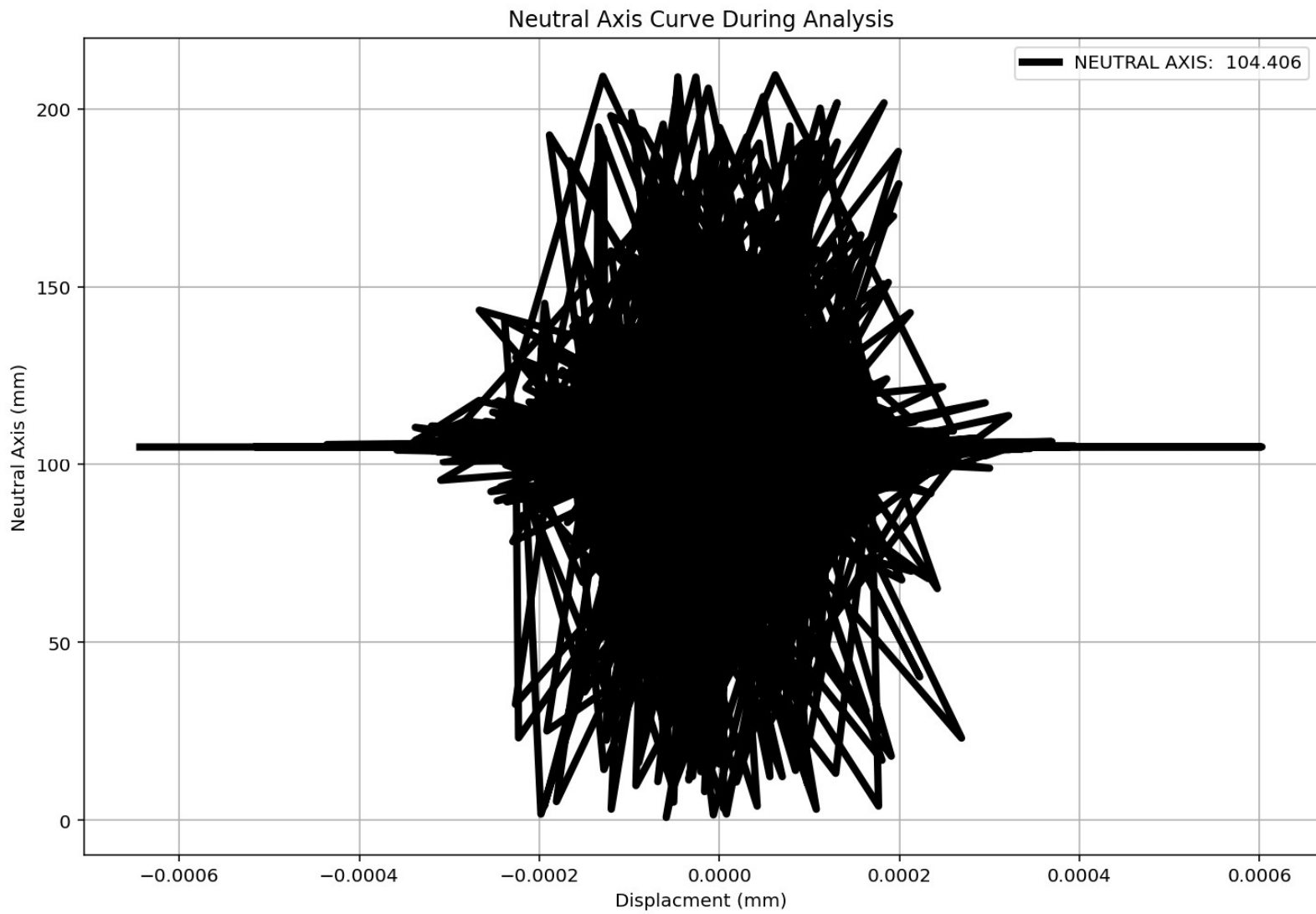




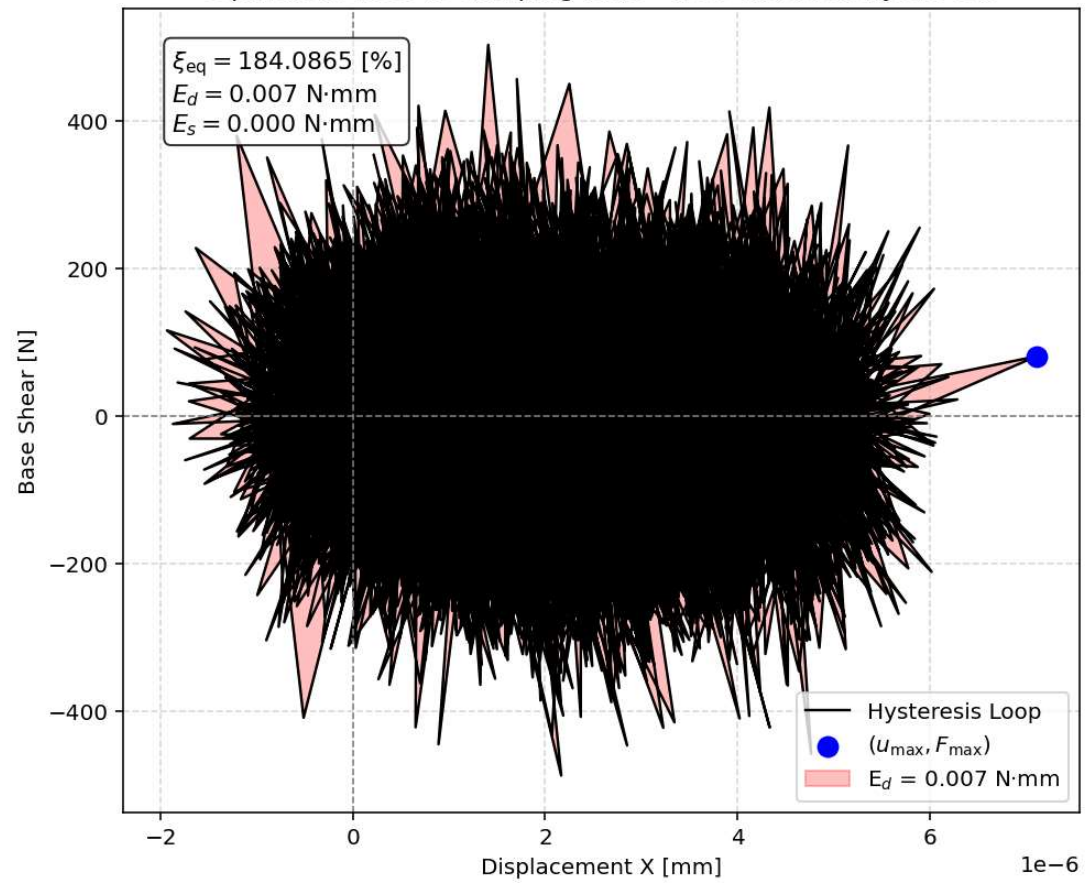




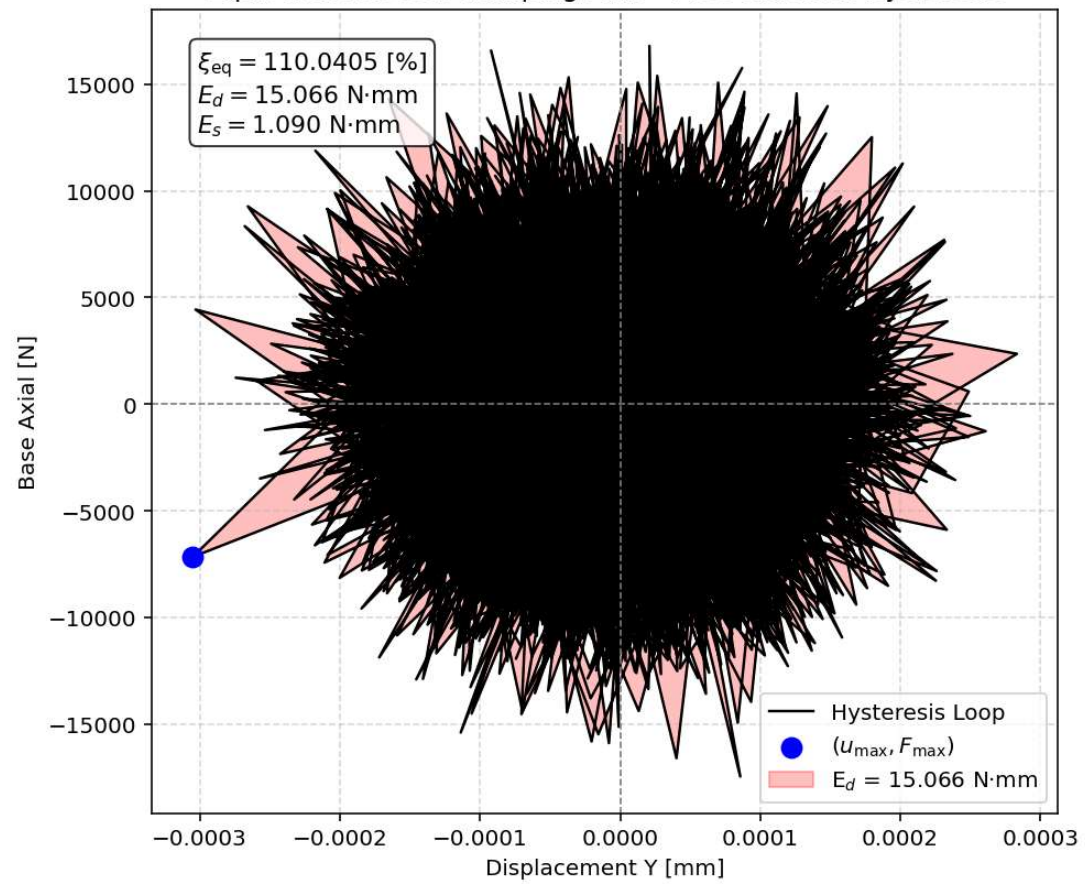


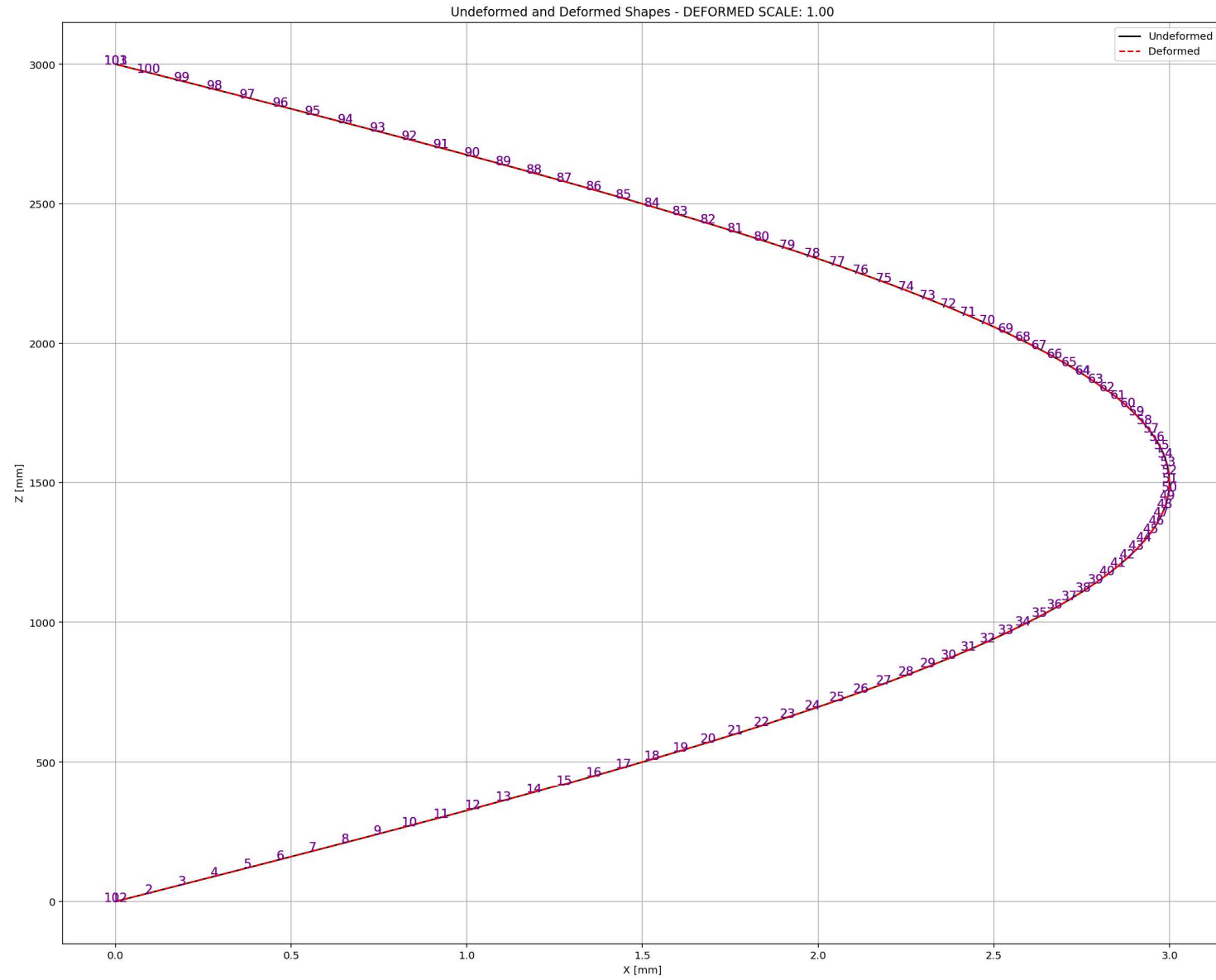


Equivalent viscous damping ratio - Free-vibration Hysteresis

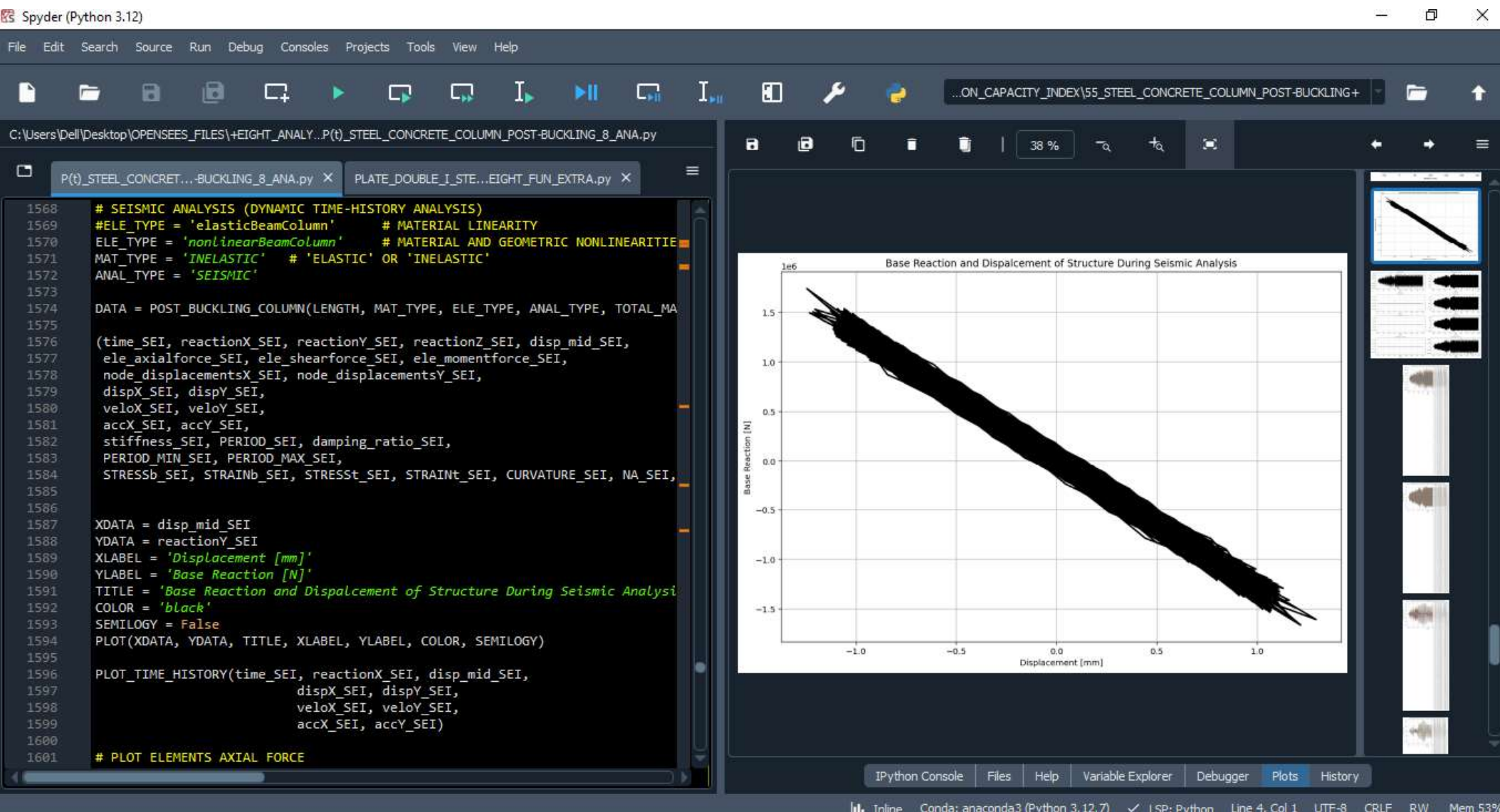


Equivalent viscous damping ratio - Free-vibration Hysteresis

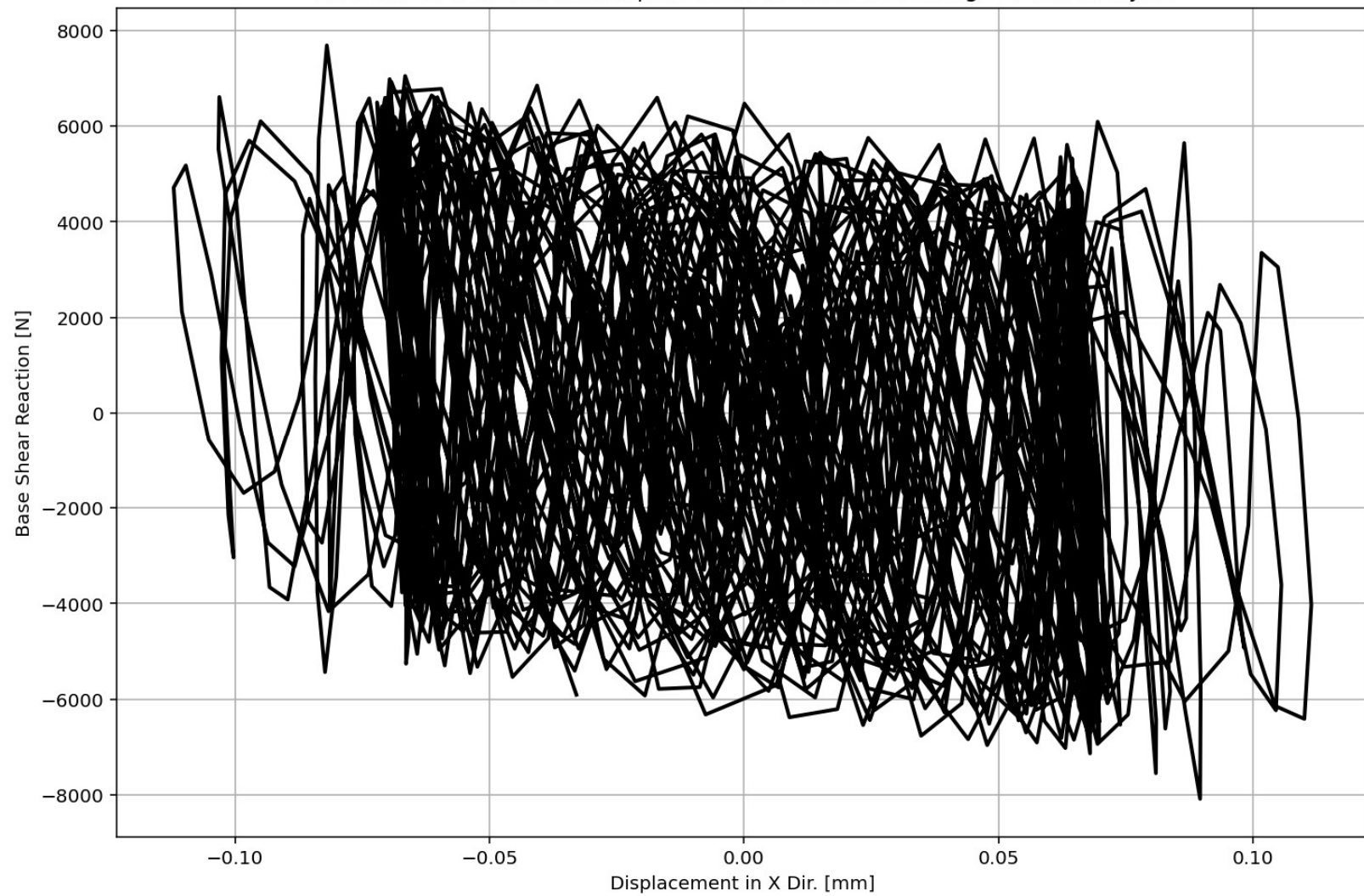


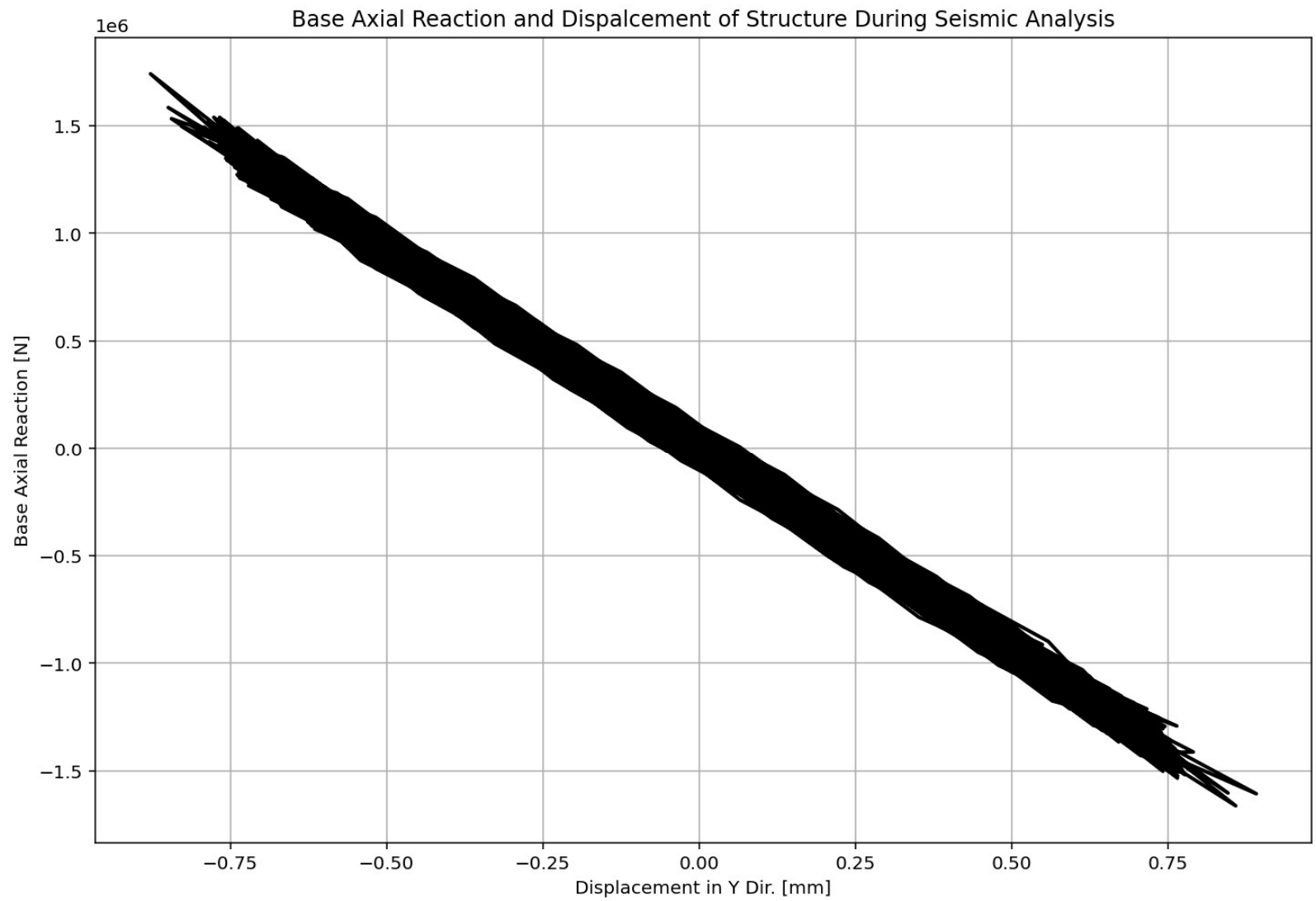


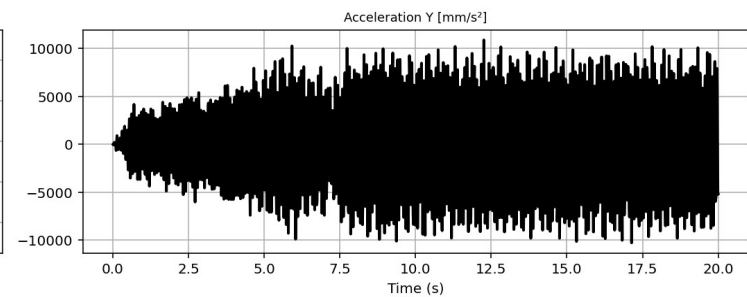
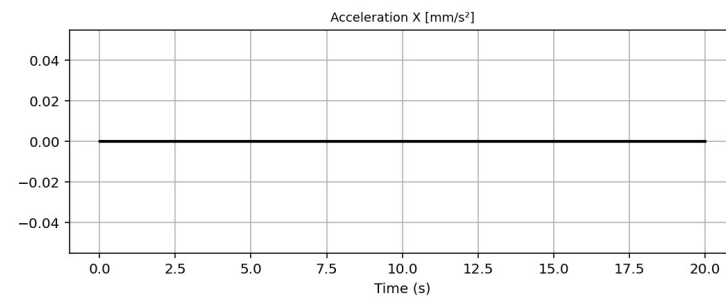
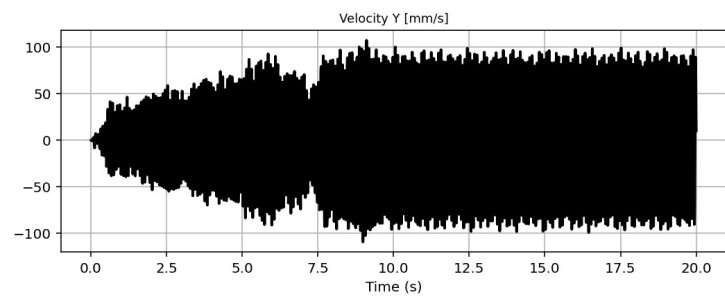
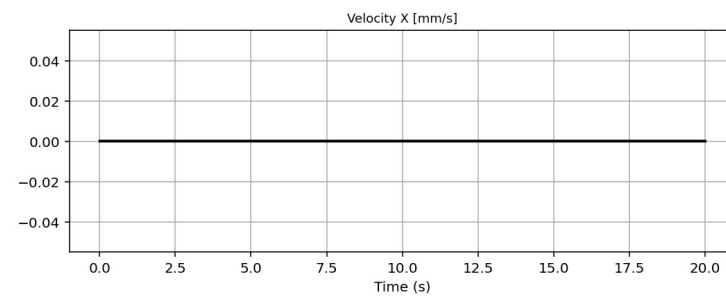
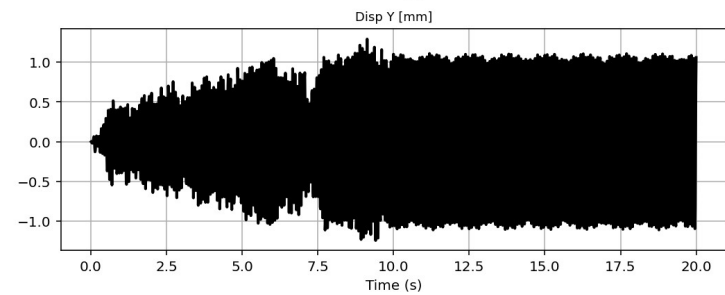
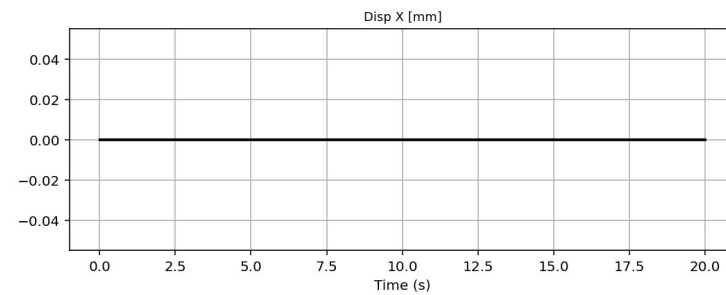
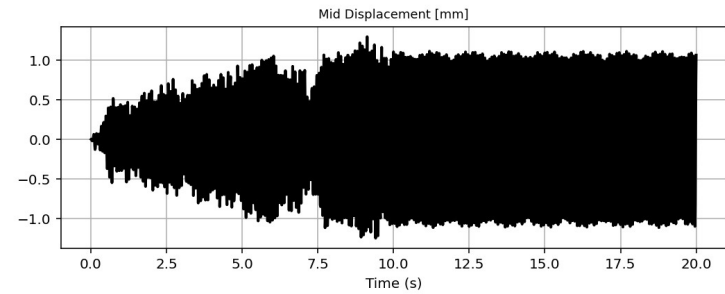
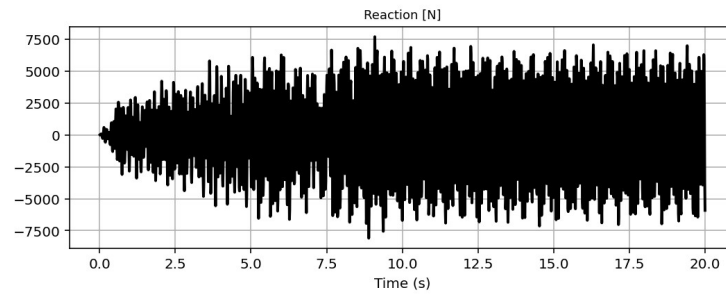
SEISMIC ANALYSIS

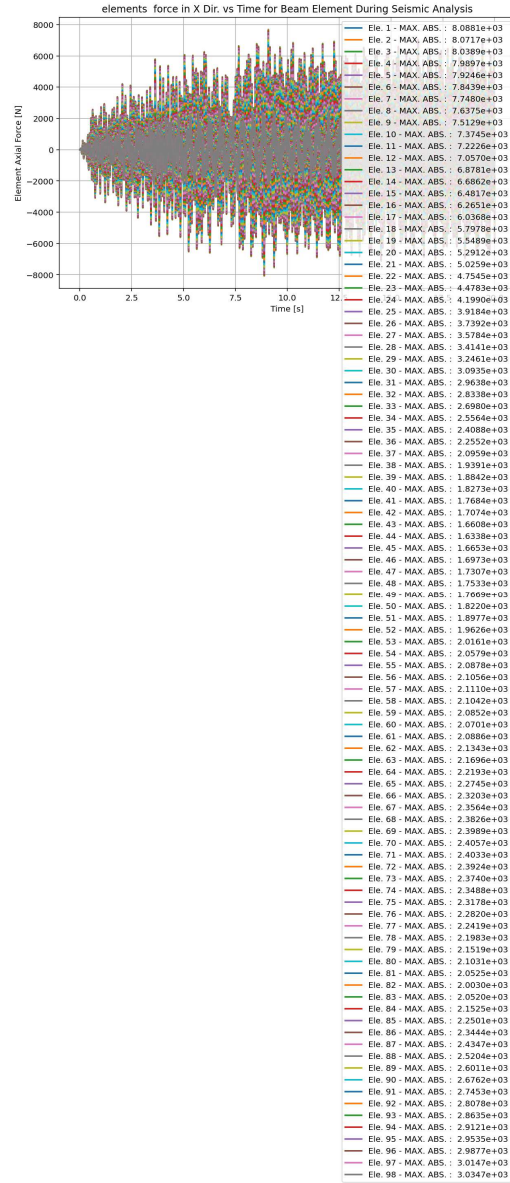


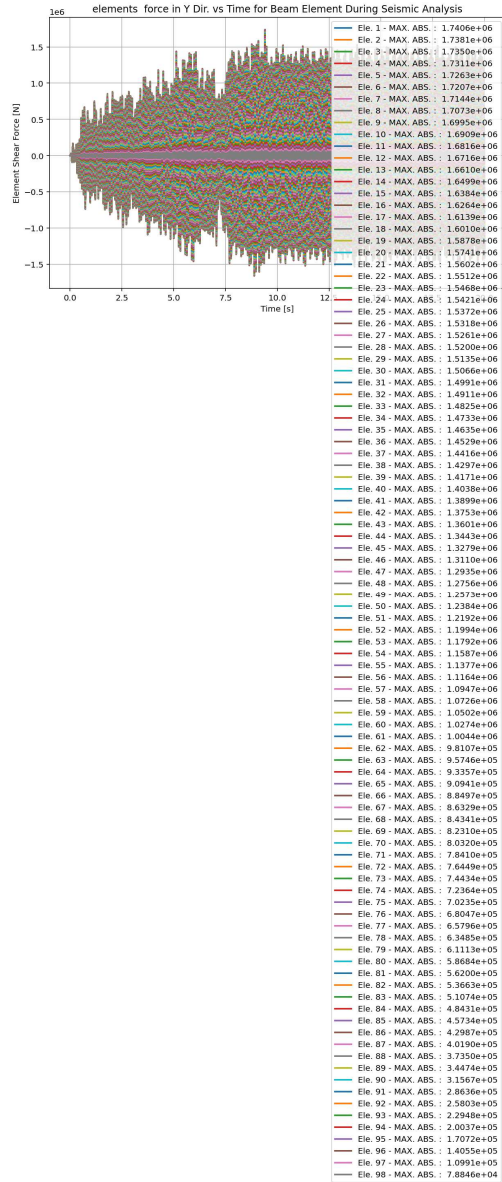
Base Shear Reaction and Displacement of Structure During Seismic Analysis

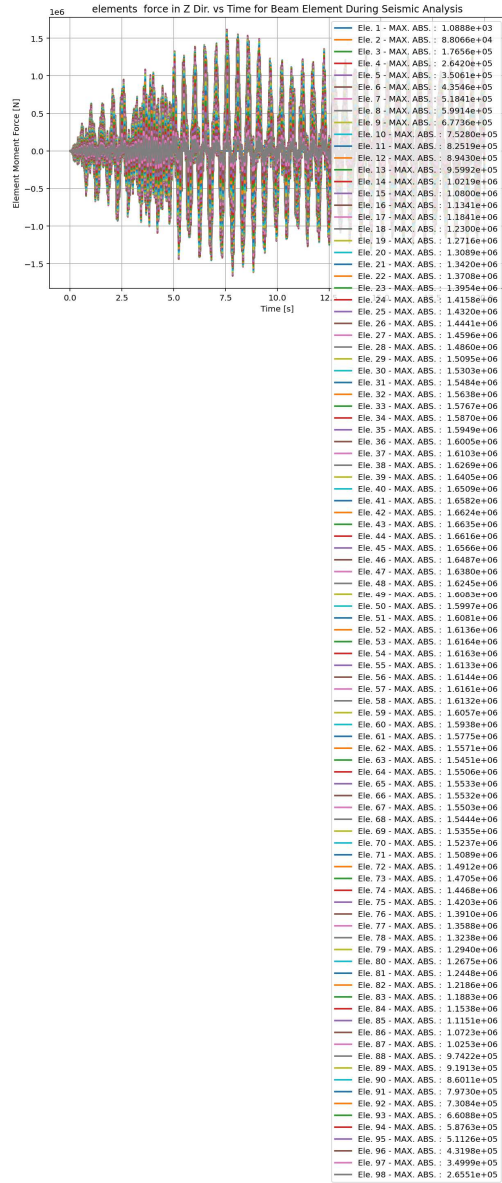


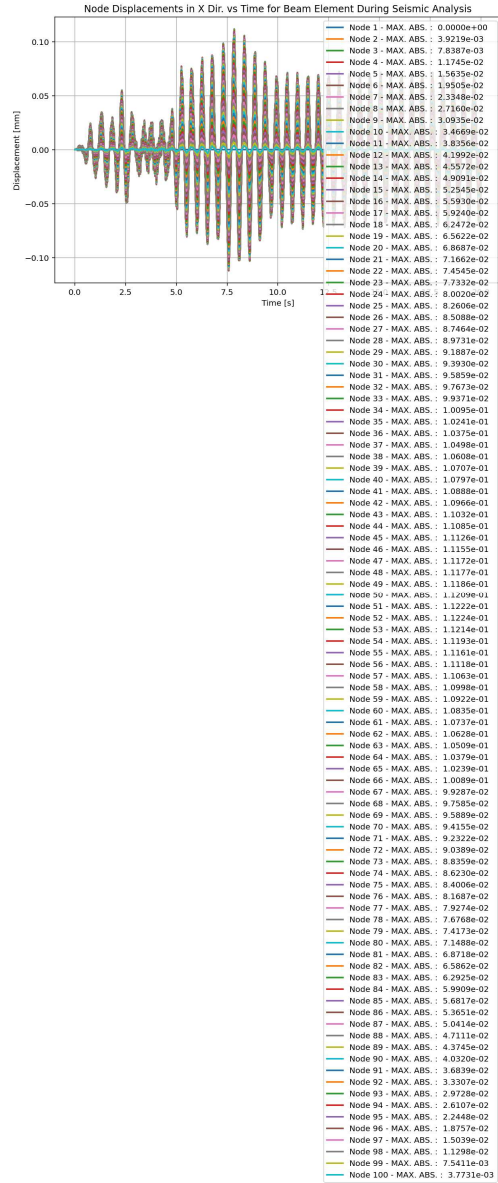


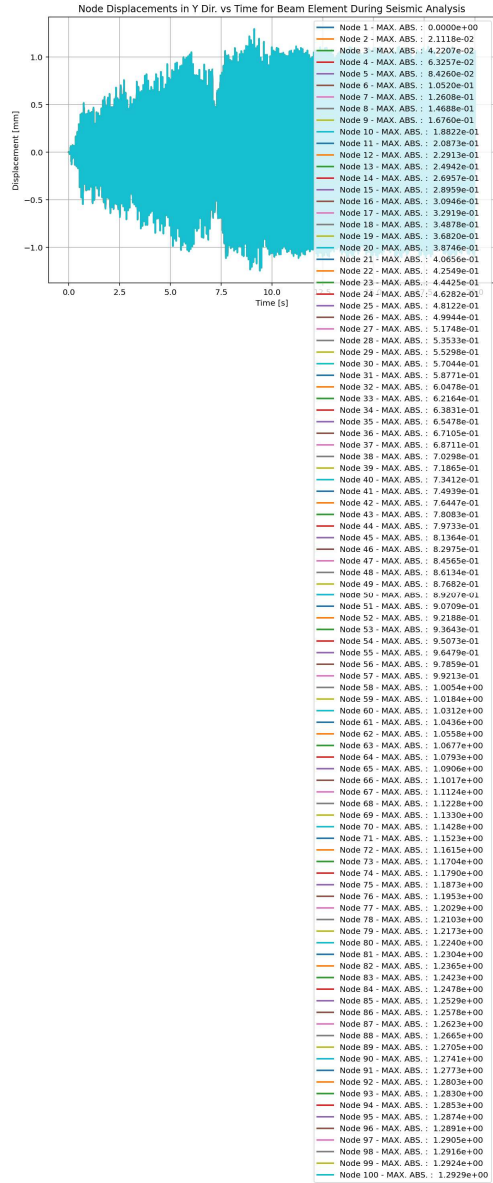




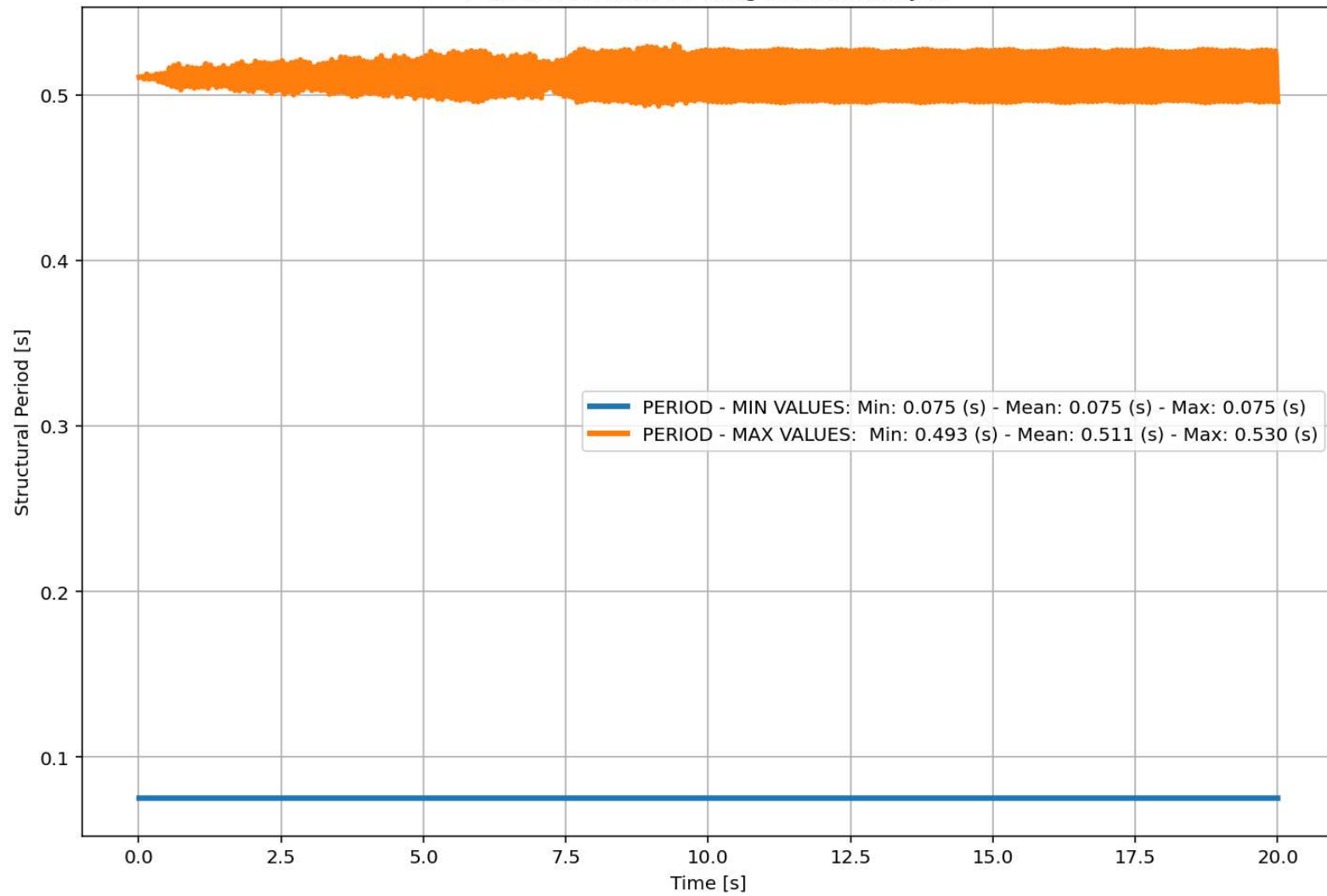


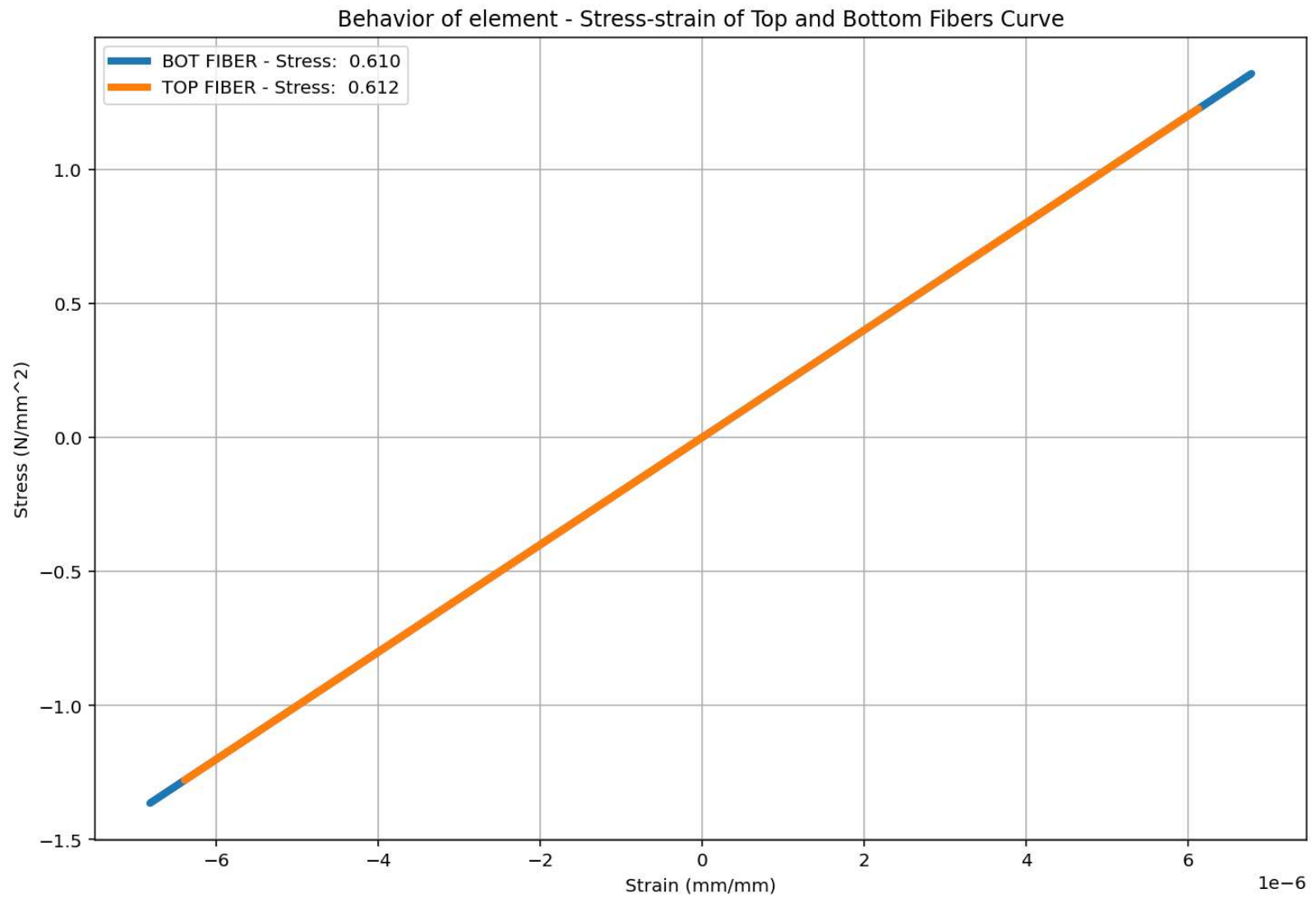


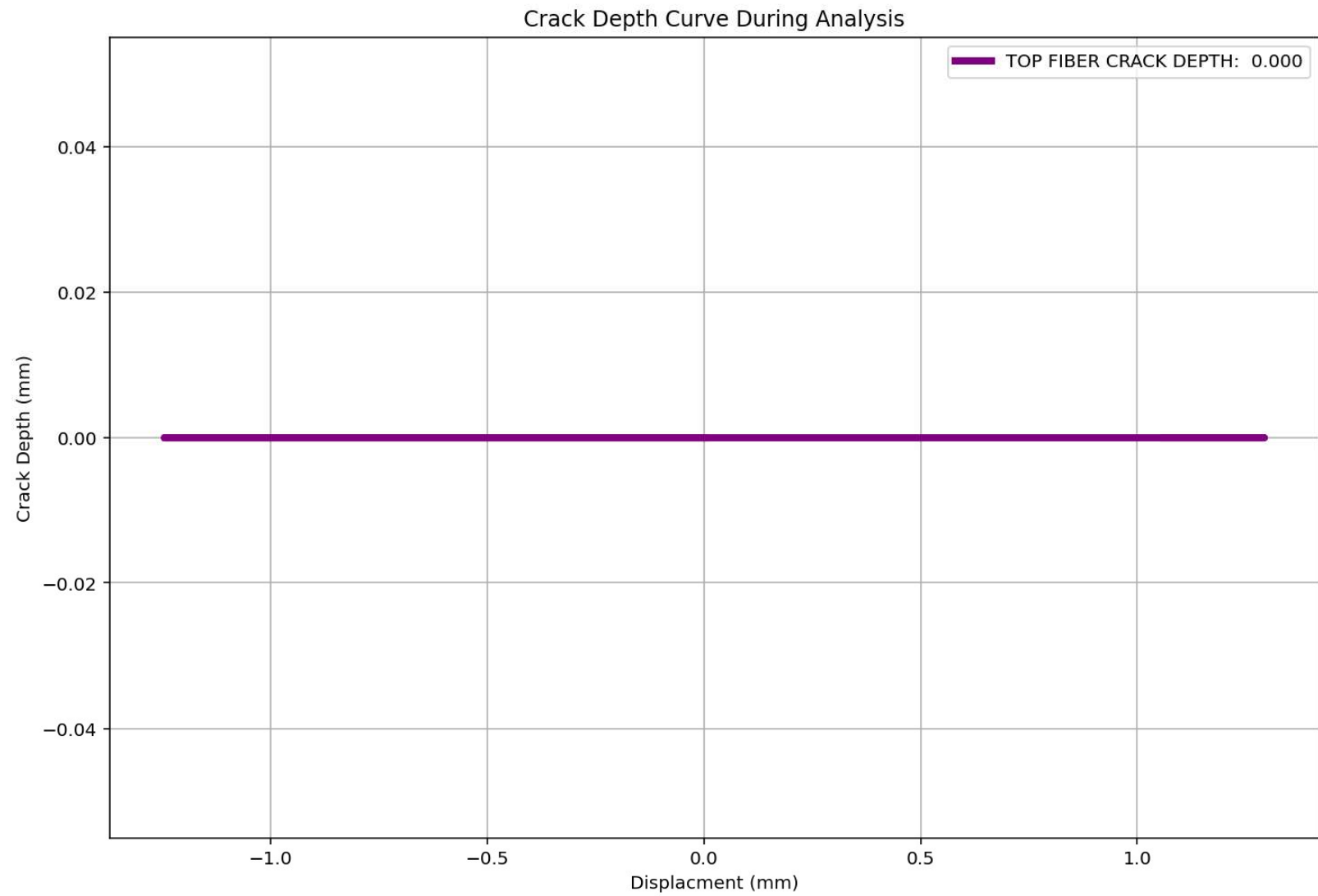


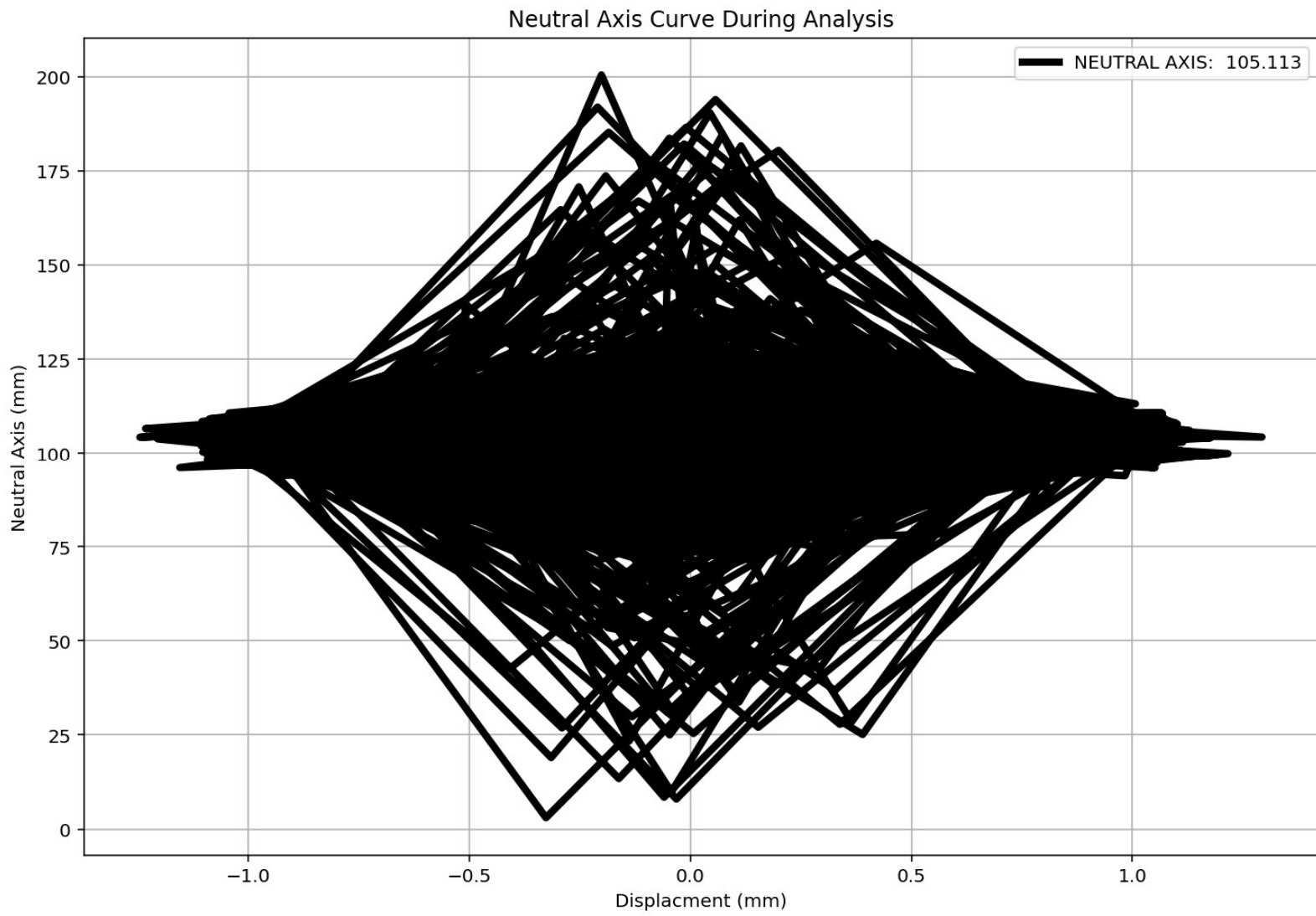


Period of Structure During Seismic Analysis

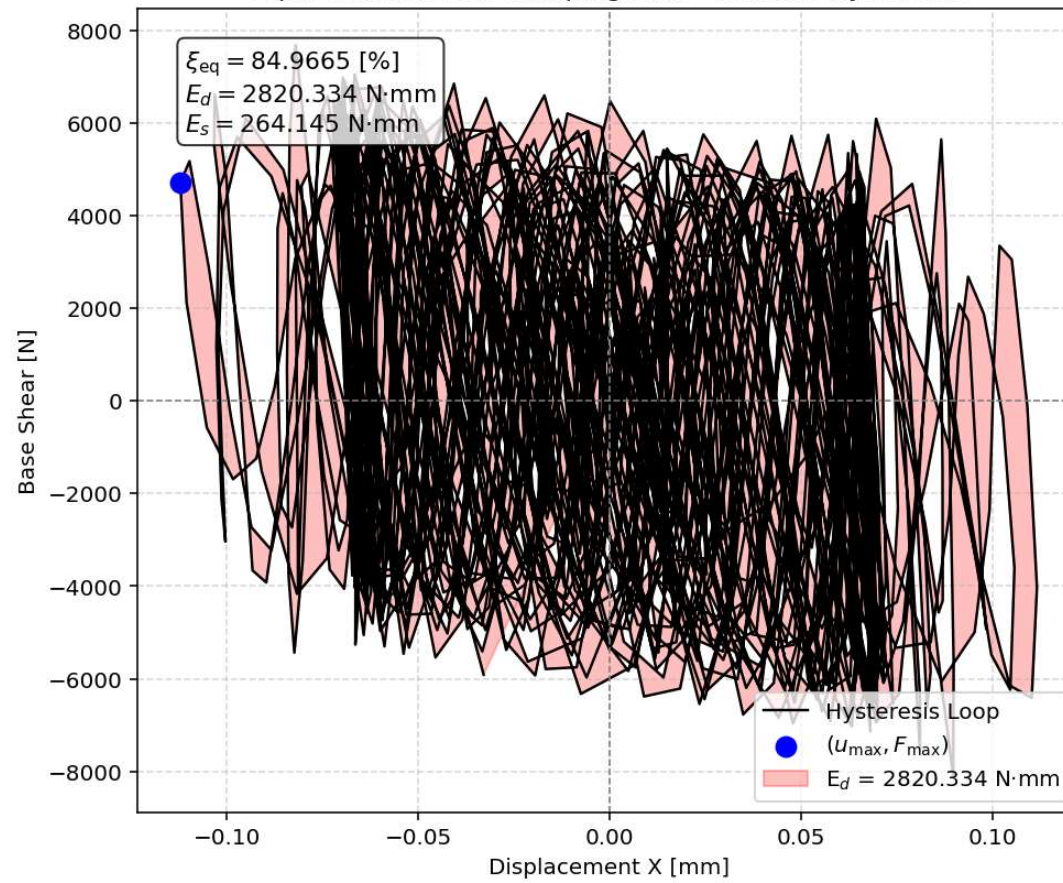


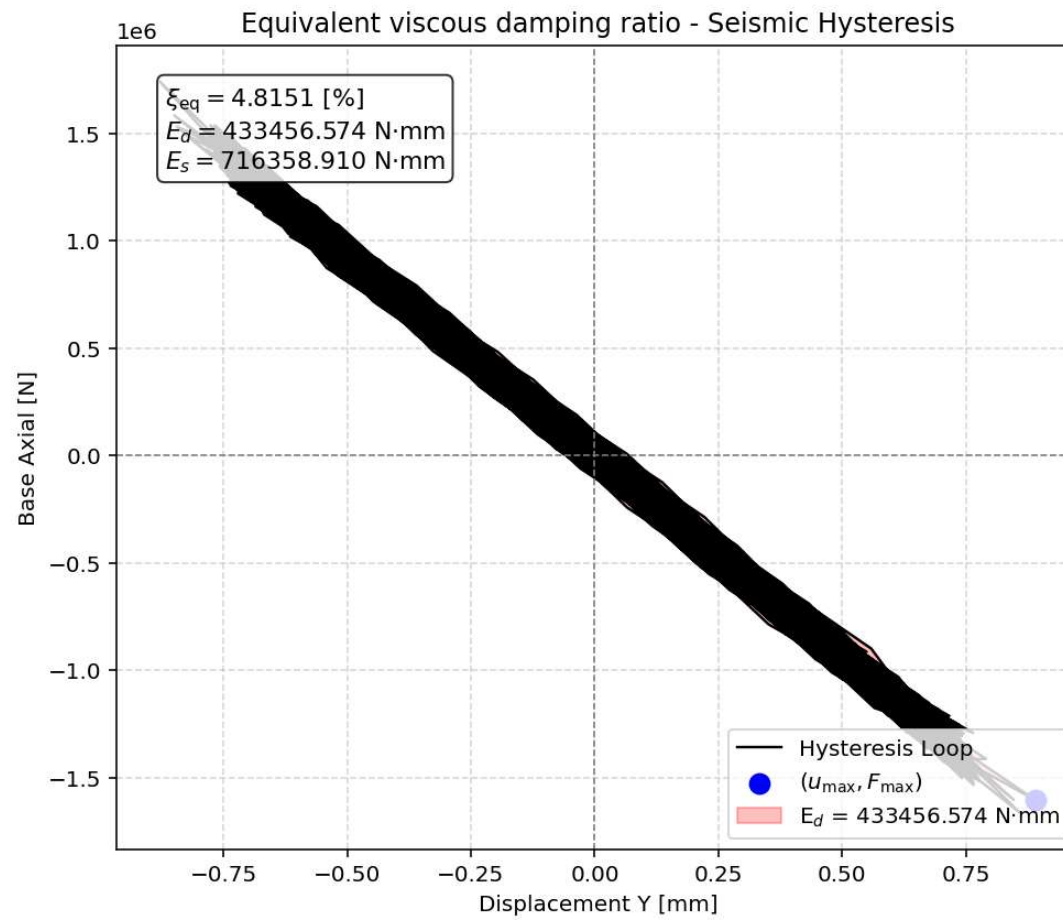


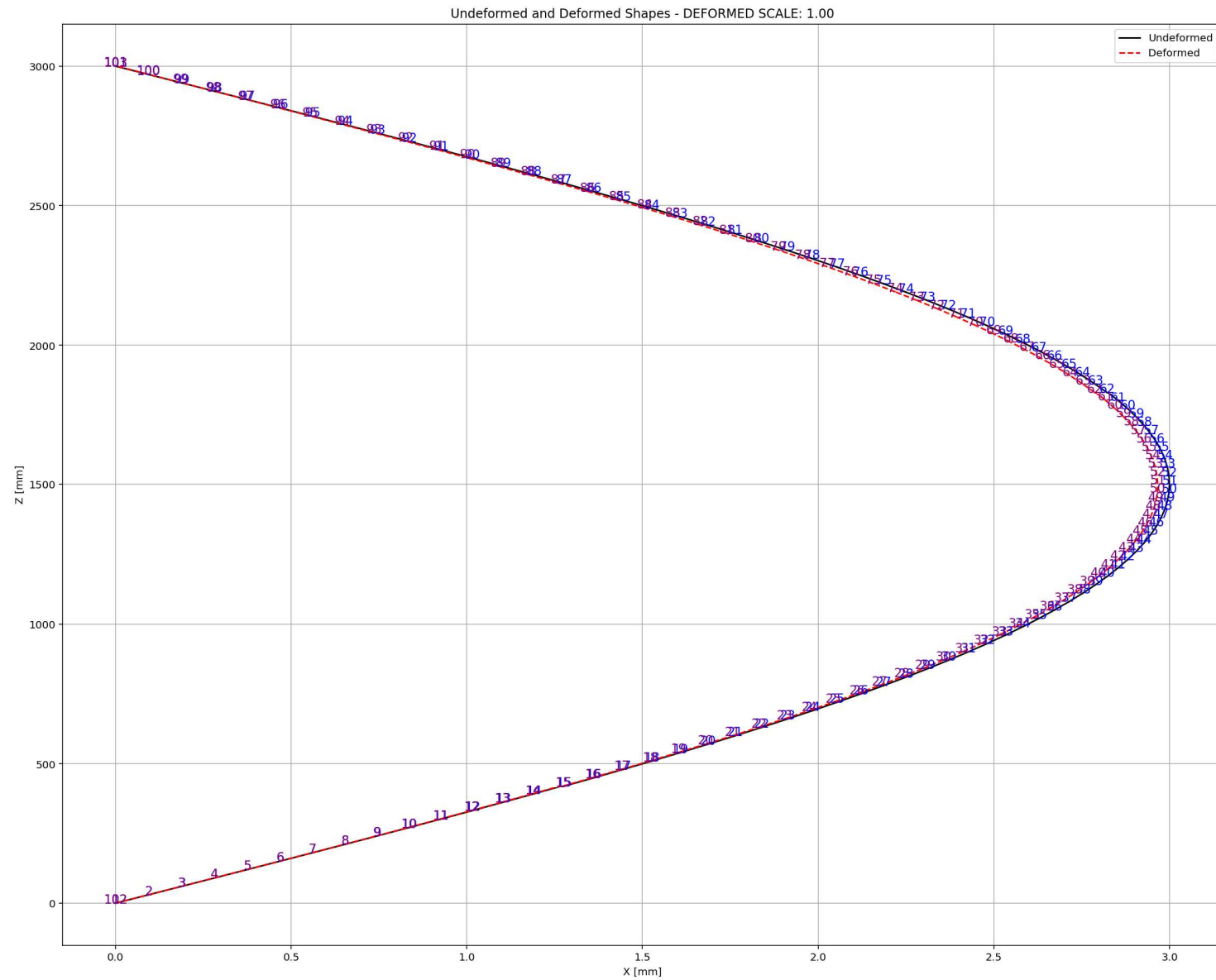


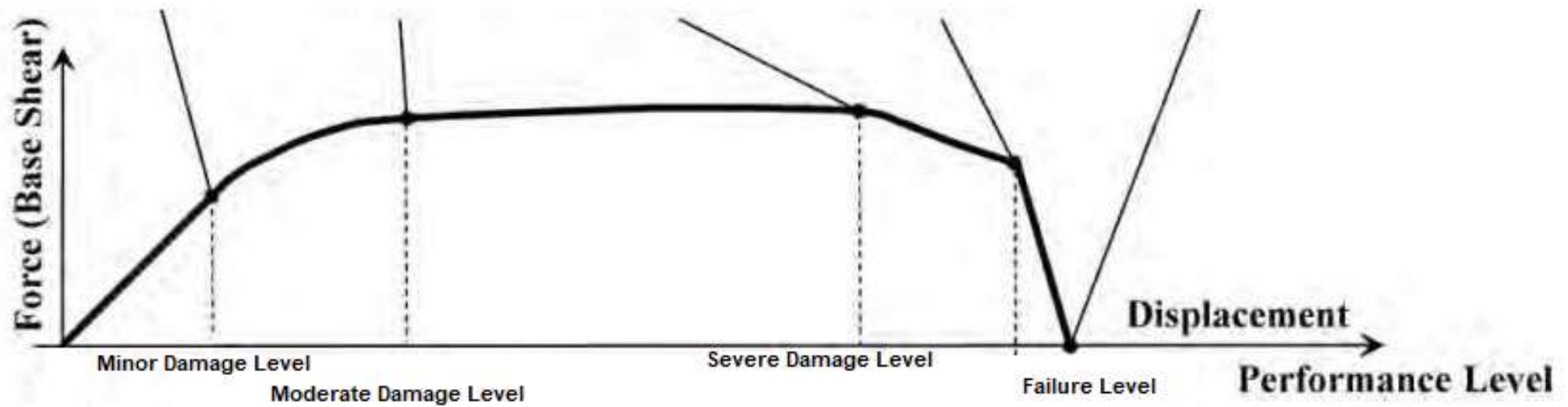
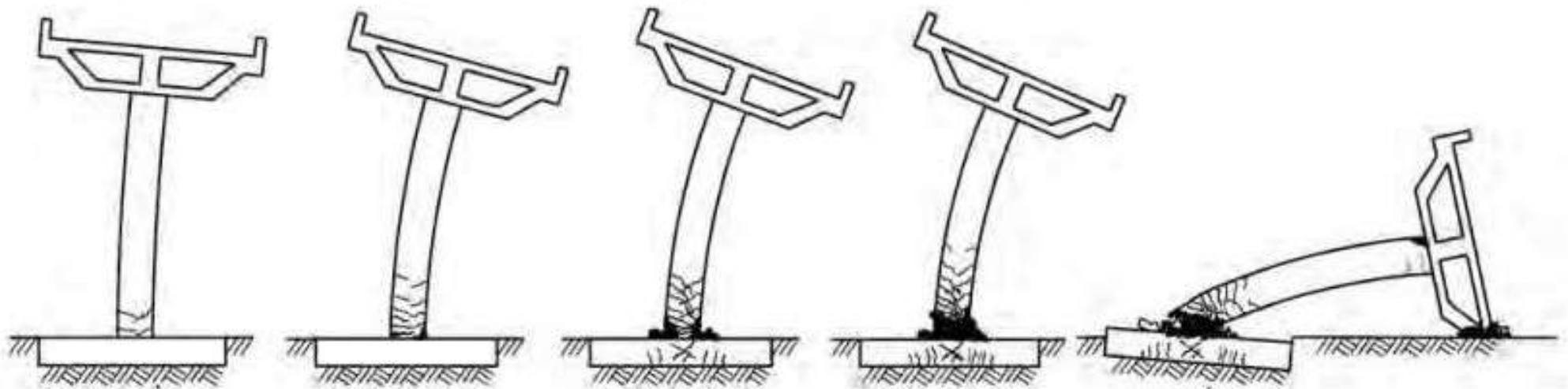


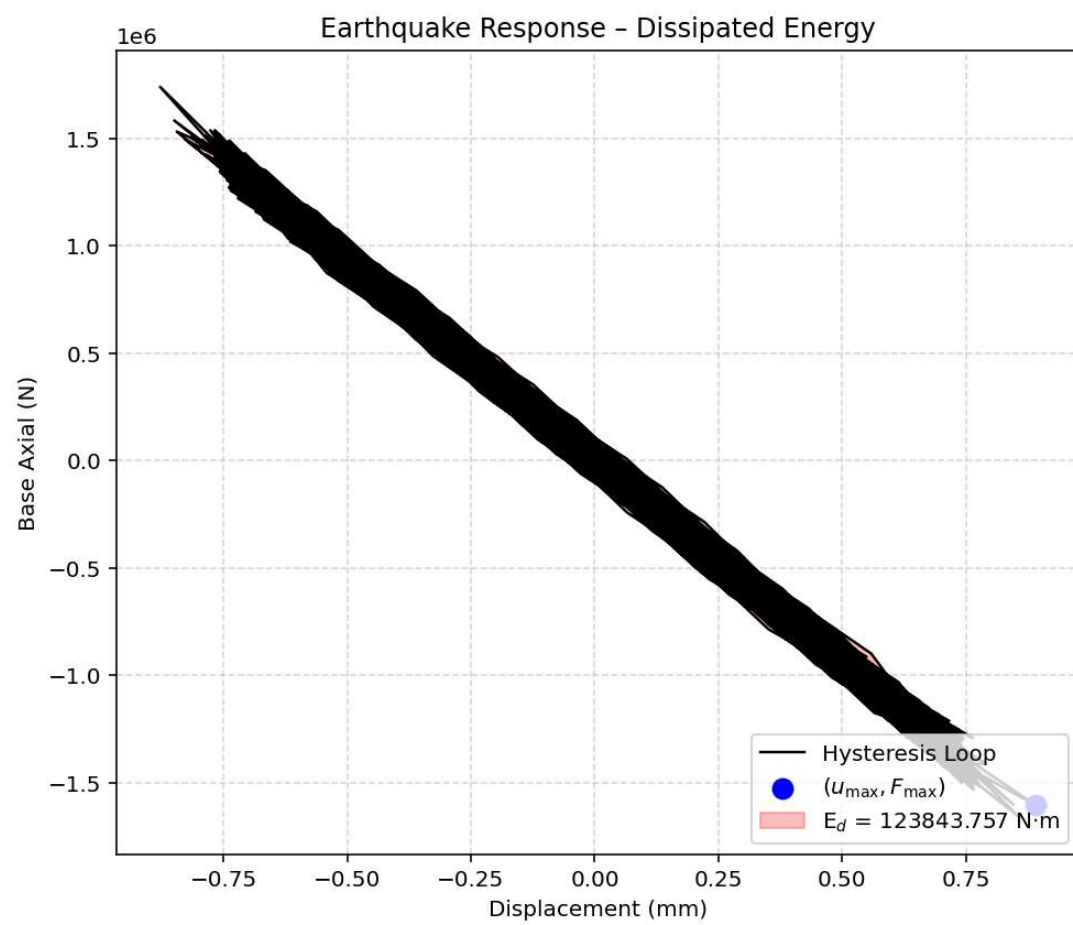
Equivalent viscous damping ratio - Seismic Hysteresis

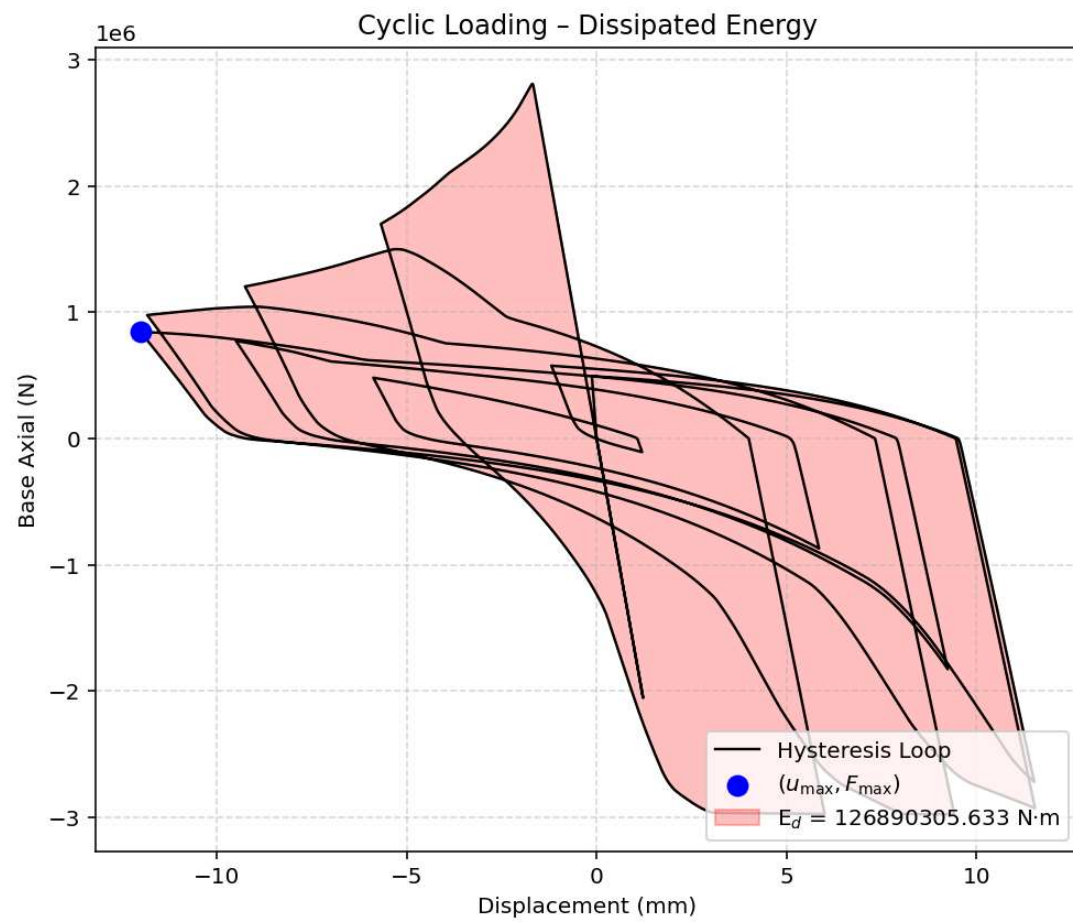










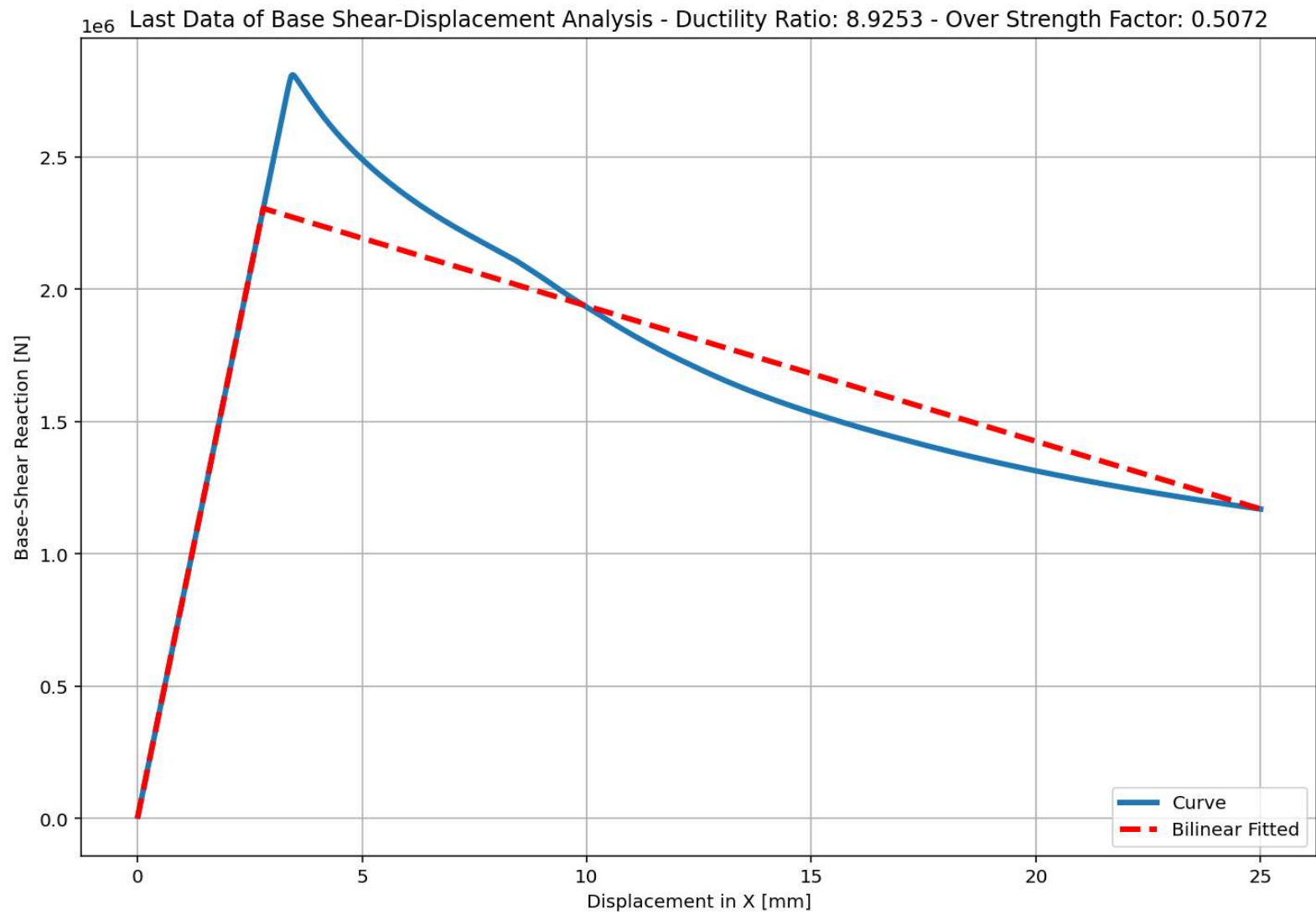


Dissipated Energy from Earthquake= 123843.76 N·mm

Dissipated Energy from Cyclic Displacement= 126890305.63 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.098 [%]

ZONE 1: VERY MINOR DAMAGE



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[2.80101456e+00 2.30592094e+06]

[2.50000000e+01 1.16951808e+06]]

+=====+

+-----+

Structure Elastic Stiffness : 823244.89

Structure Plastic Stiffness : 46780.72

Structure Tangent Stiffness : -51191.66

Structure Ductility Ratio : 8.93

Structure Over Strength Factor: 0.51

+-----+

Over Strength Coefficient (Ω_0): 0.5072

Displacement Ductility Ratio (μ): 8.9253

Ductility Coefficient (R_μ): 8.9253

Structural Behavior Coefficient (R): 4.5268

Structural Ductility Damage Index in X Direction: -6.7929 (%)

